QUANTITATIVE METHODS LEVEL 1

Trustville

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Lời giới thiệu

Tài liệu này được biên soạn và tổng hợp chủ yếu từ nguồn học liệu Curriculum level 1, với sự hỗ trợ tích cực từ AI (phụ trách tự động hóa các phần việc nhàm chán). Bên cạnh những trích dẫn, công thức và biểu đồ, đôi khi bạn sẽ gặp một số đoạn văn tiếng Việt in nghiêng và màu xanh là phần diễn giải bổ sung của Giảng viên biên soạn, hy vọng giúp giải thích rõ ràng hơn.

Năm 2027 đánh dấu sự thay đổi đáng kể trong môn Quantitative Methods, khi CFA đưa trở lại những khung kiến thức cơ bản, đồng thời mở rộng những ý hiểu sâu sắc giúp học viên xây dựng nền tảng kiến thức về định lượng và thống kê một cách vững vàng. Vì vậy, Giảng viên Trustville đánh giá nội dung mới của môn học là tương đối thách thức cho học viên so với giáo trình những năm trước đó.

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Mọi ý kiến đóng góp xin gửi về email: dongkhautu@gmail.com.

Chúc các bạn có những giờ học bổ ích và hoàn thành mục tiêu!

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Module 1

Returns of Financial Assets and Instruments

LEARNING OUTCOMES	
a	describe, compare, and interpret returns
b	describe, compare, and interpret required rates of return, risk-free rates, risk premia, and inflation

Financial Returns

Total Return
Arithmetic and Geometric Holding Period Return
Compounding: Daily, Monthly, Quarterly, and Annual Returns
Continuous Compounding

Common Return Measures

The Required Rate of Return
Excess Returns
Real Returns
Gross and Net Return
Pre-Tax and After-Tax Nominal Return
Leveraged Returns

1.1 Financial Returns

a | describe, compare, and interpret returns

Lãi suất là đối tượng thống kê chính trong hoạt động đầu tư. Có thể hiểu, thị trường tài chính là nơi buôn bán một loại hàng hóa đặc biệt, đó là Vốn. Lãi suất chính là mức giá của loại hàng hóa đặc biệt đó.

1.1.1 Total Return

$$\text{Total Return} = \text{Capital Appreciation Return} + \text{Capital Distribution Return}$$

$$\begin{aligned} r &= r_{\text{price}} + r_{\text{distribution}} \\ &= \frac{P_1 - P_0}{P_0} + \frac{\text{Inc}}{P_0} \end{aligned} \tag{1.1}$$

For capital distributions on equities, we focus on the dividend yield; for bonds, we look at current yield.

- Some expressions of Return:

Expected returns (ex ante returns): are the anticipated returns on an investment over a specified period.

Actual returns (ex post returns): are the returns on an investment after the investment period has ended.

Unrealized returns: changes in value represent while an investor holds an investment.

Realized returns: when the investment is ultimately sold, gains and losses would be realized at the current market value.

1.1.2 Arithmetic and Geometric Holding Period Return

To ensure accurate comparisons across different investments and time periods, it's crucial that these returns are calculated using a uniform time period.

- **arithmetic mean return**:

$$\bar{r}_A = \frac{r_1 + r_2 + \cdots + r_T}{T} = \frac{1}{T}\sum_{t=1}^{T} r_t \tag{1.2}$$

- **geometric mean return**:

$$\bar{r}_G = \sqrt[T]{(1+r_1)\times(1+r_2)\times\cdots\times(1+r_T)} - 1 = \left[\prod_{t=1}^{T}(1+r_t)\right]^{1/T} - 1 \tag{1.3}$$

Geometric mean is a more accurate reflection of investment performance over time. Arithmetic average return calculations assume that the amount invested at the beginning of the period does not change, but the base amount changes due to capital appreciation or depreciation. The arithmetic-geometric mean inequality highlights the impact of volatility on investment returns.

Exam Tip

Remember: - **Arithmetic Mean** is the best estimator of return for a single future period (ex-ante). - **Geometric Mean** represents the actual historical compound rate of return over multiple periods (ex-post). - Unless all period returns are identical, **Arithmetic Mean is always strictly greater than Geometric Mean** due to volatility ($\bar{r}_A > \bar{r}_G$).

1.1.3 Compounding: Daily, Monthly, Quarterly, and Annual Returns

With c is the number of compounding periods in a year,

$$r_{\text{annualized}} = (1 + r_{\text{period}})^c - 1 \tag{1.4}$$

One major limitation of annualizing returns is the implicit assumption that earned returns can be repeated.

1.1.4 Continuous Compounding

The more frequently we compound returns, the higher the annual return will be; however, there is a limit. As shown in Exhibit 1.1:

- **continuously compounded return**:

$$\tilde{r}_{t,t+1} = \ln\left(\frac{P_{t+1}}{P_t}\right) = \ln(1 + r_{t,t+1}) \tag{1.5}$$

The additive property of continuously compounded returns provides calculation advantages:

$$\begin{aligned}
\frac{P_T}{P_0} &= \frac{P_T}{P_{T-1}} \cdot \frac{P_{T-1}}{P_{T-2}} \cdots \frac{P_2}{P_1} \cdot \frac{P_1}{P_0} \\
\Rightarrow \ln\left(\frac{P_T}{P_0}\right) &= \ln\left(\frac{P_T}{P_{T-1}}\right) + \ln\left(\frac{P_{T-1}}{P_{T-2}}\right) + \cdots + \ln\left(\frac{P_2}{P_1}\right) + \ln\left(\frac{P_1}{P_0}\right) \\
\Rightarrow \tilde{r}_{0,T} &= \tilde{r}_{T-1,T} + \tilde{r}_{T-2,T-1} + \cdots + \tilde{r}_{1,2} + \tilde{r}_{0,1} = \sum_{t=1}^{T} \tilde{r}_{t-1,t}
\end{aligned}$$

Log return (lãi liên tục) được ưa chuộng do tính cộng gộp thời gian (time-additivity): tổng log return các ngày bằng log return cả năm, đơn giản hóa mô hình toán học.

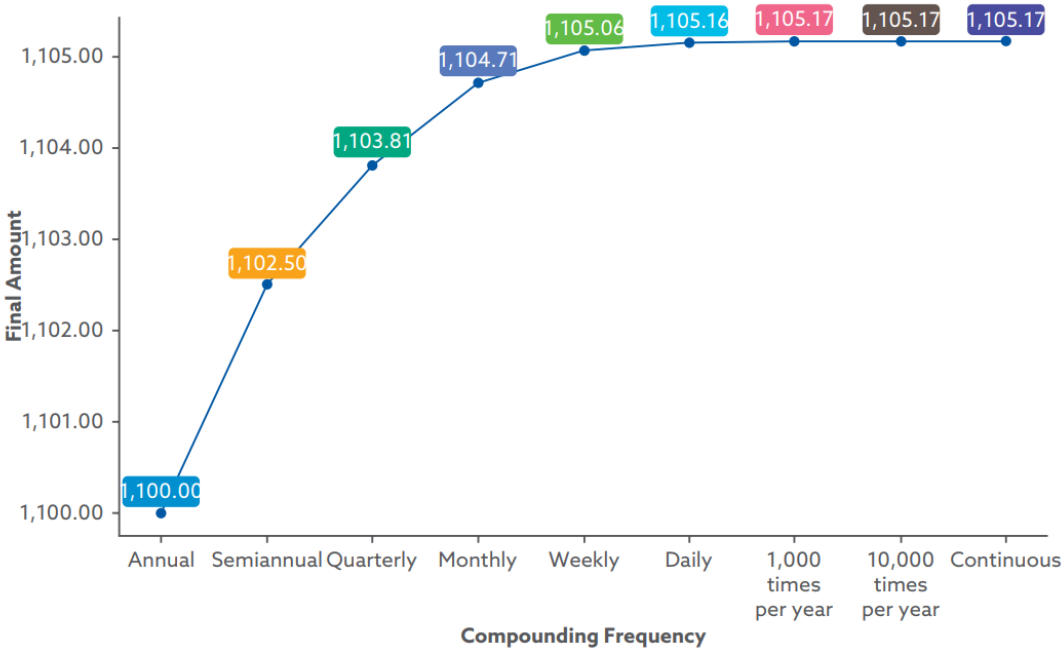


Figure 1.1: The Impact of Compounding

1.2 Common Return Measures

b | describe, compare, and interpret required rates of return, risk-free rates, risk premia, and inflation

1.2.1 The Required Rate of Return

- **opportunity cost**: that incurs when delaying consumption, or the value that investors forgo by choosing a course of action.
- **required rate of return**: the minimum rate of return an investor must receive to accept the investment.

Components of Required Rate of Return:

- **risk-free rate**: is the return on an investment without any default or reinvestment risk.

  It refers to returns on short-term government bonds, such as the 30-day US Treasury bill. The bonds have a short maturity, which reduces their sensitivity to interest rates changes.

  real risk-free rate reflects the time preferences of individuals for current versus future real consumption.

$$r_{\text{real_rf}} = \frac{1 + r_{\text{rf}}}{1 + \text{infl}} - 1 \quad \text{or quick approximation: } r_{\text{real_rf}} \approx r_{\text{rf}} - \text{infl} \tag{1.6}$$

- Total required returns for any asset can then be approximated as:

$$r = r_{\text{rf}} + \text{infl} + \sum_{j=1}^{n} (f_j \times \text{rp}_j) \tag{1.7}$$

  n is the number of risk factors

  f_j is the exposure of the security to the jth risk factor

  rp_j is the specific risk premium associated with the jth risk factor

  $\sum_{j=1}^{n} (f_j \times \text{rp}_j)$ is the sum of all risk premia.

As shown in Exhibit 1.2:
Examples of risk premia for bonds:

- **Default risk premium**: the additional return for buying a riskier corporate bond instead of a less-risky government bond.
- **Liquidity premium**: the additional return for holding a bond that may be harder to sell quickly.

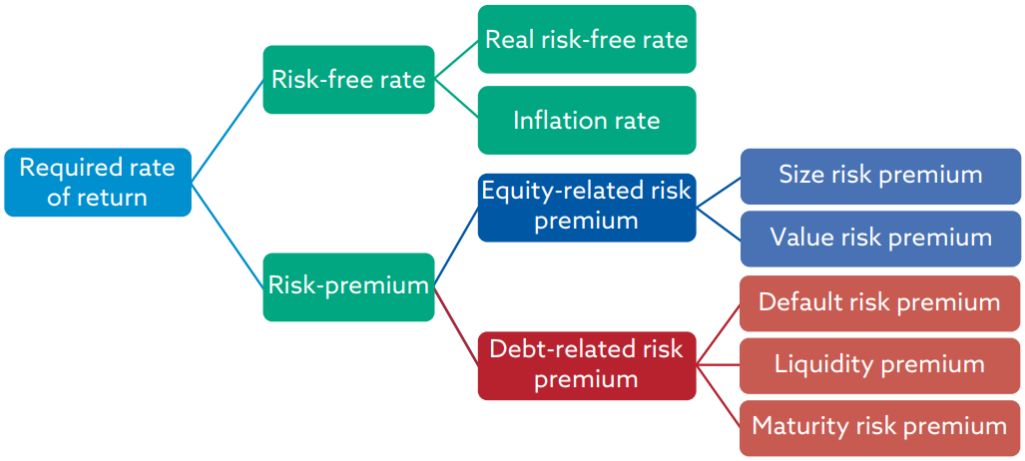


Figure 1.2: Components of Required Rate of Return and Examples of Different Risk Premia

- **Maturity risk premium**: the additional return for holding more interest rate sensitive, longer maturity bonds.

Examples of risk premia for stocks:

- **Size risk premium**: the return spread between small and large market-capitalization stocks.
- **Value risk premium**: the return spread between value stocks (high book-to-market) and growth stocks (low book-to-market).

1.2.2 Excess Returns

Excess returns: the difference between an asset's return and the return on some reference rate or benchmark.

$$\dot{r}_i = \frac{1 + r_i}{1 + r_B} - 1 \quad \text{or quick approximation: } \dot{r}_i \approx r_i - r_B \tag{1.8}$$

1.2.3 Real Returns

Real returns are inflation-adjusted returns.

$$r_{\text{real}} = \frac{1 + r_i}{1 + \text{infl}} - 1 \quad \text{or quick approximation: } r_{\text{real}} \approx r_i - \text{infl} \tag{1.9}$$

1.2.4 Gross and Net Return

Gross return = Total return - Transaction costs

An appropriate measure for evaluating and comparing the investment skill of managers.

Excludes indirect expenses related to the management and administration of an investment.

Net return = Gross return - Management fees - Administrative expenses

The return the investor actually earns.

Exhibit 1.3 depicts the relationship and shows how costs erode investment returns:

1.2.5 Pre-Tax and After-Tax Nominal Return

After-tax nominal return:

$$r_{\text{net-tax}} = r_{\text{price}} \times (1 - \text{tax}_{\text{capital gains}}) + r_{\text{capital}} \times (1 - \text{tax}_{\text{capital distribution}}) \tag{1.10}$$

Capital gains tax is a tax on a positive price return. Two forms of capital gains tax rate:

Short-term capital gains are levied on capital gains on short-term holdings.

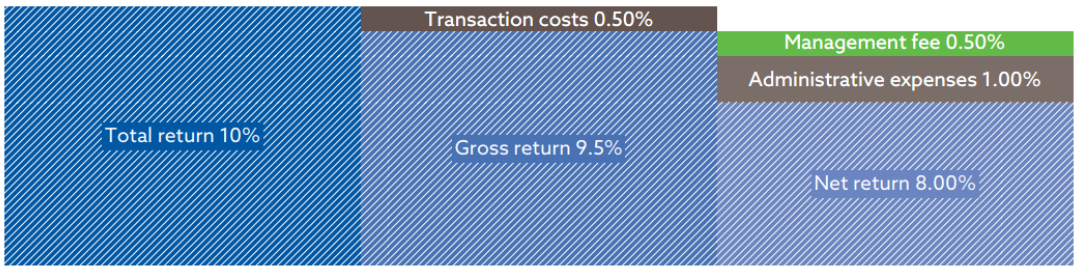


Figure 1.3: The Relationship between Total Return, Gross Return, and Net Return

Long-term capital gains are levied on longer-term holdings and receive preferential tax treatment in a number of countries.

Forms of capital distribution tax rate:

Interest income is taxed as ordinary income in most countries.

Dividend income may be taxed as ordinary income, at a lower rate, or be exempt from taxes depending on the jurisdiction.

1.2.6 Leveraged Returns

Using leverage to increase the total amount of investible assets is a common feature of investments, whether through borrowing or derivatives. Leverage increases investible capital, magnifying both returns and risks.

$$R_L = R_A + \frac{D}{E}(R_A - R_D) \tag{1.11}$$

R_L is the leveraged return (Return on Equity)

R_A is the return on total assets (unleveraged return)

R_D is the cost of debt (borrowing interest rate)

D/E is the debt-to-equity ratio.

Đòn bẩy là dao hai lưỡi. Khi $R_A > R_D$, đòn bẩy phóng đại suất sinh lời vốn tự có R_L. Ngược lại nếu $R_A < R_D$, tỷ suất sinh lời sẽ sụt giảm rất nhanh!

Module 2

Types of Returns for Financial Assets, Instruments, and Indicators

LEARNING OUTCOMES	
a	calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators

Types of Returns

Types of Returns: Financial Assets

Types of Returns: Financial Instruments

Types of Returns: Financial Indicators

Long-Term Financial Asset Returns

Historical Long-Term Asset Returns

Long-Term Asset Return Components

Risks and Returns

Systematic and Unsystematic Risk

The Risk–Return Trade-Off

Portfolio Return Calculation

2.1 Types of Returns

a	calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators

Cập nhật chương trình 2027

Nội dung phân loại chi tiết suất sinh lời thành: Tài sản tài chính (Financial Assets), Công cụ tài chính (Financial Instruments) và Chỉ báo tài chính (Financial Indicators) là phần được cập nhật và làm rõ mới trong giáo trình 2027. Học viên cần lưu ý cách phân biệt và đặc trưng sinh lời của từng loại.

2.1.1 Matrix Mapping of Return Types

To provide a scientific and intuitive overview, Table 2.1 maps the characteristics of price return and capital distribution return across all assets, instruments, and indicators.

Table 2.1: Matrix Mapping of Return Types across Assets, Instruments, and Indicators

Category	Price Return	Capital Distribution Return
Financial Assets		
- Cash	None (zero price change).	None (unless held in interest-bearing accounts).
- Equities	Highly volatile and uncertain (Capital Appreciation).	Variable and discretionary (Dividends).
- Bonds (Debt)	Inversely related to interest rates; higher volatility for long-term bonds.	Fixed and contractual (Coupon payments).
- Hybrids	Moderate appreciation potential.	Fixed or structured (Preferred dividends).
Financial Instruments		
- Standardized legal agreements	It depends on what it is (equity, bonds, derivatives or securitized debts).	It depends.
- Derivatives	Determined by price movements of underlying assets.	None (does not distribute cash flows).
Financial Indicators		
- Interest Rates	Inverse relation to debt instrument prices.	None (acts as valuation driver).
- FX Rates	Currency appreciation/depreciation (r_{FX}).	None (acts as return multiplier: $1 + r_{\text{FX}}$).
- Market Indexes	Reflects aggregate price movements.	Reflects weighted cash distributions of constituents.

2.1.2 Types of Returns: Financial Assets

Let's detail the characteristics across major financial assets:

- **Cash:** does not generate any return unless invested. In the short term, excess cash is typically placed in interest-bearing bank accounts or short-term debt instruments.
- **Equities:** generate returns from capital appreciation and discretionary dividends (variable capital distributions). Rapidly growing companies often choose not to distribute dividends and instead reinvest earnings to fuel growth.
- **Bonds (Debt):** represent contractual lending agreements that pay fixed interest coupons (fixed capital distributions). Shorter-term bonds are dominated by capital distributions, whereas longer-term bonds exhibit substantial price volatility due to market interest rate movements.
- **Hybrids:** preferred stock and convertible bonds blend fixed-income features of debt with the growth potential of equity.

2.1.3 Types of Returns: Financial Instruments

- **Financial instruments** are standardized legal agreements for financial assets (such as equity and bonds), as well as derivatives and securitized debt. The standardization facilitates public trading. For example, Cash Instruments represent direct ownership claims on financial assets (e.g., actual shares of stock or actual corporate bonds).
- **Derivatives:** financial instruments whose value is derived from the performance of another underlying financial asset or indicator (e.g., stock price, interest rate, index, exchange rate). A call option is a typical example.

2.1.4 Types of Returns: Financial Indicators

Financial indicators do not directly generate cash flows, but are critical to financial market operations.

- **Interest Rates:** traded through risk-free government bonds. Their level varies inversely with bond prices.
- **The Risk-Free Rate:** represented by short-term, risk-free government debt (e.g., US Treasury bills).

- **Foreign Exchange (FX) Rates:** represent the value of one currency expressed in terms of another currency.

When investing internationally, the home-currency return (r_{home}) depends on both the investment return in foreign currency (r_{foreign}) and the currency appreciation/depreciation rate (r_{FX}):

$$r_{\text{home}} = (1 + r_{\text{foreign}})(1 + r_{\text{FX}}) - 1 \approx r_{\text{foreign}} + r_{\text{FX}} \tag{2.1}$$

Where:

$$r_{\text{FX}} = \frac{\text{Ending FX Rate} - \text{Beginning FX Rate}}{\text{Beginning FX Rate}} \tag{2.2}$$

Example: Foreign Exchange FX Adjustment on Returns

A US dollar-based investor purchases a bond denominated in Euros (EUR).

- **Foreign asset return:** $r_{\text{foreign}} = -2.91\%$.
- **FX rate change (EUR to USD):** $r_{\text{FX}} = -6.90\%$.
- **Home-currency return calculation:**

$$r_{\text{home}} = (1 - 0.0291)(1 - 0.0690) - 1 = -9.61\%$$

Tác động kép từ trái phiếu giảm giá và EUR mất giá khuếch đại khoản lỗ bằng USD.

- **Market Indexes:** reflect the aggregate value of a basket of assets (e.g., S&P 500, OMX Copenhagen 25 Index) to indicate market trends.

2.2 Long-Term Financial Asset Returns

2.2.1 Historical Long-Term Asset Returns

Based on historical data for selected major US asset classes between 1979 and 2023 (large-cap stocks, small-cap stocks, corporate bonds, government bonds, and Treasury bills), several trends emerge:

- **Equities Outperform Bonds:** driven primarily by capital appreciation, equities deliver higher long-term average returns than fixed-income indexes.
- **Small-Cap Premium:** small-capitalization stocks have periodically outperformed large-capitalization stocks (size premium), though this premium varies over time and can occasionally be negative.
- **Corporate vs. Government Debt:** corporate bonds offer higher coupon yields on paper, but government bonds can occasionally match or exceed corporate bond performance over specific horizons due to the impact of default losses.

2.2.2 Long-Term Asset Return Components

The relative contribution of capital appreciation and capital distributions has evolved significantly:

- **Equities:** dividend yields have trended downward over time, particularly following a 1982 regulatory change that established a safe harbor for share buybacks. Companies increasingly repurchase shares to return cash to shareholders, which reduces share count, increases stock value, and provides tax advantages (capital gains tax is generally lower than dividend tax).
- **Government Bonds:** income return (coupon payments) remains the dominant driver of total returns, historically accounting for approximately 85% of total government bond returns. Capital appreciation plays a minor role, and bond prices suffer capital losses during periods of rising interest rates (such as the early 2020s).

2.3 Risks and Returns

Risk represents the uncertainty of investment returns and can be categorized into macro and micro risks:

- **Systematic Risk (Macro Risks):** broad, economy-wide risks that affect all financial assets (e.g., inflation risk, market interest rate risk, and market risk). Systematic risk cannot be diversified away.

- **Unsystematic Risk (Micro Risks):** risks specific to a particular issuer or sector (e.g., credit/default risk, business risk, financial leverage, sector/industry risk). Unsystematic risk can be mitigated through diversification.

The Risk–Return Trade-Off: Investors demand a higher expected return (risk premium) to compensate for taking on higher volatility and uncertainty. Exhibit 2.1 shows that historically, higher average annual returns are generally associated with a wider range of annual returns (higher volatility).

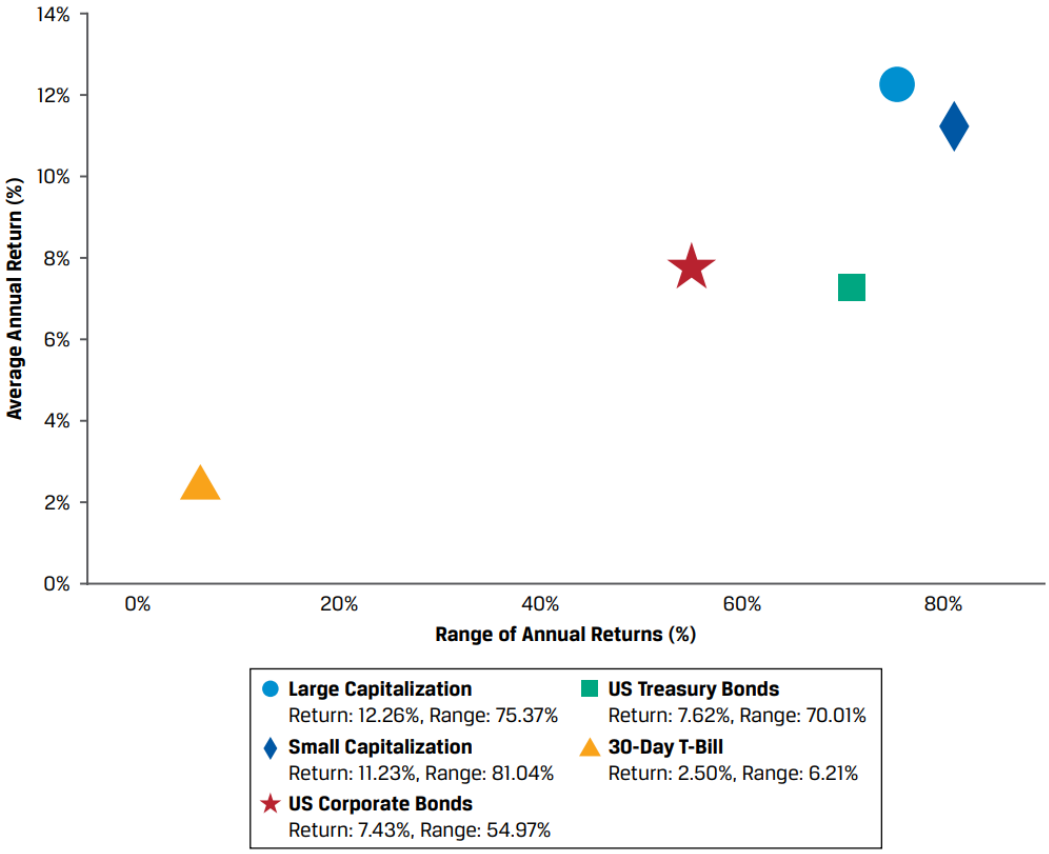


Figure 2.1: Historical Geometric Average Return for Different Assets (1979–2023)

Module 3

Benchmarking Returns

LEARNING OUTCOMES	
a	calculate and compare money-weighted and time-weighted rates of return
b	describe the choices and the implications of the different weighting methods used in index construction and management, and calculate, interpret, and explain the value and the returns of an index

Money-Weighted and Time-Weighted Rates of Return

Index Definition and Calculation

Price Return and Total Return Indexes
Index Weightings
Index Management
Comparing Index Performance

3.1 Money-Weighted and Time-Weighted Rates of Return

a | calculate and compare money-weighted and time-weighted rates of return

Cần phân biệt TWR (hiệu quả quản lý quỹ) và MWR (hiệu quả đầu tư thực tế của cá nhân chịu ảnh hưởng từ dòng tiền).

3.1.1 Time-Weighted Rate of Return (TWR)

The time-weighted rate of return measures the compounded rate of growth of an investment portfolio over a specified horizon. To eliminate the distorting effects of external cash flows (which the investment manager typically does not control), the total investment horizon is broken down into subperiods at each cash flow event. The returns for each subperiod are calculated and then geometrically linked:

$$r_{\text{TW}} = \prod_{t=1}^{T}(1 + r_t) - 1 \tag{3.1}$$

Note that r_{TW} gauges a return for the whole time of all subperiod. If you want to estimate a mean return for **each** subperiod, you have to use r_{G}

$$r_{\text{G}} = \sqrt[T]{\prod_{t=1}^{T}(1 + r_t)} - 1 \tag{3.2}$$

Where:

r_{TW} is the time-weighted rate of return.

T is the total number of subperiods.

r_t is the holding period return for subperiod t, calculated between two successive cash flow events.

Because TWR measures performance between periods and filters out investor-driven contributions or withdrawals, it is the standard metric used to evaluate the investment skill of public market fund managers.

3.1.2 Money-Weighted Rate of Return (MWR)

The money-weighted rate of return (or dollar-weighted rate of return) measures the internal rate of return (IRR) of the investment. It accounts for both the size and timing of all cash flows. The MWR is the discount rate that sets the net present value of all cash flows equal to zero:

$$0 = \sum_{t=0}^{T} \frac{\mathrm{CF}_t}{(1 + r_{\mathrm{MW}})^t} \tag{3.3}$$

Where:

r_{MW} is the money-weighted rate of return.

CF_t is the net cash flow at time t.

Quy ước dấu dòng tiền: Dầu tư ban đầu/góp thêm mang dấu âm (−); cổ tức, rút vốn/giá trị cuối kỳ mang dấu dương (+). Sử dụng máy tính tài chính (chức năng CF) hoặc Excel để giải MWR.

3.1.3 Case Study

Case Study: MWR vs TWR

Table 3.1: Money-Weighted Return for a Dividend-Paying Stock

Time	Outflows
0	USD200 to purchase the first share
1	USD225 to purchase the second share
Time	**Inflows**
1	USD5 dividend received from first share (and not reinvested)
2	USD10 dividend (USD5 per share × 2 shares) received
2	USD470 received from selling two shares at USD235 per share

$$\Rightarrow CF_0 = -200 \qquad CF_1 = -220 \qquad CF_2 = +480$$

$$\Rightarrow \frac{CF_0}{(1+IRR)^0} + \frac{CF_1}{(1+IRR)^1} + \frac{CF_2}{(1+IRR)^2} = \frac{-200}{1} + \frac{-220}{(1+IRR)^1} + \frac{+480}{(1+IRR)^2}$$

$$\Rightarrow IRR = 9.39\%$$

Importantly, two investors in the same mutual fund or with the same portfolio of underlying investments may have different money-weighted returns because they invested different amounts in different years.

Exam Tip

Texas Instruments BA II Plus Keystrokes for MWR:

1. Press **[CF]** → **[2nd]** **[CE|C]** → **[CLR WRK]** to clear worksheet memory.
2. Enter $\mathrm{CF}_0 = -200$: Press **200** → **[+/-]** → **[ENTER]**.
3. Press [↓] to display **C01**.
4. Enter $\mathrm{CF}_1 = -220$: Press **220** → **[+/-]** → **[ENTER]**.
5. Press [↓] → [↓] to display **C02** (F01 defaults to 1).
6. Enter $\mathrm{CF}_2 = 480$: Press **480** → **[ENTER]**.
7. Press **[IRR]** then **[CPT]**.

$HPR_1 = 15\%, HPR_2 = 6.7\%$, trong khi đó $MWR = IRR = 9.39\%, TWR = 10.76\%$. Bằng một ước tính gần đúng, ta có: (bạn có thể tự tính lại để kiểm tra)

$$TWR \approx \frac{HPR_1 + HPR_2}{2}$$

$$MWR \approx \frac{|CF_1| \cdot HPR_1 + |CF_2| \cdot HPR_2}{|CF_1| + |CF_2|}$$

Như vậy, dựa trên quan sát kết quả, ta nhận thấy, lợi suất của tài sản trong kỳ đầu tiên là cao hơn tương đối so với kỳ thứ hai ($HPR_1 > HPR_2$).
Dựa trên công thức xấp xỉ ở trên, ta có nhận xét rằng lợi suất bình quân tính theo MWR sẽ có trọng số dồn vào kỳ thứ hai nhiều hơn (do có dòng tiền phát sinh lớn hơn $|CF_2| > |CF_1|$, ở đây ta chỉ xét về độ lớn của dòng tiền, không xét về dấu). Vậy nên, có thể hiểu rằng, dòng tiền lớn hơn đang dồn về kỳ có lợi suất thấp, nên nhìn chung sẽ hạ mức lợi suất trung bình xuống (vì vậy mà mức lợi suất này có tên gọi là money-weighted).
Còn nếu sử dụng TWR, ta coi mức lợi suất bình quân tính theo lợi suất từng kỳ với tỷ trọng như nhau, mà không phụ thuộc vào hiệu ứng dòng tiền đổ vào mỗi kỳ nhiều hay ít.

3.1.4 Summary of Differences

Table 3.2 summarizes the key comparisons between the two return methodologies.

Table 3.2: Comparison of Money-Weighted and Time-Weighted Returns

Dimension	Money-Weighted Return (MWR)	Time-Weighted Return (TWR)
Core Objective	Measures the actual growth rate of the investor's funds.	Measures the compound growth rate of the portfolio assets.
Cash Flow Sensitivity	Highly sensitive to the size and timing of external cash flows.	Completely unaffected by external cash flows.
Calculation Base	Internal Rate of Return (IRR) equation.	Geometric linkage of subperiod returns.
Performance Bias	Can misrepresent manager skill if they cannot control cash flow timings.	Provides an objective measure of the manager's investment selection skill.
Primary Suitability	Recommended for private equity (where managers call capital) and individual investors.	Mandated by GIPS for public market mutual funds and active asset managers.

3.2 Index Definition and Calculation

b | describe the choices and the implications of the different weighting methods used in index construction and management, and calculate, interpret, and explain the value and the returns of an index

Market Index làm Benchmark và cơ sở cho ETF. Khi thiết kế chỉ số, cần lưu ý: (1) Cách tính (Price/Total Return) và (2) Phương pháp phân bổ tỷ trọng (Weighting Scheme).

Cập nhật chương trình 2027

Toàn bộ phần "Index Definition and Calculation" (xây dựng, quản lý chỉ số, các phương pháp trọng số giá/vốn hóa và tái cơ cấu danh mục chỉ số) là nội dung hoàn toàn mới được đưa vào cấu phần Quantitative Methods năm 2027. Trước đây nội dung này nằm ở các môn học khác (như Portfolio Management hay Equity).

3.2.1 Price Return vs. Total Return Indexes

- **Price Return Index ($r_{\mathbf{PR}}$):** reflects only the prices of the constituent securities and measures capital appreciation:

$$r_{\text{PR}} = \frac{P_{t+1}}{P_t} - 1 \tag{3.4}$$

- **Total Return Index ($r_{\mathbf{TR}}$):** reflects both the prices and the reinvestment of all capital distributions (dividends/interest):

$$r_{\text{TR}} = \frac{P_{t+1} + \text{Div}_{t+1}}{P_t} - 1 \tag{3.5}$$

Over time, the total return index will significantly outperform the price return index due to the compounding effect of reinvested cash distributions.

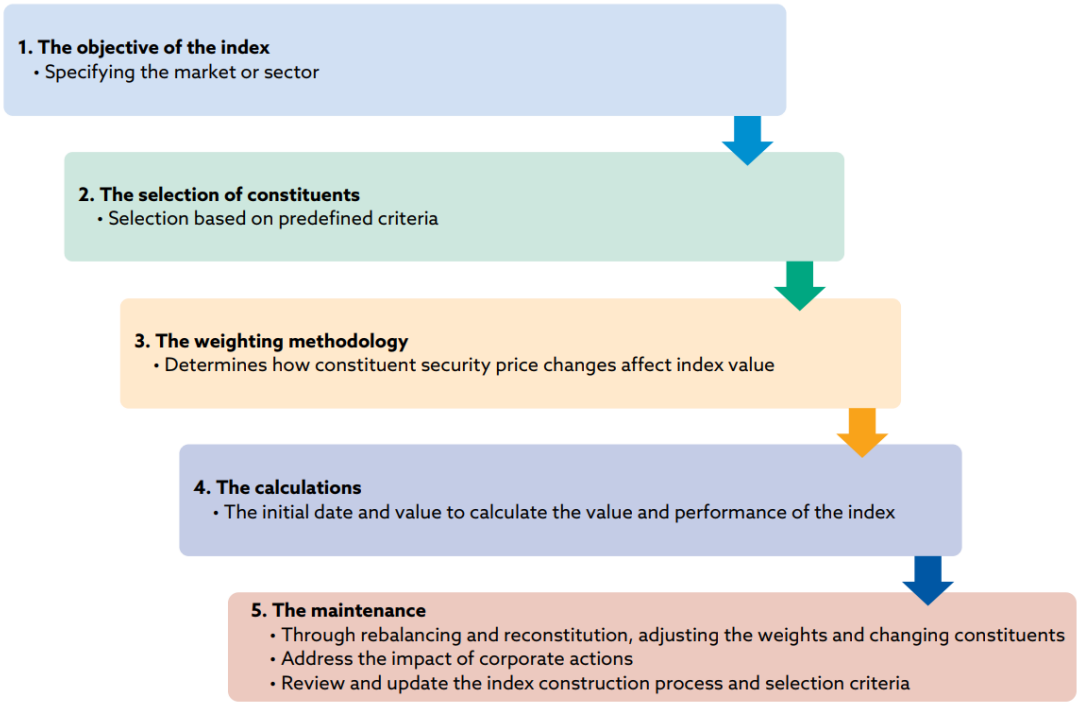


Figure 3.1: Steps in the Index Construction, Calculation, and Management Process

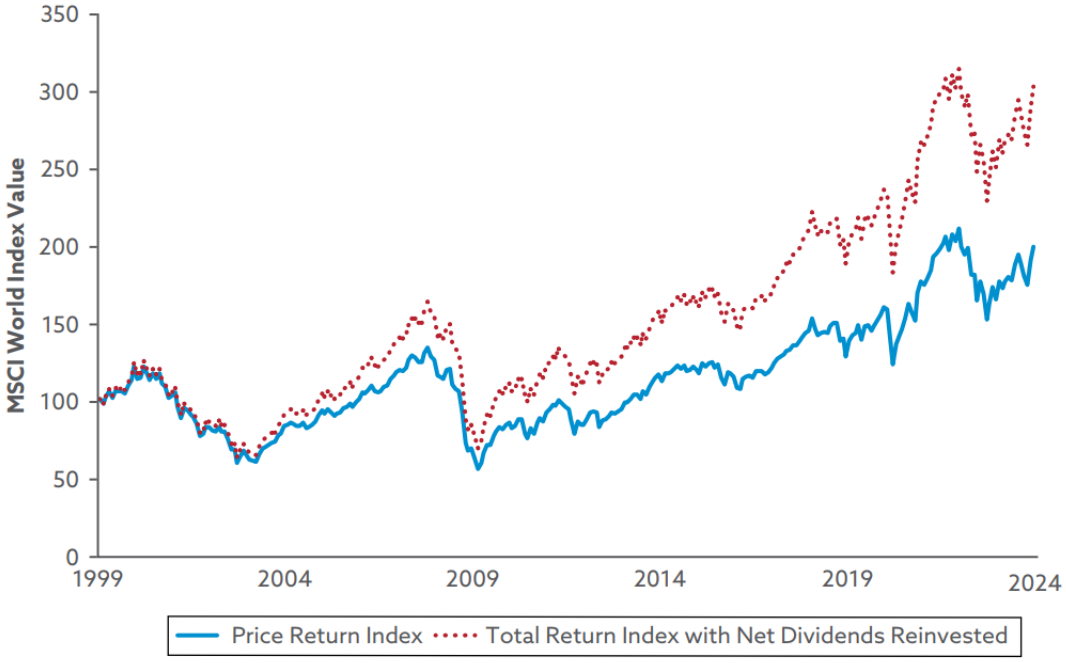


Figure 3.2: MSCI World Index Price Return vs. Total Return (1999–2023)

3.2.2 Index Weighting Methodologies

The general formula for calculating an index value at time t is:

$$\text{Index Value}_t = \sum_{i=1}^{N} w_{i,t} \times P_{i,t} \tag{3.6}$$

Where:

$w_{i,t}$ is the weight of constituent i at time t.

$P_{i,t}$ is the price of constituent i at time t.

Let's review the five primary weighting methodologies:

1. **Market Capitalization Weighting:** Weights are proportional to the total market capitalization ($Q_i P_i$):

$$w_i^M = \frac{Q_i P_i}{\sum_{j=1}^{N} Q_j P_j} \tag{3.7}$$

The index value is often calculated using a divisor (D_t) to normalize the value:

$$V_t = \frac{\sum_{i=1}^{N} Q_{i,t} P_{i,t}}{D_t} \tag{3.8}$$

2. **Float-Adjusted Market Cap Weighting:** Accounts only for shares available to the public ($f_i Q_i$), excluding strategic holdings:

$$w_i^{M,\text{Float}} = \frac{f_i Q_i P_i}{\sum_{j=1}^{N} f_j Q_j P_j} \tag{3.9}$$

3. **Fundamental Weighting:** Weights are based on economic or accounting indicators (e.g., book value, revenues, earnings, or dividends) instead of market prices:

$$w_i^F = \frac{F_i}{\sum_{j=1}^{N} F_j} \tag{3.10}$$

4. **Price Weighting:** Weights are proportional to the security price per share:

$$w_i^P = \frac{P_i}{\sum_{j=1}^{N} P_j} \tag{3.11}$$

The index value is calculated by dividing the sum of constituent prices by a divisor (D_t):

$$V_{\text{PRI},t} = \frac{\sum_{i=1}^{N} P_{i,t}}{D_t} \tag{3.12}$$

5. **Equal Weighting:** Each security is assigned an equal share at the inception/rebalancing date:

$$w_i^E = \frac{1}{N} \tag{3.13}$$

The index value is:

$$V_{\text{EQI},t} = \frac{\sum_{i=1}^{N} n_i P_{i,t}}{D_t} \tag{3.14}$$

3.2.3 Summary of Weighting Methodologies

Table 3.3 compares the characteristics of these methodologies.

Chỉ số vốn hóa tự điều chỉnh tỷ trọng (Self-Rebalancing), giúp giảm chi phí giao dịch. Tuy nhiên, nó bị ảnh hưởng bởi đà tăng giá (momentum bias), ép mua nhiều các cổ phiếu đang bị định giá quá cao.

Table 3.3: Comparison of Index-Weighting Methodologies

Index Type	Weighting Basis	Examples	Pros	Cons
Market Cap	Proportional to market capitalization.	S&P 500, MSCI World, S&P BSE SENSEX	Reflects relative sizes; low turnover; automatic rebalancing.	Concentrated in top firms; overweights overpriced stocks.
Equal Weighted	Identical weight $(1/N)$.	S&P 500 Equal Weight, MSCI USA Equal Weight	Reduces concentration risk; enhances exposure to smaller firms.	Requires frequent rebalancing; high turnover and transaction costs.
Fundamental	Economic metrics (sales, dividends, book value).	FTSE RAFI US 1000	Focuses on firm size over market sentiment; value bias.	Sector bias; excludes firms with zero earnings or dividends.
Price Weighted	Share price (P_i).	Dow Jones Industrial Average (DJIA), Nikkei 225	Simple to calculate and interpret.	High price stocks dominate index; stock splits distort weights.

3.2.4 Index Management: Rebalancing and Reconstitution

- **Periodic Rebalancing:** adjusts the weights of the constituents back to their target levels. This is critical for equal-weighted and fundamentally weighted indexes because price variations cause weights to drift. Equal-weighted indexes require counter-trend trading (selling winners and buying losers), which increases transaction costs and portfolio turnover.
- **Reconstitution:** the periodic process of adding or removing constituent securities from the index based on eligibility rules. Widely followed indexes often apply a *banding strategy* (e.g., Russell 2000's 5% band) to reduce unnecessary index turnover at the breakpoints.

3.2.5 Divisor Adjustment for Stock Splits

In a price-weighted index, when a constituent stock splits, its price per share drops mechanically without any change in its actual economic value or the firm's market capitalization. To prevent the index value from dropping artificially, the divisor (D) must be adjusted downward:

$$D_{\text{new}} = \frac{\sum P_{\text{new}}}{V_{\text{PRI, before}}} \tag{3.15}$$

Example: Divisor Adjustment for Stock Splits

Consider a price-weighted index with 5 stocks: A (50), B (40), C (30), D (25), E (5). Divisor $D_{\text{old}} = 1.5$.

- **Before Split:** The index value is:

$$V_{\text{PRI}} = \frac{50 + 40 + 30 + 25 + 5}{1.5} = 100$$

- **After 2-for-1 Split of A:** Stock A's price drops from 50 to 25. The new price sum is $25 + 40 + 30 + 25 + 5 = 125$.
- **Divisor Adjustment:** To keep the index value unchanged at 100:

$$D_{\text{new}} = \frac{125}{100} = 1.25$$

Khi cổ phiếu chia tách làm giá giảm, Divisor phải giảm tương ứng để giữ điểm số chỉ số không bị thay đổi ảo.

Module 4

The Time Value of Money in Finance

LEARNING OUTCOMES	
a	calculate and interpret the present value of fixed-income and equity instruments based on expected future cash flows
b	calculate and interpret the implied return of fixed-income instruments and required return and implied growth of equity instruments given their present value and cash flows
c	explain the cash flow additivity principle and its importance for the condition of no arbitrage, and explain its use in calculating implied forward interest rates, forward exchange rates, and option values

Time Value of Money in Fixed Income and Equity

Fixed-Income Instruments

Equity Instruments

Implied Return and Growth

Implied Return for Fixed-Income Instruments

Equity Instruments, Implied Return, and Implied Growth

Cash Flow Additivity

Implied Forward Rates Using Cash Flow Additivity

Forward Exchange Rates Using No Arbitrage

Option Pricing Using Cash Flow Additivity

4.1 Time Value of Money in Fixed Income and Equity

a | calculate and interpret the present value of fixed-income and equity instruments based on expected future cash flows

The relationship between a current or present value, PV_0, and future value, FV_t, of a cash flow—where r is the stated discount rate per period and t is the number of compounding periods—is:

$$FV_t = PV_0(1+r)^t \tag{4.1}$$

If the compounding frequency approaches infinity (continuous compounding), we calculate the future value as follows:

$$FV_t = PV_0 e^{r\times t} \tag{4.2}$$

Conversely, we can solve for the present value of a future cash flow under simple and continuous compounding:

$$PV_0 = FV_t(1+r)^{-t} = \frac{FV_t}{(1+r)^t} \tag{4.3}$$

$$PV_0 = FV_t e^{-r\times t} \tag{4.4}$$

4.1.1 Fixed-Income Instruments and the Time Value of Money

Cash flows associated with fixed-income instruments (debt) usually follow one of three general patterns:

1. **Discount (Zero-Coupon) Bonds:** Single payment representing both interest and principal repayment at maturity.
2. **Coupon Bonds:** Regular periodic interest coupon payments with the principal repaid at maturity.
3. **Annuities / Fully Amortizing Loans:** Equal periodic payments containing both interest and principal over the life of the instrument.

Discount Instruments

The cash flow pattern of a discount bond is illustrated in Exhibit 4.1.

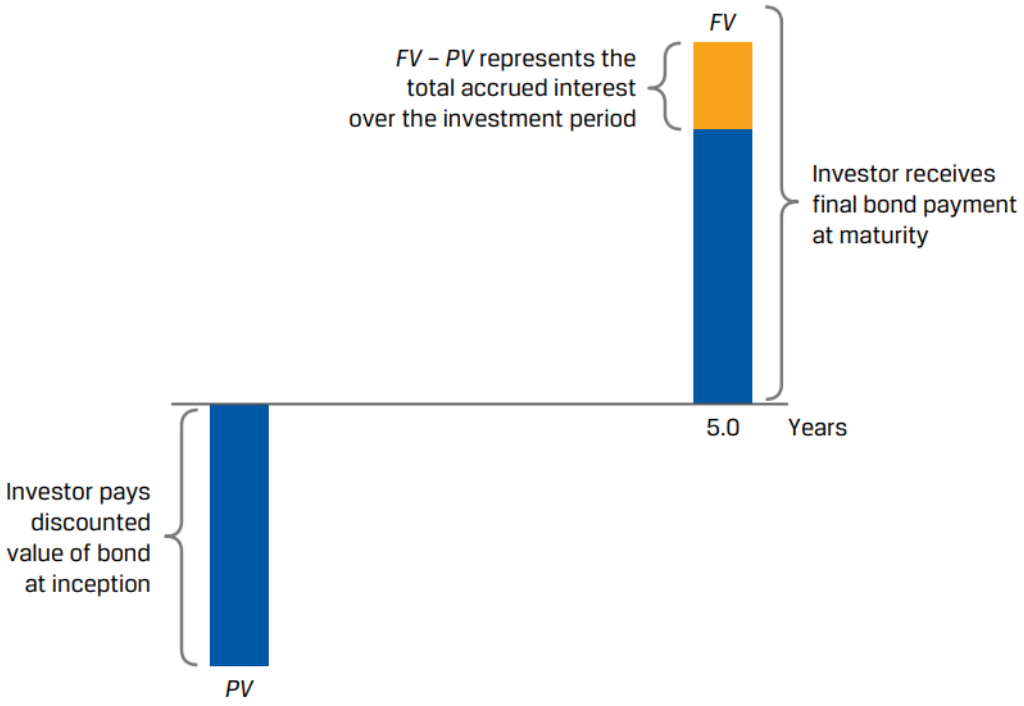


Figure 4.1: Discount Bond Cash Flows

The present value calculation for a discount bond with principal FV_t repaid fully after t periods with a periodic market discount rate of r per period is:

$$PV_{\text{Discount}} = \frac{FV_t}{(1+r)^t} \tag{4.5}$$

Coupon Instruments

The cash flow pattern of a coupon bond is illustrated in Exhibit 4.2.

The price of a coupon bond is the sum of the present values of all periodic coupon payments and the present value of the final principal repayment:

$$PV_{\text{Coupon}} = \sum_{t=1}^{T} \frac{PMT}{(1+r)^t} + \frac{FV}{(1+r)^T} \tag{4.6}$$

Using the sum of a geometric series, this simplifies to:

$$PV_{\text{Coupon}} = \frac{PMT}{r}\left[1 - \frac{1}{(1+r)^T}\right] + \frac{FV}{(1+r)^T} \tag{4.7}$$

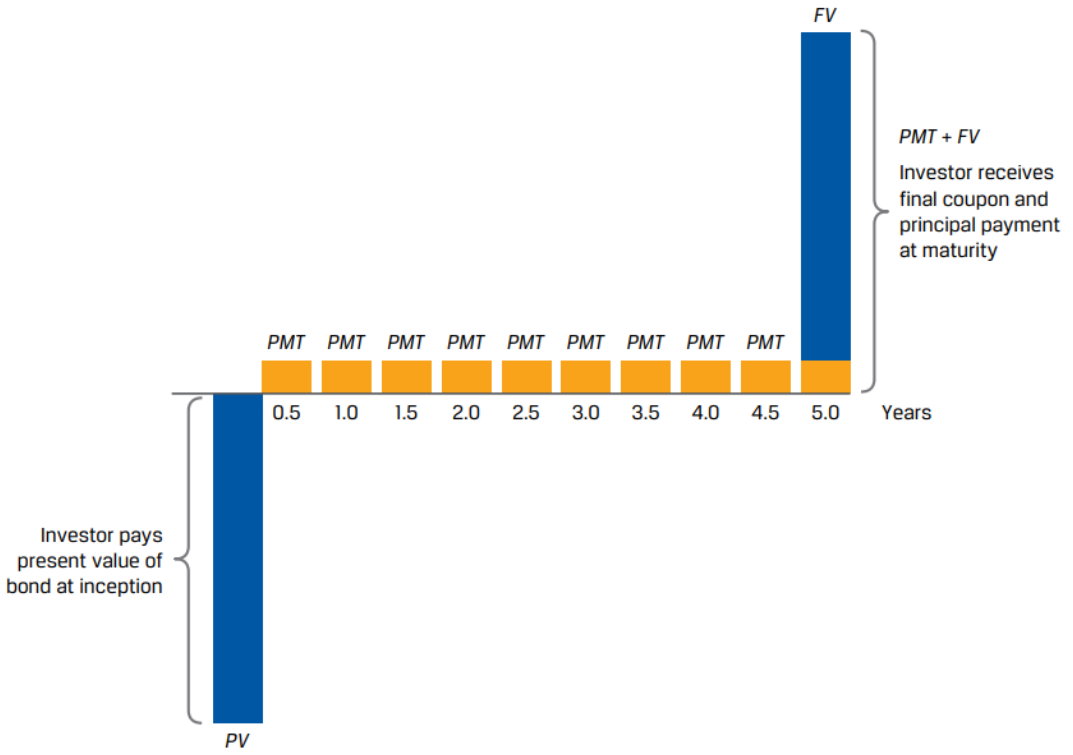


Figure 4.2: Coupon Bond Cash Flows

Floating-Rate and Perpetual Bonds

- **Floating-Rate Bonds:** Adjust their coupon payments periodically based on a reference rate (e.g., NIBOR, SOFR) plus a fixed credit margin.
- **Perpetual Bonds:** Have no stated maturity date ($T \to \infty$), paying coupons indefinitely. The valuation simplifies to:

$$PV_{\text{Perpetual}} = \frac{PMT}{r} \tag{4.8}$$

Annuities and Fully Amortizing Loans

The cash flow pattern for fully amortizing loans (mortgages) is illustrated in Exhibit 4.3.

For level-payment instruments (like mortgages), the periodic level cash flow, PMT, containing both interest and principal repayment is:

$$PMT = \frac{r \times PV_0}{1 - (1+r)^{-t}} \tag{4.9}$$

Case Study: Mortgage Cash Flow Amortization

A 30-year residential mortgage is used to finance 80% of a USD $1,000,000$ building. Loan amount $PV_0 =$ USD $800,000$. Mortgage rate is 5.25% annual monthly compounded ($r = 0.4375\%$ per month, $t = 360$):

$$PMT = \frac{0.004375 \times \text{USD } 800,000}{1 - (1.004375)^{-360}} \approx \text{USD } 4,417.63 \text{ per month}$$

The amortization breakdown for the first two months is:

- **Month 1:**
  - Interest payment: USD $800,000 \times 0.4375\% =$ USD $3,500.00$
  - Principal repayment: USD $4,417.63 -$ USD $3,500.00 =$ USD 917.63
  - Remaining balance: USD $800,000 -$ USD $917.63 =$ USD $799,082.37$
- **Month 2:**
  - Interest payment: USD $799,082.37 \times 0.4375\% =$ USD $3,495.99$
  - Principal repayment: USD $4,417.63 -$ USD $3,495.99 =$ USD 921.64
  - Remaining balance: USD $799,082.37 -$ USD $921.64 =$ USD $798,160.73$

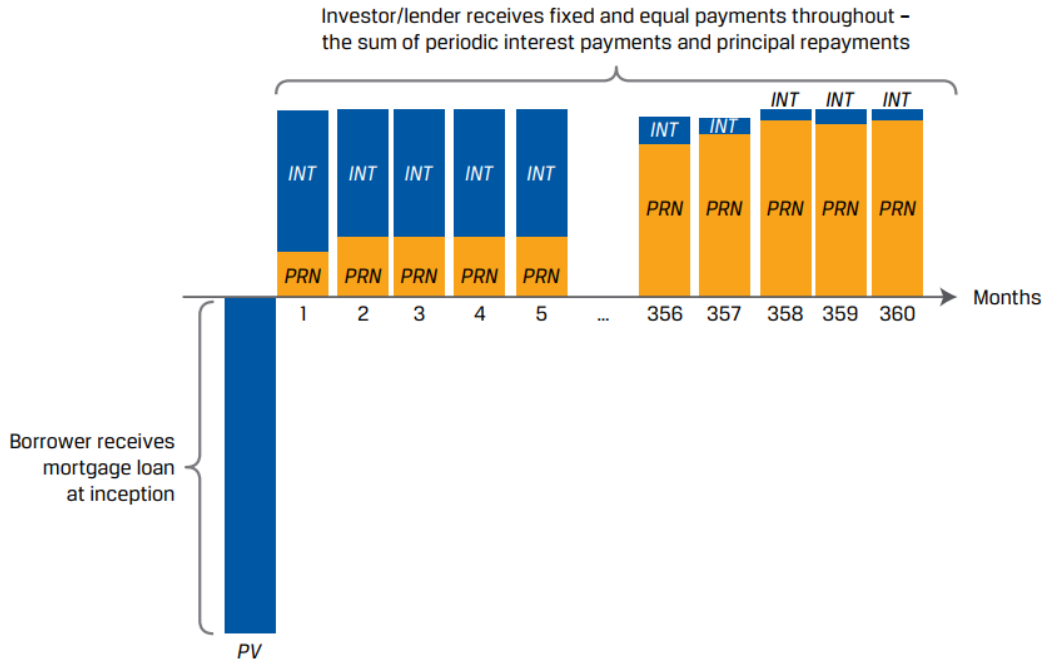


Figure 4.3: Mortgage Cash Flows

Dư nợ giảm dần: Dù tổng tiền trả hàng tháng cố định, cấu phần lãi giảm dần theo dư nợ gốc thực tế, và cấu phần gốc tăng dần tương ứng.

4.2 Equity Instruments and the Time Value of Money

Equity shares have no maturity date and are assumed to remain outstanding indefinitely. Share valuation is based on discounting expected discretionary cash dividends.

4.2.1 Constant Dividends

The cash flow pattern for equity paying constant dividends is illustrated in Exhibit 4.4.

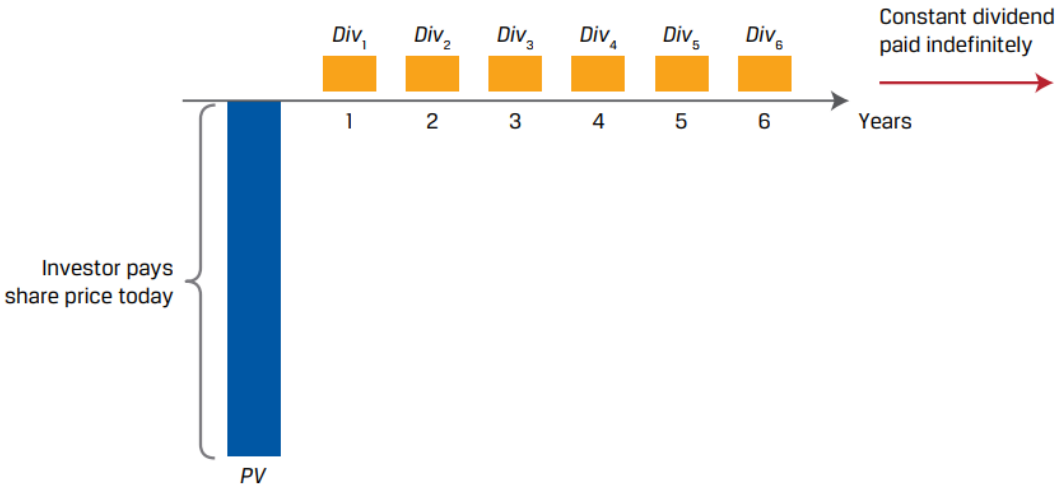


Figure 4.4: Equity Cash Flows with Constant Dividends

The valuation of equity paying constant dividends in perpetuity simplifies to:

$$PV_t = \sum_{i=1}^{\infty} \frac{Div_{t+1}}{(1+r)^i} = \frac{Div_{t+1}}{r} \tag{4.10}$$

4.2.2 Constant Dividend Growth (Gordon Growth Model)

The cash flow pattern for equity paying constant growth dividends is illustrated in Exhibit 4.5.

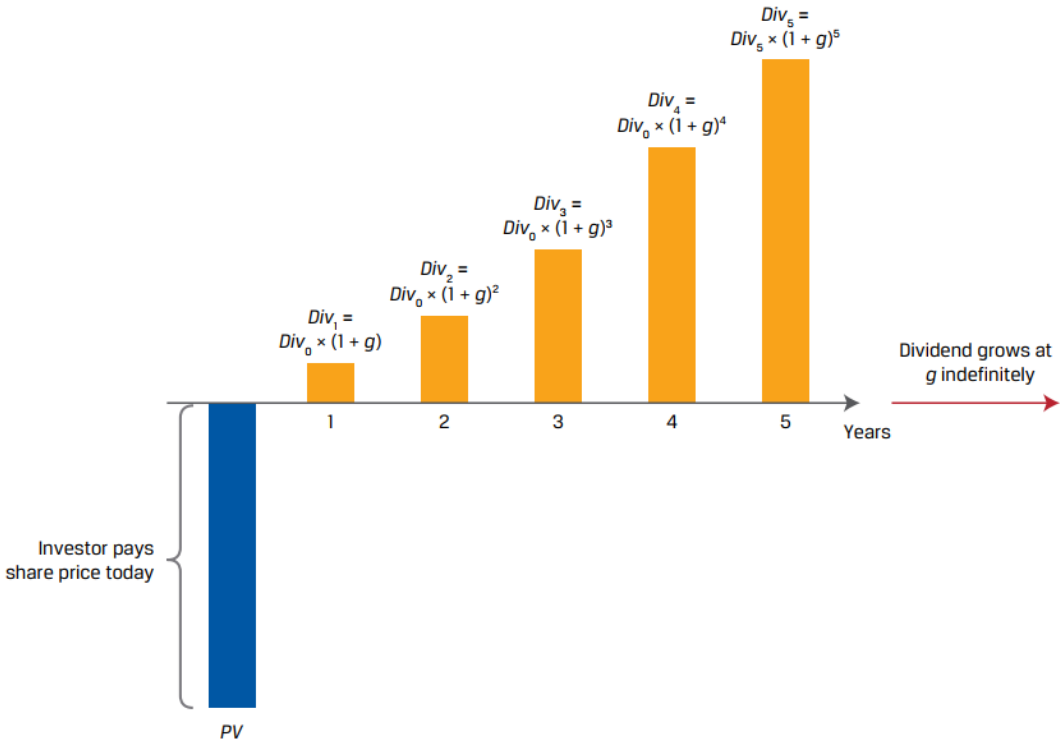


Figure 4.5: Equity Cash Flows with Constant Dividend Growth

If dividends grow at a constant rate g in perpetuity:

$$PV_t = \frac{Div_t(1+g)}{r-g} = \frac{Div_{t+1}}{r-g} \tag{4.11}$$

where $r - g > 0$.

4.2.3 Two-Stage Dividend Growth Model

The cash flow pattern of a two-stage dividend growth model is illustrated in Exhibit 4.6.

For companies experiencing a rapid initial growth phase (g_s) for n periods, followed by a stable long-term growth rate (g_l):

$$PV_0 = \sum_{i=1}^{n} \frac{Div_0(1+g_s)^i}{(1+r)^i} + \frac{E[S_n]}{(1+r)^n} \tag{4.12}$$

Where the expected terminal value at the end of the high-growth phase $E[S_n]$ is:

$$E[S_n] = \frac{Div_{n+1}}{r-g_l} \tag{4.13}$$

Example: Shipline PLC Two-Stage Dividend Growth Valuation

Assume Shipline grows its dividend by $g_s = 6\%$ for three years, followed by $g_l = 2\%$ constant growth. $r = 15\%$, $Div_0 = \text{GBP } 1.50$:

- **Step 1: PV of high-growth dividends (first 3 years):**

$$Div_1 = 1.5900 \implies PV(Div_1) = \frac{1.5900}{1.15} \approx 1.3826$$

$$Div_2 = 1.6854 \implies PV(Div_2) = \frac{1.6854}{1.15^2} \approx 1.2744$$

$$Div_3 = 1.7865 \implies PV(Div_3) = \frac{1.7865}{1.15^3} \approx 1.1747$$

Sum of PV of high-growth phase ≈ 3.8317

- **Step 2: PV of terminal value at Year 3:**

$$Div_4 = Div_3 \times (1+g_l) = 1.7865 \times 1.02 \approx \text{GBP } 1.8222$$

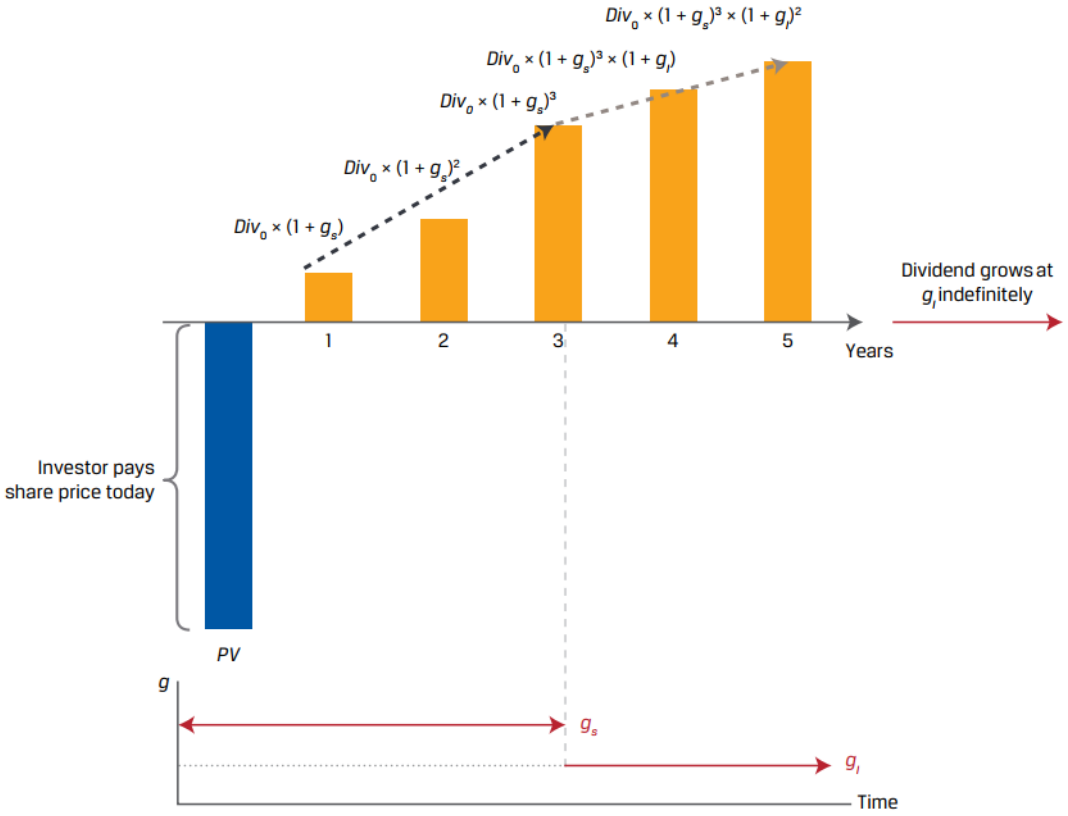


Figure 4.6: Equity Cash Flows with Two-Stage Dividend Growth

$$Terminal\ Value\ E[S_3] = \frac{1.8222}{0.15 - 0.02} \approx \text{GBP } 14.0169$$

$$PV\ of\ E[S_3] = \frac{14.0169}{1.15^3} \approx \text{GBP } 9.2163$$

- **Step 3: Total price:**

$$PV_0 = 3.8317 + 9.2163 = \text{GBP } 13.0480 \approx \text{GBP } 13.05$$

Sử dụng ví dụ ở trên, với trường hợp short-term là 3 năm, và long-term bắt đầu từ năm thứ 4 trở đi. Như vậy, theo công thức trên ta có $n = 3$, và giá trị Terminal value tương ứng tại thời điểm năm thứ 3.

$$PV = \frac{D(1+g_s)}{(1+r)} + \frac{D(1+g_s)^2}{(1+r)^2} + \frac{D(1+g_s)^3}{(1+r)^3} + \frac{D(1+g_s)^3(1+g_l)}{(1+r)^3(r-g_l)}$$

$$PV = \frac{D(1+g_s)}{(1+r)} + \frac{D(1+g_s)^2}{(1+r)^2} + \frac{D(1+g_s)^3}{(1+r)^2(r-g_l)}$$

Đến đây, chúng ta sử dụng chức năng **[CF]** trên máy tính để tính ra giá trị PV, theo 2 cách bấm: [a]

1. Dùng 3 dòng tiền: C01 = $D(1+g_s)$, C02 = $D(1+g_s)^2$, C03 = $D(1+g_s)^3 + D(1+g_s)^3(1+g_l)/(r-g_l)$
2. Hoặc 2 dòng tiền: C01 = $D(1+g_s)$, C02 = $D(1+g_s)^2 + D(1+g_s)^3/(r-g_l)$

[a] Đề xuất cách tính này từ học viên lớp TE 17 và CFA 1.25

4.3 Implied Return and Growth

b	calculate and interpret the implied return of fixed-income instruments and required return and implied growth of equity instruments given their present value and cash flows

When market prices and expected cash flows are observed, analysts can calculate the implied rate of return

and the implied growth rate to assess valuation.

4.3.1 Implied Return for Fixed-Income Instruments

1. Discount Instruments

$$\text{implied r} = \sqrt[T]{\frac{FV_T}{PV}} - 1 = \left(\frac{FV_T}{PV}\right)^{\frac{1}{T}} - 1$$

2. Coupon Instruments

implied r = YTM

$$\text{PV(Market Price of Coupon Bond)} = \frac{PMT_1}{(1+YTM)^1} + \frac{PMT_2}{(1+YTM)^2} + ... + \frac{PMT_T + FV_T}{(1+YTM)^T}$$

Use Microsoft Excel or Google Sheets YIELD function:
YTM = YIELD (settlement, maturity, rate, pr, redemption, frequency, [basis])

4.3.2 Equity Implied Return, Implied Growth, and Forward P/E Ratios

1. Constant Dividend Growth Rate

$$\text{implied r} = \frac{D(1+g)}{PV} + g = \frac{D_1}{PV} + g$$
$$\text{implied g} = \frac{r \times PV - D}{PV} = r - \frac{D_1}{PV}$$

Given a **price-to-earnings ratio** and **dividend payout ratio**, we can solve for either required return or implied dividend growth rate.

$$\frac{PV}{E} = \frac{\frac{D}{E} \times (1+g)}{r-g}$$

In practice, the **forward price-to-earnings ratio** or ratio of its share price to an estimate of its next period (t + 1) earnings per share is commonly used.

$$\frac{PV}{E_1} = \frac{\frac{D_1}{E_1}}{r-g}$$

4.4 Cash Flow Additivity

c	explain the cash flow additivity principle and its importance for the condition of no arbitrage, and explain its use in calculating implied forward interest rates, forward exchange rates, and option values

The cash flow additivity principle states that the present value of any sum of future cash flows equals the sum of their individual present values. This principle is vital in maintaining the no-arbitrage condition (meaning it is impossible to make a riskless profit).

The comparison between two investment strategies under this principle is illustrated in Exhibit 4.7.

Case Study: Basic Cash Flow Additivity

Consider two strategies with an initial investment of GBP 100 and $r = 10\%$:

- **Strategy 1:** Cash flows are $-100, +45, +45, +45$.

$$PV_1 = -100 + \frac{45}{1.1} + \frac{45}{1.1^2} + \frac{45}{1.1^3} \approx \text{GBP } 11.91$$

- **Strategy 2:** Cash flows are $-100, +60, +40, +32.35$.

$$PV_2 = -100 + \frac{60}{1.1} + \frac{40}{1.1^2} + \frac{32.35}{1.1^3} \approx \text{GBP } 11.91$$

- **Incremental Cash Flow Diff:** Strategy 1 – Strategy 2 cash flows: $0, -15, +5, +12.65$.

$$PV_{\text{Diff}} = -\frac{15}{1.1} + \frac{5}{1.1^2} + \frac{12.65}{1.1^3} \approx 0$$

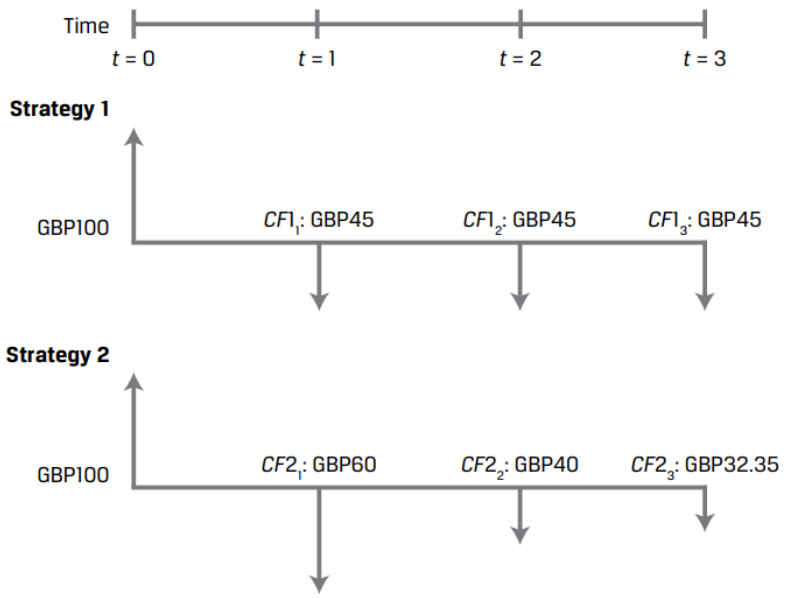


Figure 4.7: Two Investment Strategies

Tính cộng dồn dòng tiền: Hai danh mục tạo ra dòng tiền có PV bằng nhau phải có giá giao dịch giống nhau để triệt tiêu cơ hội arbitrage.

4.4.1 Implied Forward Interest Rates Using Cash Flow Additivity

The relationship between spot rates and implied forward rates is illustrated in Exhibit 4.8.

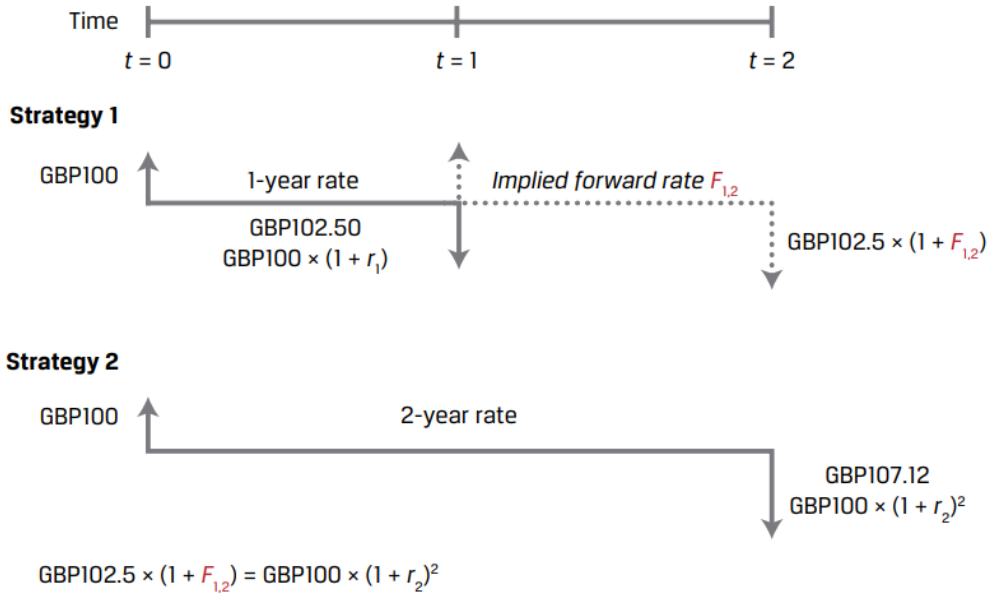


Figure 4.8: Implied Forward Rates

Under the no-arbitrage condition, investing for k periods must yield the same return as investing for j periods ($j < k$) and then reinvesting for the remaining $k - j$ periods at the forward rate:

$$(1 + S_k)^k = (1 + S_j)^j (1 + F_{j,k-j})^{k-j} \tag{4.14}$$

Case Study: Implied US Treasury forward rate calculations (STRIPS)

US Treasury STRIPS prices per USD 100 and implied spot rates:

Date	PV 1y	PV 2y	Spot S_1	Spot S_2
31 May 2022	98.028	95.109	2.012%	2.539%
15 Jun 2022	97.402	93.937	2.667%	3.177%

Using $(1+S_2)^2 = (1+S_1)(1+F_{1,1})$:

- **May 31 Forward Rate:**
$$F_{1,1} = \frac{(1+0.02539)^2}{1+0.02012} - 1 \approx 3.069\%$$
- **June 15 Forward Rate:**
$$F_{1,1} = \frac{(1+0.03177)^2}{1+0.02667} - 1 \approx 3.689\%$$
- **Forward Rate Change:** $3.689\% - 3.069\% = 62$ bps.

The differences between spot and forward curves under rising rates are illustrated in Exhibit 4.9.

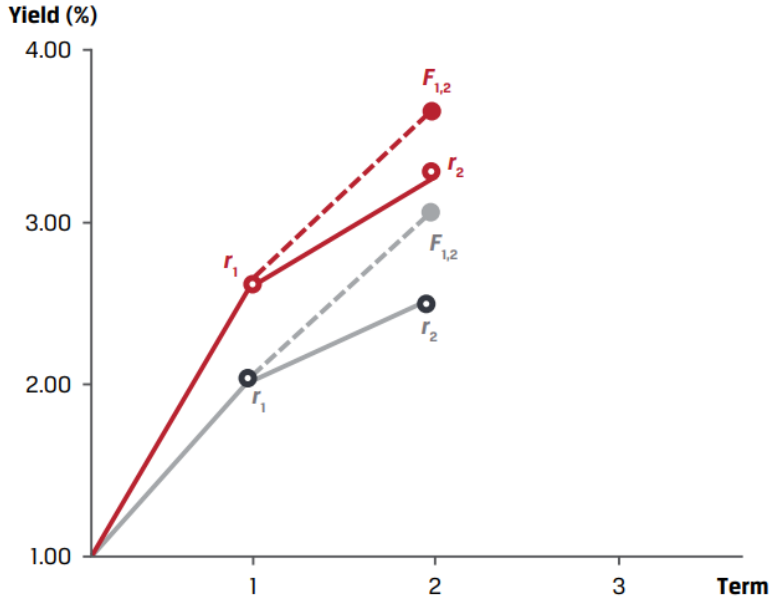


Figure 4.9: Forward Rate Differences

4.4.2 Forward Exchange Rates Using No Arbitrage

The no-arbitrage condition and the replication paths in the FX market are illustrated in Exhibit 4.10.

The forward exchange rate F (expressed as price/base) under continuous compounding interest rate parity is:

$$F = S \times e^{(r_{\text{price}} - r_{\text{base}})T} \tag{4.15}$$

Case Studies: Foreign Exchange forward rate calculations

1. **US/Japan (USD to JPY, 6 months):** Spot JPY/USD $= 134.40$. $r_{\text{JPY}} = 0.05\%$, $r_{\text{USD}} = 2.00\%$ (continuously compounded):
   - Strategy 1 (Domestic USD): USD $1,000 \times e^{0.02\times 0.5} =$ USD $1,010.05$
   - Strategy 2 (Convert JPY): USD $1,000 \to$ JPY $134,400$. Grows to JPY $134,400 \times e^{0.0005\times 0.5} \approx$ JPY $134,433.60$.
   - Forward rate required:
$$F = \frac{\text{JPY } 134,433.60}{\text{USD } 1,010.05} \approx 133.096 \text{ JPY/USD}$$
2. **GBP/USD (Interest rate differential shift):** Spot GBP/USD $= 1.2602$ (USD per GBP):
   - **May 31:** $r_{\text{USD}} = 2.012\%$, $r_{\text{GBP}} = 1.291\% \implies F = 1.2602 \times e^{0.02012-0.01291} \approx 1.2693$
   - **June 15:** $r_{\text{USD}} = 2.667\%$, $r_{\text{GBP}} = 1.562\% \implies F = 1.2602 \times e^{0.02667-0.01562} \approx 1.2742$

Khi r_{USD} tăng mạnh hơn r_{GBP}, chênh lệch nới rộng khiến USD mất giá kỳ hạn so với GBP (tỷ giá kỳ hạn tăng từ 1.2693 lên 1.2742).

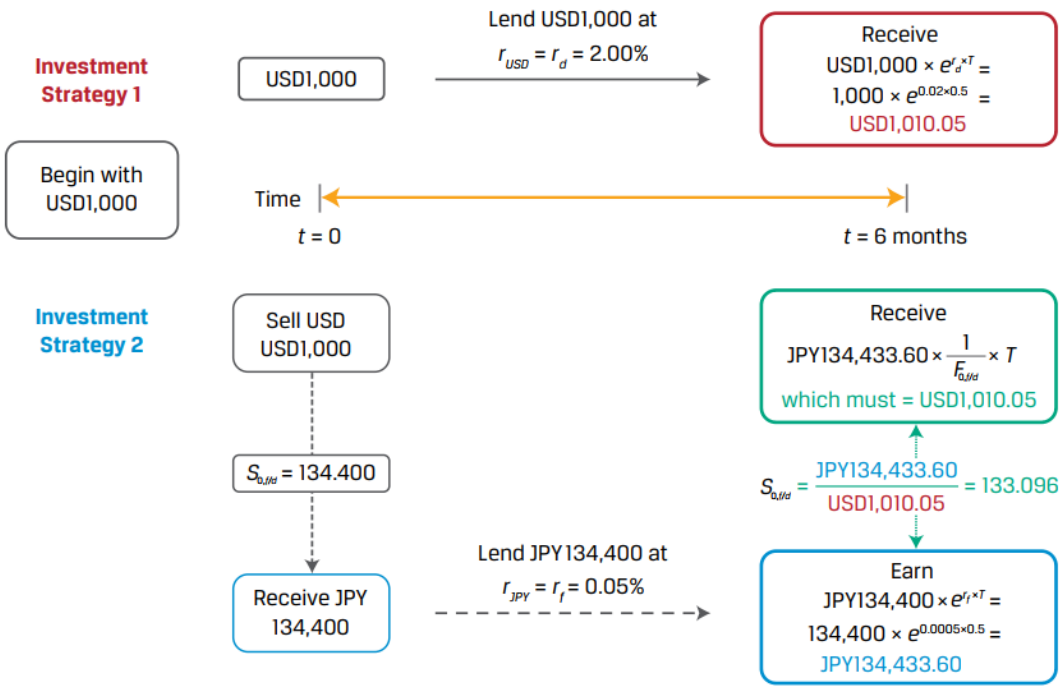


Figure 4.10: No-Arbitrage Condition in the Foreign Exchange Market

4.4.3 Option Pricing Using Cash Flow Additivity

The binomial tree model for the underlying asset's price is illustrated in Exhibit 4.11, and the contract value under the two scenarios is shown in Exhibit 4.12.

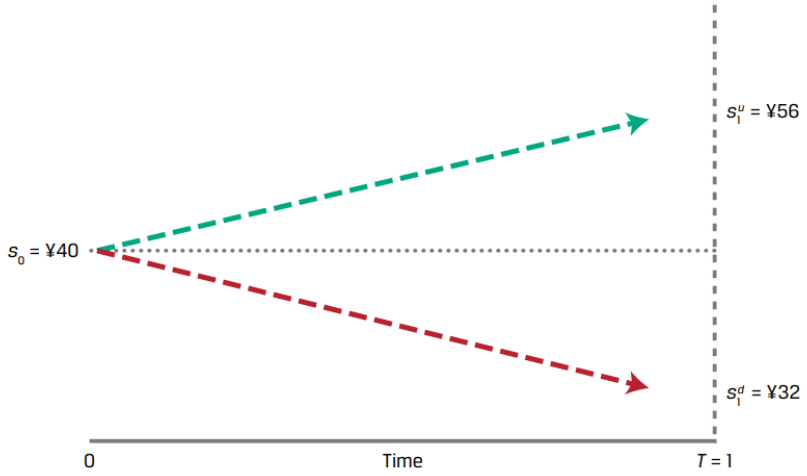


Figure 4.11: One-Period Binomial Tree for the Asset's Price

We can calculate the no-arbitrage price of derivative options using a replicating portfolio consisting of a position in the underlying asset (Δ shares) and risk-free lending/borrowing.

Case Study: Call and Put Option replication pricing

Stock price $S_0 = 40$. In one year, it can go up to $S_1^u = 56$ or down to $S_1^d = 32$. Risk-free rate is 5%. Option strike price $X = 50$.

1. **Call Option ($c_1^u = 6$, $c_1^d = 0$):** Replicate with Δ stock and short 1 call option.

$$\text{Hedge Ratio } \Delta = \frac{c_1^u - c_1^d}{S_1^u - S_1^d} = \frac{6 - 0}{56 - 32} = 0.25$$

- Portfolio Value in up state: $0.25 \times 56 - 6 = 8$
- Portfolio Value in down state: $0.25 \times 32 - 0 = 8$

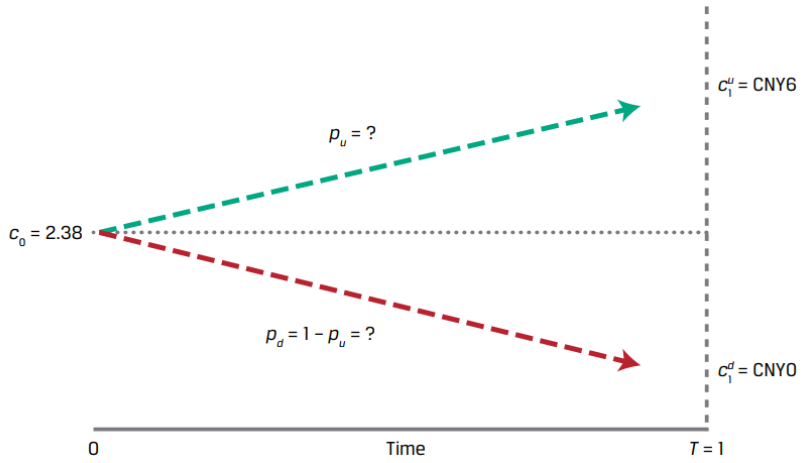


Figure 4.12: One-Period Binomial Tree for the Contract's Price

Since the portfolio is risk-free, its initial value must equal the discounted payoff of 8:

$$0.25(40) - c_0 = \frac{8}{1.05} \approx 7.62 \implies c_0 = 10 - 7.62 = \text{CNY } 2.38$$

2. **Put Option ($p_1^u = 0$, $p_1^d = 18$):** Replicate with Δ stock and long 1 put option.

$$\text{Hedge Ratio } \Delta = \frac{p_1^d - p_1^u}{S_1^u - S_1^d} = \frac{18 - 0}{56 - 32} = 0.75$$

- Portfolio Value in up state: $0.75 \times 56 + 0 = 42$
- Portfolio Value in down state: $0.75 \times 32 + 18 = 42$

Setting initial value equal to discounted payoff of 42:

$$0.75(40) + p_0 = \frac{42}{1.05} = 40 \implies p_0 = 40 - 30 = \text{CNY } 10$$

The replication payoffs for the call option under the up and down scenarios are shown in Exhibit 4.13.

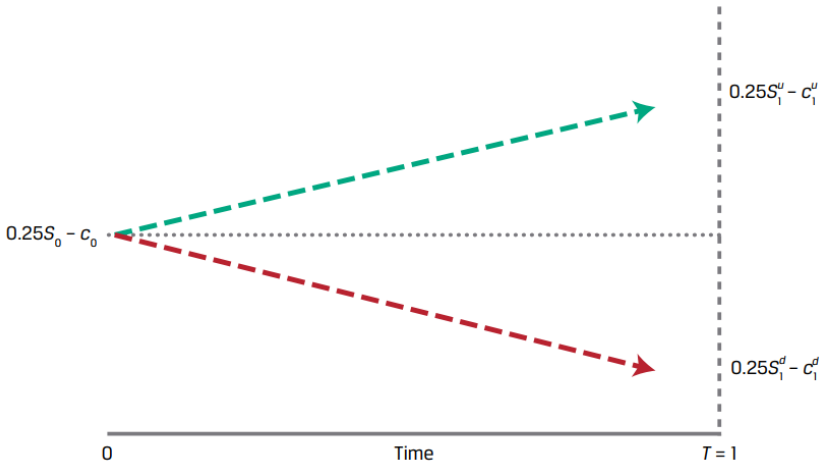


Figure 4.13: Call Option Replication

Case Study: Real Option Pricing in Project Valuation

A desalination project in Malaysia (MYR 10.5 million initial investment) has a lifetime of 20 years. In Year 10, the manager has the option to invest an additional MYR 0.5 million to lock in costs.

- **Project value with option exercised:** MYR 1.03 million (Case 1)
- **Project value without option exercised:** −MYR 2.12 million (Case 2)

- **Gross value of the option (at Year 10):**

$$\text{MYR } 1.03 \text{ million} - (-\text{MYR } 2.12 \text{ million}) = \text{MYR } 3.15 \text{ million}$$

- **PV of option cost (discounted 10 years at 6%):**

$$PV(\text{Cost}) = \frac{\text{MYR } 0.5 \text{ million}}{1.06^{10}} \approx \text{MYR } 0.28 \text{ million}$$

- **Net Value of Real Option:**

$$\text{MYR } 3.15 \text{ million} - \text{MYR } 0.28 \text{ million} = \text{MYR } 2.87 \text{ million}$$

Real Option tạo sự linh hoạt trong đầu tư. Việc thực hiện quyền chọn ở năm 10 giúp tăng giá trị ròng dự án thêm MYR 2.87 triệu.

The outcomes and decision tree of the desalination project with and without the real option are illustrated in Exhibit 4.14.

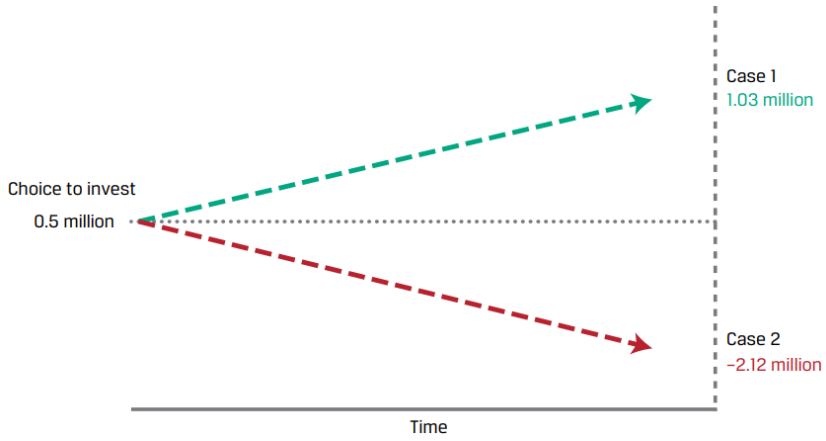


Figure 4.14: Project Outcomes

Module 5

Statistical Characteristics of Asset Returns

LEARNING OUTCOMES	
a	calculate, interpret, and evaluate various measures of (1) central tendency and location and (2) dispersion
b	describe, interpret, and evaluate measures of skewness and kurtosis
c	calculate, interpret, and evaluate covariance and correlation
d	calculate, interpret, and evaluate semi-deviation and coefficient of variation

Measures of Central Tendency and Dispersion

Measures of Central Tendency

Measures of Dispersion

Measures of Shape of a Distribution

Skewness

Kurtosis

Covariance and Correlation between Variables

Covariance

Correlation

Other Dispersion Measures

Semi-Deviation

Coefficient of Variation

5.1 Measures of Central Tendency and Dispersion

a	calculate, interpret, and evaluate various measures of (1) central tendency and location and (2) dispersion

To describe a dataset systematically, statisticians rely on two concepts: the **population** (the entire set of observations, such as the entire exchange rate history) and the **sample** (a subset of that population, such as daily exchange rates during a specific period).

5.1.1 Measures of Central Tendency

A measure of central tendency is a single numerical value that describes the center or typical position of a dataset.

1. **Arithmetic Mean:** The sum of all values divided by the number of observations.

$$\text{Population Mean} = \mu = \frac{\sum_{i=1}^{N} X_i}{N} \tag{5.1}$$

$$\text{Sample Mean} = \bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \tag{5.2}$$

   The arithmetic mean is measured in the same units as the underlying data. It is highly sensitive to outliers.

2. **Median:** The middle value in a dataset when the values are arranged in ascending or descending order.
   - If the number of observations n is odd, the median is the value at position $\frac{n+1}{2}$:

$$\text{Med} = X_{\left(\frac{n+1}{2}\right)} \tag{5.3}$$

   - If n is even, the median is the average of the two middle observations:

$$\text{Med} = \frac{X_{\left(\frac{n}{2}\right)} + X_{\left(\frac{n}{2}+1\right)}}{2} \tag{5.4}$$

   The median is robust to outliers, but it does not use all the information in the dataset.

3. **Mode:** The value that appears most frequently in a dataset. A dataset may be unimodal (one mode), multimodal (multiple modes), or have no mode at all.

Dealing with Outliers

Outliers can occur due to measurement errors, data entry issues, or natural variability. When outliers are valid, analysts use three primary strategies to manage them:

- **Leave the outliers unchanged:** Best if it is important to reflect the entire actual distribution.
- **Trimmed Mean:** Excludes a stated percentage of the lowest and highest values before computing the mean. For example, a 1% trimmed mean discards the bottom 0.5% and top 0.5% of values.
- **Winsorized Mean:** Substitutes extreme values with specified percentiles (e.g., setting the lowest 2.5% to the 2.5th percentile and the highest 2.5% to the 97.5th percentile value).

5.1.2 Measures of location

Quantile (or **fractile**): is a value at, or below which a stated proportion of the observations in a data set lie.

- **quartiles**, which divide the distribution in quarters or four equal parts.
- **quintiles**, which divide the distribution into fifths.
- **deciles**, which divide the data into tenths.
- **percentiles**, which divide the distribution into hundredths.

The position L_y of the y-th percentile in a sorted dataset of size n is:

$$L_y = (n+1)\frac{y}{100} \tag{5.5}$$

If L_y is not an integer, we linearly interpolate between the two closest values.

- The **interquartile range** (IQR) is the difference between the third quartile and the first quartile, or $IQR = Q_3 - Q_1$.

- A **box and whisker plot**: consists of a "box" with "whiskers" connected to the box. The "box" represents the lower bound of the second quartile and the upper bound of the third quartile, with the median or arithmetic average noted as a measure of central tendency of the entire distribution. The whiskers are the lines that run from the box and are bounded by the "fences", which represent the lowest and highest values of the distribution.

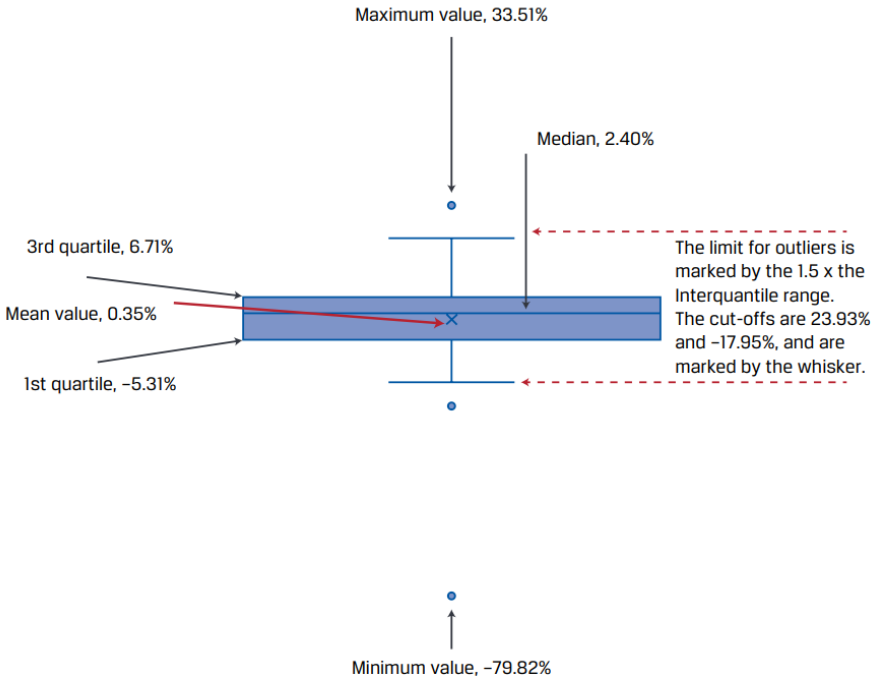


Figure 5.1: box and whisker

Outliers are typically identified as points beyond $1.5 \times$ IQR from the quartiles.

Table 5.1: Excel and Google Sheets Functions for Measures of Central Tendency

Calculation Type	Excel Formula	Google Sheets Formula
Mean	`=AVERAGE(A1:A5)`	`=AVERAGE(A1:A5)`
Median	`=MEDIAN(A1:A5)`	`=MEDIAN(A1:A5)`
Mode	`=MODE.SNGL(A1:A5)`	`=MODE(A1:A5)`
Trimmed Mean	`=TRIMMEAN(A1:A5, percent)`	`=TRIMMEAN(A1:A5, percent)`
1st Quartile	`=QUARTILE.INC(A1:A5, 1)`	`=QUARTILE(A1:A5, 1)`
3rd Quartile	`=QUARTILE.INC(A1:A5, 3)`	`=QUARTILE(A1:A5, 3)`
Percentile	`=PERCENTILE.INC(A1:A5, K)`	`=PERCENTILE(A1:A5, K)`

5.1.3 Measures of Dispersion

Dispersion quantifies the variability of data around its central tendency, serving as a measure of investment risk. Below are **absolute dispersion**, some of **relative dispersion** are refered to 5.4.2.

- **Range:** The difference between the highest and lowest values:

$$\text{Range} = \text{Maximum value} - \text{Minimum value} \tag{5.6}$$

- **Mean Absolute Deviation (MAD):** The arithmetic average of absolute deviations from the mean. It is less sensitive to outliers than standard deviation:

$$\text{MAD} = \frac{\sum_{i=1}^{n} |X_i - \bar{X}|}{n} \tag{5.7}$$

- **Variance (σ^2 or s^2):** The average of the squared deviations from the mean. Sample variance uses $n-1$ (Bessel's correction) to prevent underestimating the population variance:

$$\text{Population Variance} = \sigma^2 = \frac{\sum_{i=1}^{N} (X_i - \mu)^2}{N} \tag{5.8}$$

$$\text{Sample Variance} = s^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1} \tag{5.9}$$

- **Standard Deviation (σ or s):** The square root of variance ($s = \sqrt{s^2}$). It is measured in the same units as the underlying data.

Root of time law: The variance of returns scales linearly with time, while the standard deviation scales with the square root of time.

$$\sigma^2_{Monthly} \approx T \times \sigma^2_{Daily}$$

$$\sigma_{Monthly} \approx \sqrt{T \times \sigma^2_{Daily}} = \sqrt{T}\sigma_{Daily}$$

5.2 Measures of Shape of a Distribution

b | describe, interpret, and evaluate measures of skewness and kurtosis

To describe a distribution's shape completely, analysts evaluate its higher-order moments, particularly its asymmetry (**skewness**) and tail heaviness (**kurtosis**).

5.2.1 Skewness

Skewness is a measure of the asymmetry of a distribution.

- **Symmetric Distribution:** Skewness is zero. Mean, Median, and Mode are equal.
- **Positively Skewed (Right-skewed):** Has a long tail on the right side. Extreme positive values pull the mean upward, making Mean > Median > Mode.
- **Negatively Skewed (Left-skewed):** Has a long tail on the left side. Extreme negative values pull the mean downward, making Mean < Median < Mode.

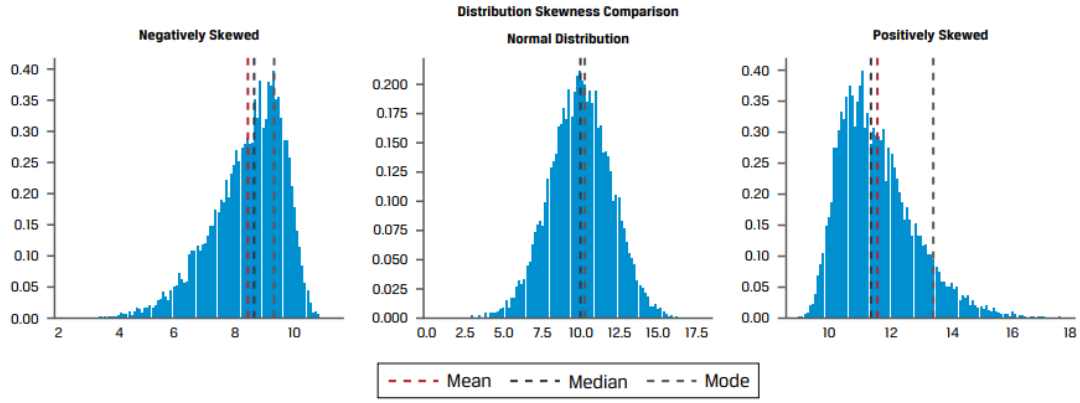


Figure 5.2: Skewness in Distributions

5.2.2 Kurtosis

Kurtosis measures the thickness or heaviness of the tails of a distribution relative to a normal distribution.

- **Leptokurtic (Fat-tailed):** Kurtosis > 3, Excess Kurtosis > 0. Taller peaks and fatter tails, representing higher probability of extreme outcomes.
- **Platykurtic (Thin-tailed):** Kurtosis < 3, Excess Kurtosis < 0. Flatter peaks and thinner tails.
- **Mesokurtic (Normal-like):** Kurtosis = 3, Excess Kurtosis = 0.

Excess Kurtosis is defined as:

$$\text{Excess Kurtosis} = \text{Kurtosis} - 3 \tag{5.10}$$

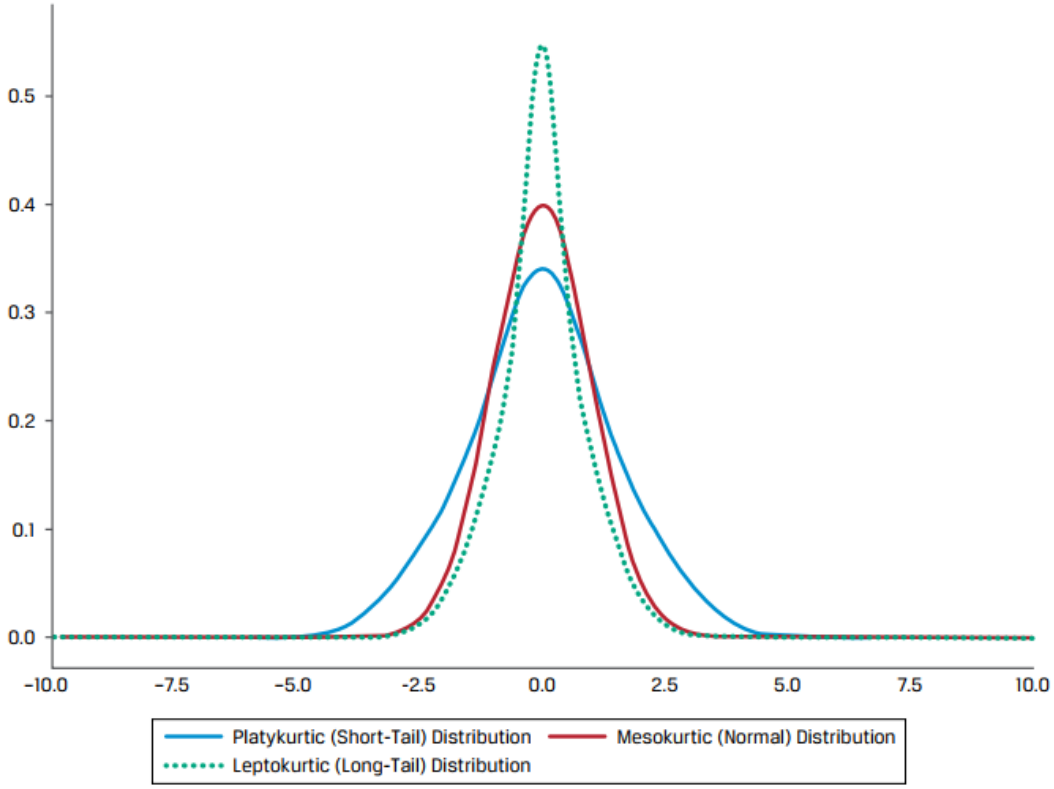


Figure 5.3: Types of Kurtosis

Skewness and Kurtosis Mathematical Approximations

For a sample of n observations, sample mean $\bar{X}$, and sample standard deviation s, the approximations for sample skewness and kurtosis are:

$$\text{Sample Skewness} \approx \frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_i - \bar{X}}{s}\right)^3 \tag{5.11}$$

$$\text{Sample Kurtosis} \approx \frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_i - \bar{X}}{s}\right)^4 \tag{5.12}$$

Lưu ý: Không cần ghi nhớ công thức toán học chi tiết của Skewness và Kurtosis. Kỳ thi chỉ tập trung vào việc đọc hiểu ý nghĩa thực tế và so sánh các giá trị này.

Table 5.2: The Four Moments of Statistical Distributions

Moment	Moment Name / Description	Population Formula	Sample Approximation
First	Mean (Central Tendency)	$\mu = \frac{\sum x_i}{N}$	$\bar{X} = \frac{\sum x_i}{n}$
Second	Variance (Dispersion)	$\sigma^2 = \frac{\sum(x_i-\mu)^2}{N}$	$s^2 = \frac{\sum(x_i-\bar{X})^2}{n-1}$
Third	Skewness (Asymmetry)	$\text{Skew} = \frac{1}{N}\sum\left(\frac{x_i-\mu}{\sigma}\right)^3$	$\text{Skew} = \frac{1}{n}\sum\left(\frac{x_i-\bar{X}}{s}\right)^3$
Fourth	Kurtosis (Tail Heaviness)	$\text{Kurt} = \frac{1}{N}\sum\left(\frac{x_i-\mu}{\sigma}\right)^4$	$\text{Kurt} = \frac{1}{n}\sum\left(\frac{x_i-\bar{X}}{s}\right)^4$

Table 5.3: Excel and Google Sheets Functions for Skewness and Kurtosis

Calculation	Excel Formula	Google Sheets Formula
Population Skewness	`=SKEW.P(A1:A5)`	`=SKEW.P(A1:A5)`
Sample Skewness	`=SKEW(A1:A5)`	`=SKEW(A1:A5)`
Excess Kurtosis	`=KURT(A1:A5)`	`=KURT(A1:A5)`

5.3 Covariance and Correlation

c | calculate, interpret, and evaluate covariance and correlation

To quantify the co-movements and linear relationships between two variables, analysts use covariance and correlation.

Cảnh báo: Hệ số tương quan bằng 0 chỉ có nghĩa là không có mối quan hệ TUYẾN TÍNH. Do đó, trực quan hóa dữ liệu bằng đồ thị (Scatter Plot) luôn là bước bắt buộc.

5.3.1 Covariance

Covariance measures the joint variability of two random variables, indicating how they move in relation to each other.

$$\sigma_{X,Y} = \frac{\sum_{i=1}^{N}(X_i - \mu_X)(Y_i - \mu_Y)}{N} \tag{5.13}$$

$$s_{X,Y} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{n-1} \tag{5.14}$$

If covariance is positive, the variables tend to move in the same direction relative to their means. If negative, they tend to move in opposite directions.

5.3.2 Correlation

Correlation standardizes covariance into a dimensionless measure that ranges between -1 and $+1$:

$$\rho_{X,Y} = \frac{\sigma_{X,Y}}{\sigma_X \sigma_Y} \quad \text{or} \quad r_{X,Y} = \frac{s_{X,Y}}{s_X s_Y} \tag{5.15}$$

Table 5.4: Correlation Levels and Interpretation

Correlation Value	Correlation Type	Description
$\rho_{X,Y} = 0$	No correlation	Absence of any linear relationship.
$\rho_{X,Y} > 0$	Positive correlation	Moves in the same direction; $+1$ represents a perfect linear relationship.
$\rho_{X,Y} < 0$	Negative correlation	Moves in opposite directions; -1 represents a perfect inverse linear relationship.

5.3.3 Anscombe's Quartet

Anscombe's Quartet consists of four datasets that have nearly identical simple statistical properties (mean, variance, correlation), yet have completely different distributions and look completely different when graphed. This underscores the vital importance of visualizing data prior to performing correlation analysis.

5.4 Other Dispersion Measures

d | calculate, interpret, and evaluate semi-deviation and coefficient of variation

5.4.1 Semi-Deviation and Target Downside Deviation

While variance and standard deviation capture both upside and downside volatility, investors are primarily concerned with downside risk.

- **Semi-deviation (Semi-variance):** Measures variability for observations below the mean.
- **Target Downside Deviation:** Measures dispersion only for observations below a specific target or threshold B:

$$s_{\text{Target}} = \sqrt{\frac{\sum_{X_i \le B}(X_i - B)^2}{n-1}} \tag{5.16}$$

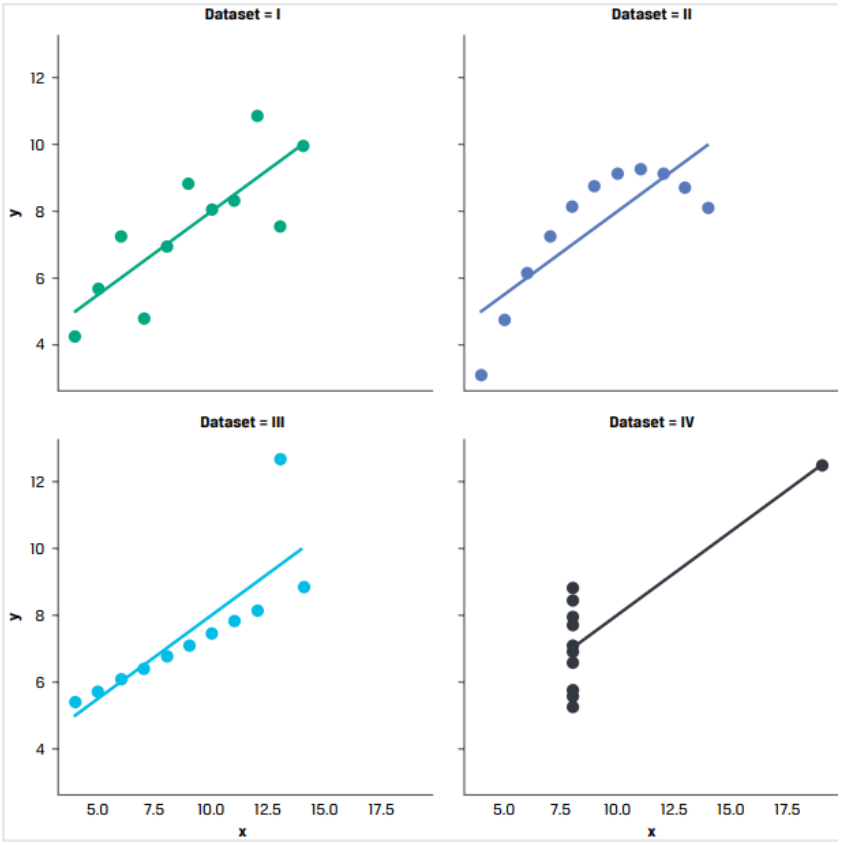


Figure 5.4: The Four Cases for Anscombe's Quartet Scatter Plots

5.4.2 Relative dispersion

1. **Coefficient of variation**
   the ratio of the standard deviation of the data set to its mean. In the investment arena, CV is used to measure the *risk per unit of return* in various investments.

$$\text{CV} = \frac{s}{\bar{X}} \tag{5.17}$$

   A lower CV represents a more favorable risk-to-return ratio.

2. **Sharpe ratio**
   the ratio of excess return over the risk-free rate from an investment to its standard deviation of returns. It basically measures *excess return per unit of risk*.

$$\text{Sharpe ratio} = \frac{R_i - r_f}{\sigma_i} \tag{5.18}$$

Module 6

Statistical Distributions for Financial Asset Prices and Returns

LEARNING OUTCOMES	
a	calculate, interpret, and evaluate unconditional expected values for mean, variance, and covariance
b	calculate, interpret, and evaluate the principal moments of key statistical distributions used in finance
c	calculate, interpret, and evaluate conditional expectations, variances, and covariances
d	formulate investment problems through Bayesian updating

Random Variables and Unconditional Expectations

Random Variables and Their Properties

Unconditional Expectations, Variance, and Covariance

Key Statistical Distributions in Finance

Discrete Statistical Distributions

Continuous Statistical Distributions

Practical Implementation

Conditional Expectations, Variances, and Covariances

Joint Distributions of Random Variables

Conditional Expectations

Conditional Means, Variances, and Covariances

Bayesian Updating

6.1 Random Variables and Unconditional Expectations

a | calculate, interpret, and evaluate unconditional expected values for mean, variance, and covariance

6.1.1 Random Variables and Their Properties

A **random variable** is a function that assigns a number to each outcome of an experiment. In finance, random variables are often called **stochastic variables**, reflecting their inherent unpredictable nature. Random variables are categorized into two types:

- A **discrete random variable** has a countable number of distinct outcomes. For example, a fair coin toss results in heads or tails, each with a probability of $p = 0.50$. Similarly, rolling a fair six-sided die yields outcomes of $1, 2, 3, 4, 5$, or 6, each with a probability of $p = \frac{1}{6}$.

- A **continuous random variable** can take on an infinite number of possible values within a given range, such as picking a random number between 0 and 1 on the number line. When the number of observations is large, empirical distributions often converge to continuous distributions.

Biến rời rạc (Discrete) mô tả các sự kiện đếm được (số vụ vỡ nợ, giao dịch). Tỷ suất sinh lời cổ phiếu dùng biến liên tục (Continuous) vì có thể nhận bất kỳ giá trị thực nào.

A random variable, X, is fully described by its distribution functions:

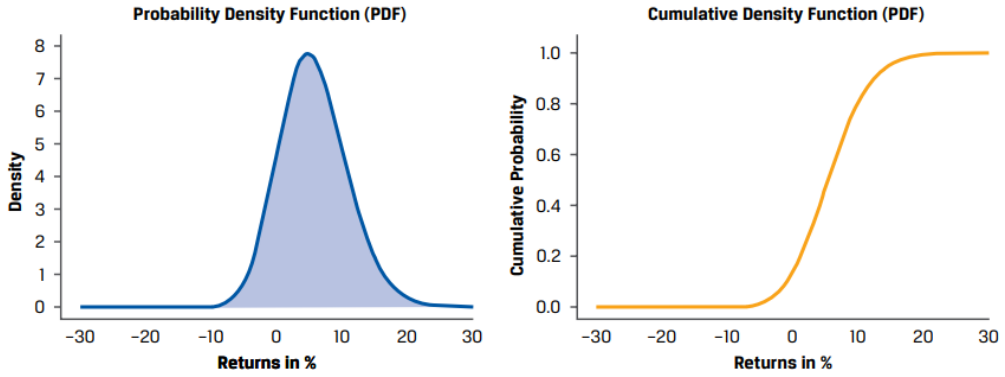


Figure 6.1: Theoretical PDF and CDF of Daily Stock Returns

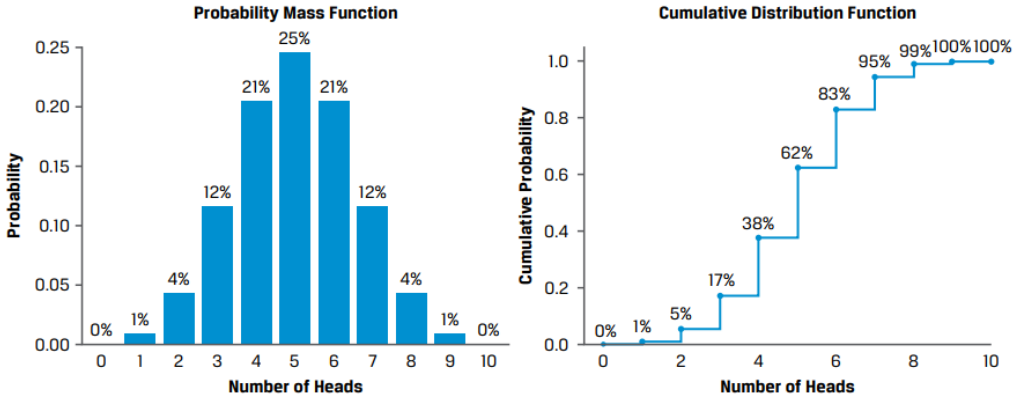


Figure 6.2: The PMF and CDF of the Coin Toss Experiment

- **Cumulative Distribution Function (CDF):** Gives the probability that the random variable X takes on a value less than or equal to x:

$$F(x) = P(X \leq x) \tag{6.1}$$

The CDF is non-decreasing with $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$. The CDF applies to both discrete and continuous random variables.

- **Probability Density Function (PDF):** For continuous random variables, the PDF is the derivative of the CDF, representing the instantaneous rate of change of the CDF:

$$f_X(x) = \frac{dF_X(x)}{dx} \tag{6.2}$$

The probability that X falls within an interval $[a, b]$ is the area under the PDF curve between a and b:

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx = F(b) - F(a) \tag{6.3}$$

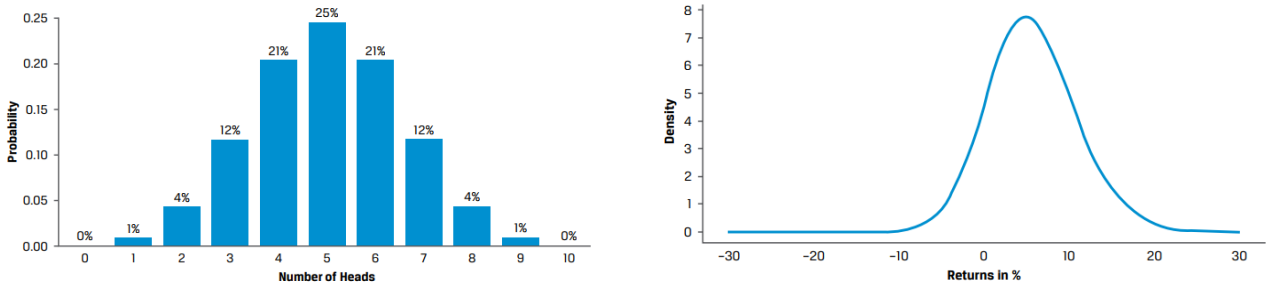
The total area under the PDF curve equals 1:

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \tag{6.4}$$

- **Probability Mass Function (PMF):** For discrete random variables, the PMF gives the probability that X is exactly equal to a specific value x, $P(X = x)$. The probability that X lies within $[a, b]$ is:

$$P(a \leq X \leq b) = \sum_{x=a}^{b} P(X = x) \tag{6.5}$$

The sum of probabilities for all possible outcomes equals 1: $\sum_x P(X = x) = 1$.



(a) Discrete PMF (Coin Tosses)

(b) Continuous PDF (Stock Returns)

Figure 6.3: Discrete vs. Continuous Probability Distributions

6.1.2 Unconditional Expectations, Variance, and Covariance

Unconditional expectations, variance, and covariance provide foundational tools for assessing the behavior of financial assets without requiring prior knowledge of specific market conditions.

- **Unconditional Mean (Expected Value):** The probability-weighted average of the outcomes of a random variable over an infinite number of trials:

$$\text{Discrete: } \mu_X = E[X] = \sum_i x_i P(x_i) \tag{6.6}$$

$$\text{Continuous: } \mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx \tag{6.7}$$

- **Unconditional Variance:** Measures the squared dispersion of a random variable around its mean:

$$\sigma_X^2 = \text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2 \tag{6.8}$$

Phương sai Var(X) (hay σ^2) đo lường mức độ biến động hoặc rủi ro của tài sản. Phương sai càng lớn nghĩa là giá trị thực tế càng dễ phân tán xa khỏi trung bình, đồng nghĩa với rủi ro càng cao.

- **Unconditional Standard Deviation:** $\sigma_X = \sqrt{\text{Var}(X)}$.

Độ lệch chuẩn đưa thước đo biến động về cùng đơn vị đo với biến gốc, giúp dễ so sánh trực tiếp.

- **Unconditional Covariance:** Quantifies the linear relationship between two random variables X and Y:

$$\sigma_{XY} = \text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y] \tag{6.9}$$

Case Study: Unconditional Mean, Variance, and Covariances of Two Assets

An analyst forecasts the performance of a stock index (X) and a bond index (Y) under different conditions:

Probability P	Stock Return X (%)	Bond Return Y (%)
0.2	−2.00	0.75
0.3	−1.00	0.50
0.3	0.00	0.25
0.2	1.00	0.00

Calculating expectations:

$$E[X] = (-2.00 \times 0.2) + (-1.00 \times 0.3) + (0 \times 0.3) + (1.00 \times 0.2) = -0.50\%$$
$$E[Y] = (0.75 \times 0.2) + (0.50 \times 0.3) + (0.25 \times 0.3) + (0 \times 0.2) = 0.375\%$$

Calculating variance:

$$E[X^2] = (0.2 \times (-2)^2) + (0.3 \times (-1)^2) + (0.3 \times 0^2) + (0.2 \times 1^2) = 1.30$$
$$\text{Var}(X) = 1.30 - (-0.50)^2 = 1.05 \implies \sigma_X = \sqrt{1.05} = 1.025\%$$
$$E[Y^2] = (0.2 \times 0.75^2) + (0.3 \times 0.50^2) + (0.3 \times 0.25^2) + 0 = 0.2063$$
$$\text{Var}(Y) = 0.2063 - 0.375^2 = 0.0656 \implies \sigma_Y = \sqrt{0.0656} = 0.2562\%$$

Calculating covariance:

$$E[XY] = (0.2 \times (-2 \times 0.75)) + (0.3 \times (-1 \times 0.50)) + 0 + 0 = -0.45$$
$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = -0.45 - (-0.50 \times 0.375) = -0.2625$$

The negative covariance of -0.2625 indicates that stocks and bonds tend to move in opposite directions, providing significant diversification benefits for a portfolio.

6.1.3 The Law of Large Numbers (LLN)

The **Law of Large Numbers** states that as the number of trials (sample size) increases, the empirical sample mean converges to the true population parameter (unconditional mean), assuming the sample is randomly drawn and representative of the population.

6.2 Key Statistical Distributions in Finance

b	calculate, interpret, and evaluate the principal moments of key statistical distributions used in finance

Để giúp học viên dễ dàng hình dung và phân loại, các phân phối xác suất chính trong chương trình CFA được hệ thống hóa qua Ma trận phân loại hai chiều dưới đây (phân biệt giữa biến ngẫu nhiên Rời rạc vs. Liên tục, và phân phối Đều vs. Không đều):

Table 6.1: Classification Matrix of Statistical Distributions

	Uniform (Phân phối đều)	**Non-Uniform (Phân phối không đều)**
Discrete (Biến rời rạc)	**Discrete Uniform Distribution** - Mỗi kết quả có xác suất bằng nhau $(1/n)$. - Ví dụ: Gieo xúc xắc, tung đồng xu.	**Binomial Distribution** - Số lần thành công trong n phép thử độc lập. **Poisson Distribution** - Số sự kiện xảy ra trong một khoảng xác định.
Continuous (Biến liên tục)	**Continuous Uniform Distribution** - Hàm mật độ không đổi $f(x) = \frac{1}{b-a}$. - Ví dụ: Chọn ngẫu nhiên số trong $[0, 1]$.	**Normal Distribution** (Đối xứng, hình chuông) **Log-normal Distribution** (Chặn dưới bởi 0, lệch phải) **Logistic Distribution** (Đối xứng, đuôi dày hơn Normal)

6.2.1 Discrete Statistical Distributions

Discrete Uniform Distribution

A discrete uniform distribution consists of a finite number of equally spaced outcomes, each with equal probability of occurrence. For a variable X taking n outcomes in the range $[a, b]$, each outcome occurs with probability

$\frac{1}{n}$.

$$\text{Mean: } \mu = \frac{a+b}{2} \tag{6.10}$$

$$\text{Variance: } \sigma^2 = \frac{(b-a+1)^2 - 1}{12} \tag{6.11}$$

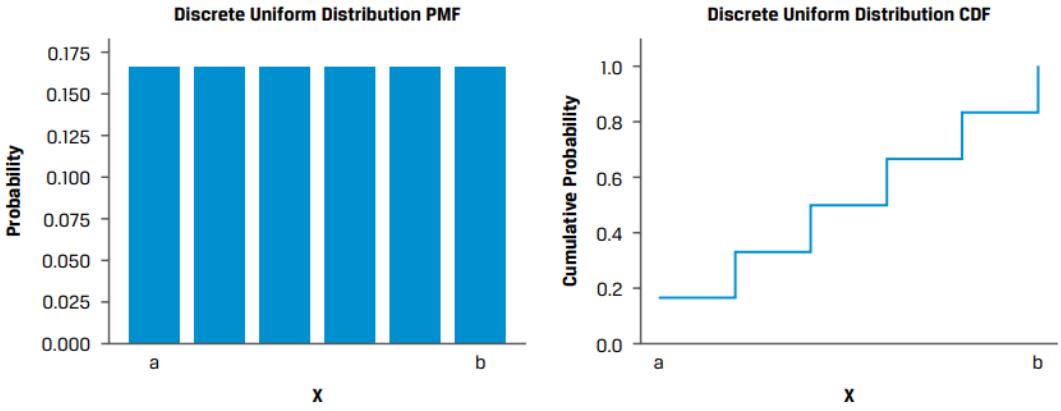


Figure 6.4: Discrete Uniform Distribution PMF and CDF

Chứng minh công thức kỳ vọng, phương sai xem thêm phần (B.1)

Binomial Distribution

The binomial distribution models the number of successes in a fixed number n of independent Bernoulli trials, where each trial has a success probability p and a failure probability $q = 1 - p$. The probability of observing exactly x successes is:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} \tag{6.12}$$

The mean and variance are:

$$\text{Mean: } \mu = n \times p \tag{6.13}$$

$$\text{Variance: } \sigma^2 = n \times p(1-p) \tag{6.14}$$

Chứng minh công thức PMF, kỳ vọng, phương sai xem thêm phần (B.2)

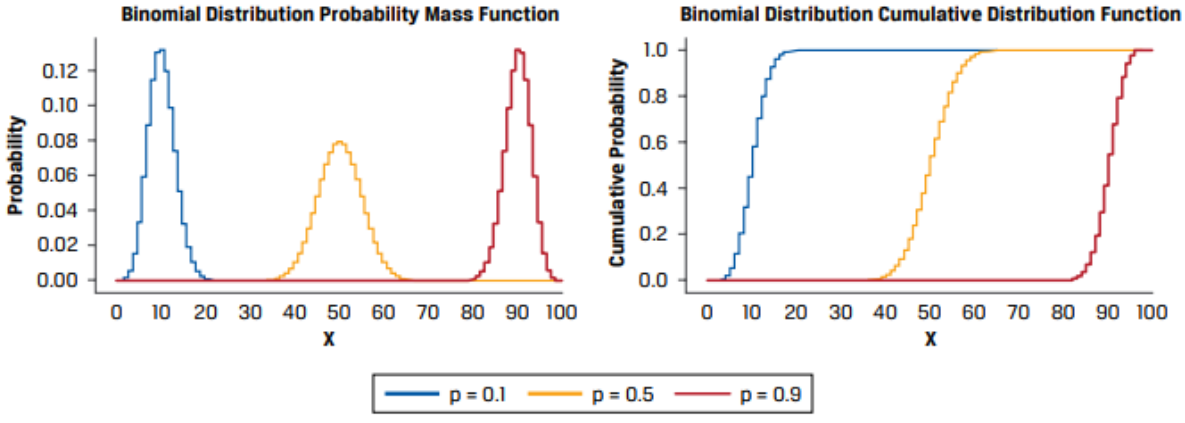


Figure 6.5: Binomial PMF and CDF Across Different Success Probabilities

If n is large, the binomial distribution can be approximated by a normal distribution:

$$X \sim N(np, np(1-p)) \tag{6.15}$$

This approximation is generally valid when both $np \geq 5$ and $n(1-p) \geq 5$.

Poisson Distribution

The Poisson distribution models the number of independent events occurring in a fixed interval of time or space at a constant average rate λ. The probability of observing k events is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{6.16}$$

The mean and variance of a Poisson distribution are identical:

$$\mu = \sigma^2 = \lambda \tag{6.17}$$

Its skewness is $\frac{1}{\sqrt{\lambda}}$ and excess kurtosis is $\frac{1}{\lambda}$.

Phân phối Poisson được dùng để đếm số lần một sự kiện xảy ra trong một khoảng thời gian/không gian xác định (ví dụ: số cuộc gọi đến tổng đài mỗi giờ, số vụ vỡ nợ của doanh nghiệp trong một năm). Đối với Binomial là tính xác suất thành công trong n lần thử, Poisson là chia thời gian thành vô số khoảng nhỏ và đếm số lần sự kiện xảy ra trong mỗi khoảng. Xác suất của sự kiện là $p = \frac{\lambda}{n}$ hay $\lambda = np$, khi n lớn và p nhỏ, phân phối Binomial tiệm cận với Poisson.

Chứng minh công thức PMF, kỳ vọng, phương sai xem thêm phần (B.3)

- Probability of no default over time t: $P(\text{No default}) = e^{-\lambda t}$
- Probability of default over time t: $P(\text{Default}) = 1 - e^{-\lambda t}$
- Expected time to first default: $E[t_d] = \frac{1}{\lambda}$

In credit risk modeling, the Poisson distribution is widely used in **reduced-form models** where default is modeled as a random event governed by a hazard rate (default intensity) λ.

Ý nghĩa trong Quản trị Rủi ro Tín dụng: Giả sử một doanh nghiệp có xác suất vỡ nợ không đổi λ mỗi năm. Phân phối Poisson cho phép tính xác suất để doanh nghiệp KHÔNG vỡ nợ sau thời gian t là $e^{-\lambda t}$. Nếu một công ty có tỷ lệ vỡ nợ trung bình $\lambda = 2\%$ một năm ($\lambda = 0.02$), thì thời gian trung bình cho tới khi công ty này thực sự vỡ nợ là $E[t_d] = \frac{1}{0.02} = 50$ năm.

Case Study: Predicting Corporate Default

Assume the hazard rate λ for a corporate bond is 0.02 per year.

- Probability of default within 1 year: $P(\text{Default}) = 1 - e^{-0.02 \times 1} \approx 1.98\%$
- Probability of no default within 1 year: $P(\text{No default}) = e^{-0.02 \times 1} \approx 98.02\%$
- Expected time until first default: $E[t_d] = \frac{1}{0.02} = 50$ years

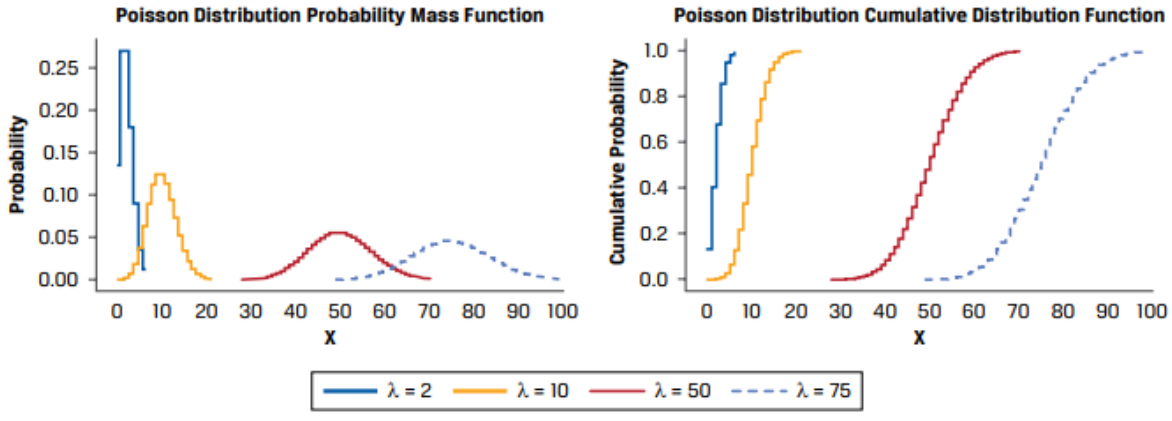


Figure 6.6: Poisson Distributions for Different Values of Lambda

6.2.2 Continuous Statistical Distributions

Continuous Uniform Distribution

The continuous uniform distribution is defined over an interval $[a, b]$, where any value within this interval is equally likely to occur.

$$\text{Mean: } \mu = \frac{a + b}{2} \tag{6.18}$$

$$\text{Variance: } \sigma^2 = \frac{(b-a)^2}{12} \tag{6.19}$$

Its skewness is 0 and kurtosis is 1.8.

Phân phối đều liên tục mô tả trường hợp một biến liên tục có khả năng xuất hiện bất kỳ giá trị nào trong khoảng $[a, b]$ với cơ hội như nhau. Mật độ xác suất $f(x)$ là một đường thẳng nằm ngang ở mức $\frac{1}{b-a}$. Do đó, diện tích dưới đường cong (chính là xác suất) trên bất kỳ phân đoạn nhỏ nào có độ dài L luôn bằng $\frac{L}{b-a}$.

Chứng minh công thức kỳ vọng, phương sai xem thêm phần (B.4)

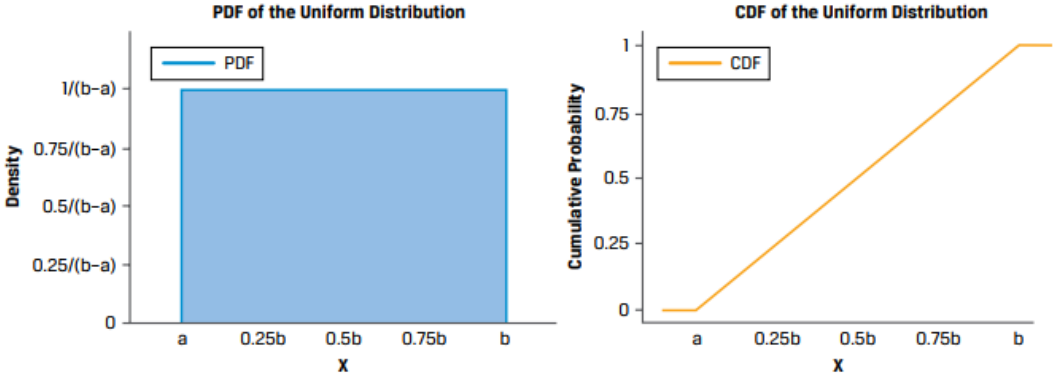


Figure 6.7: Continuous Uniform Probability Density between a and b

Continuous uniform distributions are key to generating Monte Carlo simulations. To simulate any continuous distribution, we first draw random numbers from a standard uniform distribution $U \sim [0, 1]$ and then map them to the target distribution using the inverse CDF method.

Case Study: Monte Carlo Simulation and Geometric Brownian Motion (GBM)

Stock price paths are often simulated using the GBM formula:

$$S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t}$$

where μ is drift, σ is volatility, and W_t is Brownian motion (normally distributed noise). To simulate a single daily step ($t = 1/250$):

1. Draw a uniform random value from $[0, 1]$, say $U = 0.07$.
2. Convert it into a standard normal value: using the inverse standard normal CDF, $\Phi^{-1}(0.07) \approx -1.48$.
3. Substitute $W_t = -1.48$ into the formula to compute the next stock price S_1.

Normal Distribution

The normal distribution, or Gaussian distribution, is fully defined by its mean μ and variance σ^2, denoted as $N(\mu, \sigma^2)$. It has a symmetrical bell shape where Mean = Median = Mode. Its skewness is 0, kurtosis is 3, and excess kurtosis is 0.

Phân phối Chuẩn là phân phối quan trọng nhất trong tài chính học và thống kê nhờ Định lý Giới hạn Trung tâm (Central Limit Theorem). Đường cong mật độ PDF có dạng hình chuông đối xứng hoàn toàn qua giá trị trung bình μ. Vì đối xứng nên các số đo xu hướng trung tâm gồm Kỳ vọng (Mean), Trung vị (Median) và Yếu vị (Mode) trùng nhau. Đặc điểm đặc trưng khác là Skewness = 0 và Kurtosis = 3 (Excess Kurtosis = 0).

A standard normal distribution has $\mu = 0$ and $\sigma^2 = 1$. Z-scores standardize any normal variable: $Z = \frac{X-\mu}{\sigma}$. Key probability intervals (Empirical Rule):

- $\mu \pm 1\sigma$ contains approximately 68% of the observations.
- $\mu \pm 1.96\sigma$ contains approximately 95% of the observations (often rounded to $\mu \pm 2\sigma$).
- $\mu \pm 2.58\sigma$ contains approximately 99% of the observations.

Ý nghĩa của Z-score và Quy tắc thực nghiệm: Z-score đại diện cho khoảng cách từ biến X tới trung bình tính theo số lần độ lệch chuẩn. Quy tắc thực nghiệm cho phép ta dự tính nhanh xác suất rơi vào các khoảng kỳ vọng: ví dụ có khoảng 95% khả năng kết quả thực tế rơi trong khoảng $[-1.96\sigma, +1.96\sigma]$ xung quanh giá trị trung bình tổng thể.

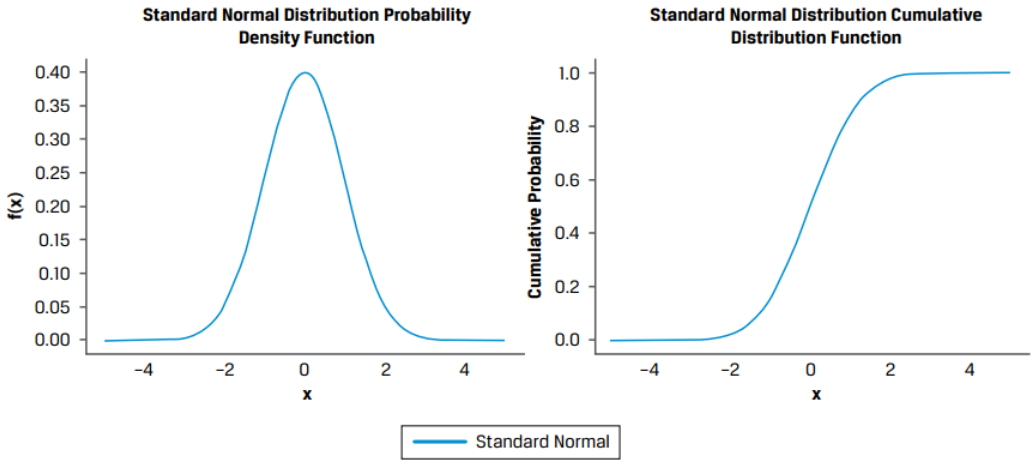


Figure 6.8: Standard Normal Distribution PDF and CDF

Log-Normal Distribution

A random variable Y is log-normally distributed if $X = \ln(Y)$ is normally distributed. Because $Y = e^X$, log-normal variables can only take strictly positive values ($Y > 0$). The distribution is right-skewed.

Biến Y có phân phối Log-normal khi logarit tự nhiên của nó ($X = \ln(Y)$) có phân phối chuẩn. Do $Y = e^X$ và hàm mũ luôn dương nên biến log-normal bị chặn dưới bởi 0 ($Y > 0$). Phân phối này lệch phải (right-skewed) với phần đuôi kéo dài sang bên phải. Trong quản lý danh mục đầu tư, giá tài sản hoặc giá cổ phiếu luôn không thể âm nên được mô hình hóa bằng phân phối Log-normal. Ngược lại, tỷ suất sinh lời của tài sản (có thể là số âm) lại phù hợp với phân phối Chuẩn.

Case Study: Modeling Asset Prices

Let $S_0 = 100$, logarithmic annual mean return $\mu = 8\%$, and volatility $\sigma = 20\%$. After 1 year, the logarithmic price is:

$$\ln(S_1) = \ln(S_0) + \left(\mu - \frac{\sigma^2}{2}\right) \times t = \ln(100) + \left(0.08 - \frac{0.04}{2}\right) \times 1 = 4.6652$$

The expected future price is $S_1 = e^{4.6652} = 106.19$.

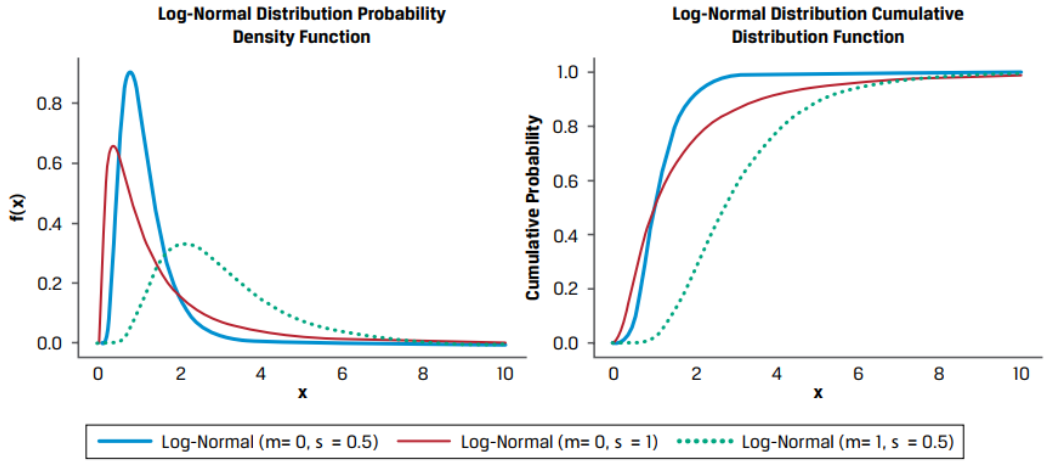


Figure 6.9: A Log-Normal Distribution Curve

Logistic Distribution

The logistic distribution is a continuous distribution defined by location μ and scale s. It resembles the normal distribution but has fatter tails.

$$\text{Mean: } \mu_X = \mu \tag{6.20}$$

$$\text{Variance: } \sigma^2 = \frac{s^2\pi^2}{3} \tag{6.21}$$

Its skewness is 0, kurtosis is 4.2, and excess kurtosis is 1.2. The logistic distribution is widely used in logistic regression to predict binary outcomes in finance (e.g., probability of default).

Phân phối Logistic có hình dạng đối xứng hình chuông cực kỳ giống phân phối Chuẩn, nhưng phần đuôi của nó dày hơn (Fatter tails) - tức là khả năng xuất hiện các biến cố cực đoan cao hơn (Excess Kurtosis = 1.2 > 0). Phân phối này là nền tảng của mô hình Hồi quy Logistic (Logistic Regression) được dùng nhiều để dự đoán xác suất vỡ nợ tín dụng hoặc phân loại rủi ro (Default vs Non-default).

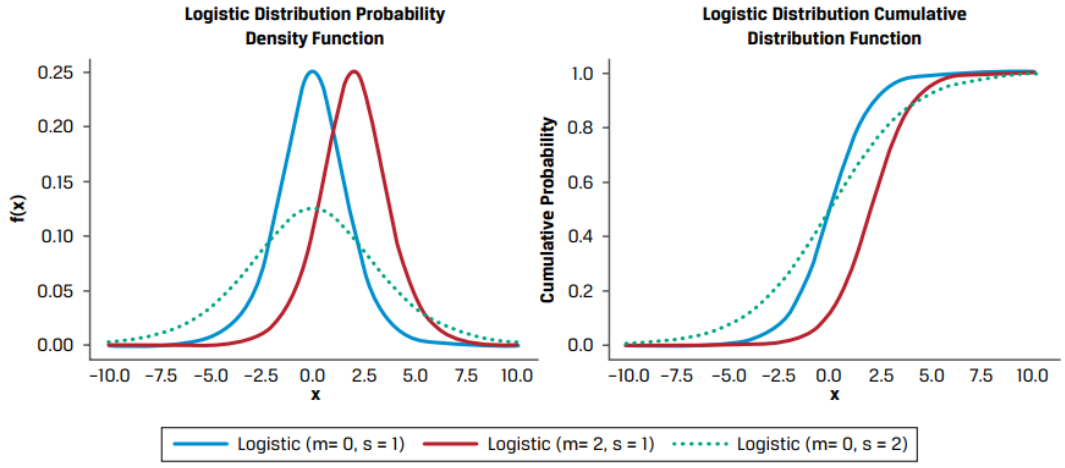


Figure 6.10: The PDF and CDF of Different Logistic Distributions

6.2.3 Practical Implementation

Table 6.2 summarizes the Excel and Google Sheets formulas used to compute these distributions in practice.

Table 6.2: Statistical Distribution Functions in Excel and Google Sheets

Distribution Function	Excel Formula	Google Sheets Formula
Uniform CDF	`BETA.DIST((x-a)/(b-a),1,1,TRUE)`	`BETA.DIST((x-a)/(b-a),1,1,TRUE)`
Uniform Random Draw	`=RAND()*(b-a)+a`	`=RAND()*(b-a)+a`
Binomial PMF	`=BINOM.DIST(x, n, p, FALSE)`	`=BINOM.DIST(x, n, p, FALSE)`
Binomial CDF	`=BINOM.DIST(x, n, p, TRUE)`	`=BINOM.DIST(x, n, p, TRUE)`
Poisson PMF	`=POISSON.DIST(x, lambda, FALSE)`	`=POISSON.DIST(x, lambda, FALSE)`
Normal PDF	`=NORM.DIST(x, mean, sd, FALSE)`	`=NORM.DIST(x, mean, sd, FALSE)`
Normal Random Draw	`=NORM.INV(RAND(), mean, sd)`	`=NORMINV(RAND(), mean, sd)`
Log-normal CDF	`=LOGNORM.DIST(x, mean, sd, TRUE)`	`=LOGNORM.DIST(x, mean, sd, TRUE)`

6.3 Conditional Expectations, Variances, and Covariances

c | calculate, interpret, and evaluate conditional expectations, variances, and covariances

6.3.1 Joint Distributions of Random Variables

Joint distributions allow for examining the simultaneous distribution of two random variables. The bivariate cumulative distribution function (CDF) is defined as:

$$F_{12}(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2) \tag{6.22}$$

For continuous variables, the joint CDF is obtained by integrating the joint PDF $f_{12}(u_1, u_2)$:

$$F_{12}(x_1, x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f_{12}(u_1, u_2) du_1 du_2 \tag{6.23}$$

If the variables are independent:

$$f_{12}(u_1, u_2) = f_1(u_1) \times f_2(u_2) \tag{6.24}$$

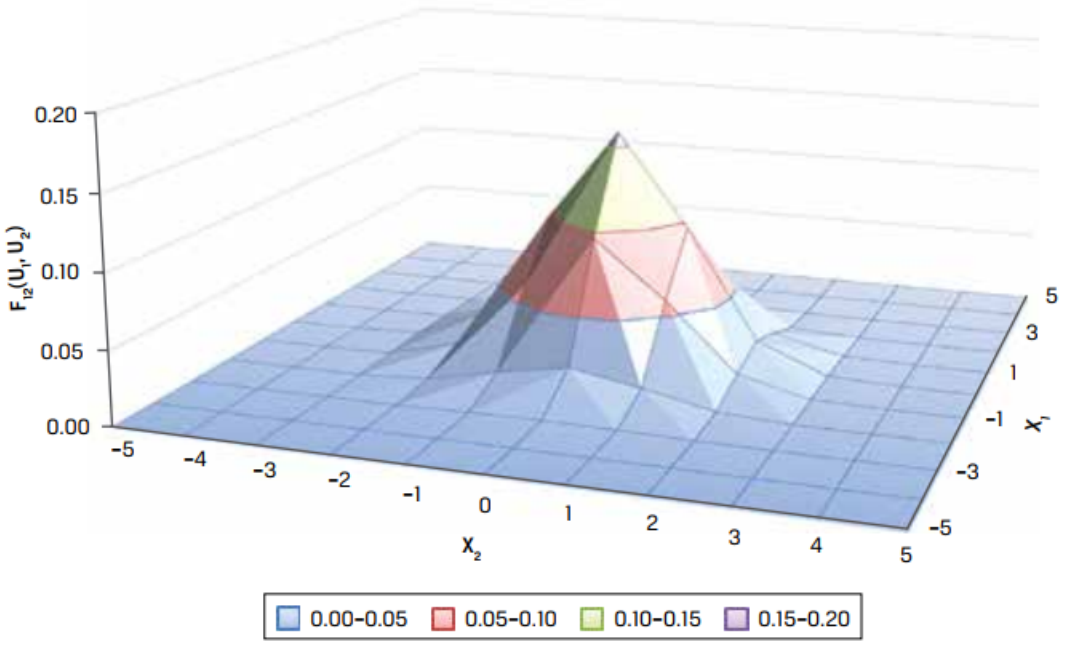


Figure 6.11: Joint Probability Density Surface Plot

The **marginal density** is the probability distribution of a single variable within a multivariate distribution. It is obtained by integrating out the other variables:

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1 X_2}(x_1, u_2) du_2 \tag{6.25}$$

For discrete variables, joint and marginal distributions are summarized in contingency tables.

Case Study: Stock Market Returns and Technology Companies

An index consists of 500 firms, 100 of which are technology firms. Among the 100 technology firms, 60 had returns $> 10\%$ and 40 had returns $\leq 10\%$. Among the 400 non-tech firms, 100 had returns $> 10\%$ and 300 had returns $\leq 10\%$. Table 6.3 displays these frequencies.

Table 6.3: Returns for Technology and Non-Technology Firms

Rate of Return R	**Non-Technology**	**Technology**	**Total**
$R > 10\%$	100	60	160
$R \leq 10\%$	300	40	340
Total	**400**	**100**	**500**

6.3.2 Conditional Expectations

The conditional expectation $E[X|Y = y]$ is the expected value of X given that Y takes on the value y:

$$\text{Discrete: } E[X|Y = y] = \sum_x xP(X = x|Y = y) \tag{6.26}$$

$$\text{Continuous: } E[X|Y = y] = \int x f_{X|Y}(x|y) dx \tag{6.27}$$

The expected conditional variance is:

$$\text{Var}(X|Y = y) = E[X^2|Y = y] - (E[X|Y = y])^2 \tag{6.28}$$

The **Total Probability Rule for Expected Value** states:

$$E[X] = \sum_{i=1}^{n} E[X|S_i] \times P(S_i) \tag{6.29}$$

where $S_1, S_2, \ldots, S_n$ are mutually exclusive and exhaustive scenarios.

Case Study: Scenario Expected Returns and Variances

An analyst estimates portfolio return forecasts across three equally likely economic scenarios (Depression, Recession, and Expansion, each with probability $P = 1/3$):

- **Depression**: -25% return (prob 50%), -10% return (prob 25%), -5% return (prob 15%), 0% return (prob 10%).
- **Recession**: -5% return (prob 30%), 0% return (prob 30%), 5% return (prob 30%), 15% return (prob 10%).
- **Expansion**: 50% return (prob 40%), 10% return (prob 25%), 15% return (prob 25%), 25% return (prob 10%).

Conditional expected returns and variances are calculated as:

$$\begin{aligned}
E[R|\text{Depression}] &= 0.50(-0.25) + 0.25(-0.10) + 0.15(-0.05) + 0.10(0) = -15.75\% \\
\text{Var}[R|\text{Depression}] &= 0.50(-0.25 - (-0.1575))^2 + 0.25(-0.10 - (-0.1575))^2 \\
&\quad + 0.15(-0.05 - (-0.1575))^2 + 0.10(0 - (-0.1575))^2 \\
&\approx 0.009319 \\
E[R|\text{Recession}] &= 0.30(-0.05) + 0.30(0) + 0.30(0.05) + 0.10(0.15) = 1.50\% \\
\text{Var}[R|\text{Recession}] &= 0.30(-0.05 - 0.015)^2 + 0.30(0 - 0.015)^2 \\
&\quad + 0.30(0.05 - 0.015)^2 + 0.10(0.15 - 0.015)^2 \\
&\approx 0.003525 \\
E[R|\text{Expansion}] &= 0.40(0.50) + 0.25(0.10) + 0.25(0.15) + 0.10(0.25) = 28.75\% \\
\text{Var}[R|\text{Expansion}] &= 0.40(0.50 - 0.2875)^2 + 0.25(0.10 - 0.2875)^2 \\
&\quad + 0.25(0.15 - 0.2875)^2 + 0.10(0.25 - 0.2875)^2 \\
&\approx 0.031719
\end{aligned}$$

If scenarios are equally likely, the unconditional expected return is:

$$E[R] = \frac{-15.75\% + 1.50\% + 28.75\%}{3} = 4.833\%$$

The unconditional variance is computed using the Law of Total Variance:

$$\text{Var}(R) = E[\text{Var}(R|S)] + \text{Var}(E[R|S]) \tag{6.30}$$

where:

$$E[\text{Var}(R|S)] = \frac{0.009319 + 0.003525 + 0.031719}{3} = 0.014854 \text{ (kỳ vọng của các phương sai có điều kiện)}$$

$$\begin{aligned}
\text{Var}(E[R|S]) &= \frac{1}{3}(-0.1575 - 0.048333)^2 + \frac{1}{3}(0.015 - 0.048333)^2 + \frac{1}{3}(0.2875 - 0.048333)^2 \\
&= 0.033560 \text{ (phương sai của các kỳ vọng có điều kiện)}
\end{aligned}$$

Do đó, phương sai vô điều kiện của danh mục là:

$$\text{Var}(R) = 0.014854 + 0.033560 = 0.048414 \text{ (hay } 4.84\%)$$

Đính chính Errata trong sách CFA Curriculum 2027 gốc:

*1. **Phát hiện lỗi:** Trong giáo trình CFA Curriculum, phương sai của các kỳ vọng có điều kiện $Var(E[R|S])$ đã bỏ quên dấu âm của kịch bản Depression (nhập nhầm $+\mathbf{0.1575}$ thay vì $-\mathbf{0.1575}$ trong phép trừ với $E[R] = 0.048333$), dẫn đến kết quả bị sai thành:*

$$\begin{aligned}\text{Var}(E[R|S])_{\text{CFA}} = \frac{1}{3}(+\mathbf{0.1575} - 0.048333)^2 + \frac{1}{3}(0.015 - 0.048333)^2 \\ + \frac{1}{3}(0.2875 - 0.048333)^2 = 0.02341 \approx \mathbf{0.0234} \quad (\mathbf{2.34}\%)\end{aligned}$$

*2. **Phép tính chuẩn xác (giữ đúng dấu âm -15.75%):***

$$\begin{aligned}\text{Var}(E[R|S])_{\text{Chuẩn}} = \frac{1}{3}(-\mathbf{0.1575} - 0.048333)^2 + \frac{1}{3}(0.015 - 0.048333)^2 \\ + \frac{1}{3}(0.2875 - 0.048333)^2 = 0.033560 \quad (\mathbf{3.36}\%)\end{aligned}$$

*3. **Phương sai vô điều kiện theo Law of Total Variance:***

$$\text{Var}(R) = E[\text{Var}(R|S)] + \text{Var}(E[R|S]) = 0.014854 + 0.033560 = \mathbf{0.048414} \quad (\mathbf{4.84}\%)$$

Xem thêm về Law of Total Variance tại phần (B.6)

6.3.3 Conditional Means, Variances, and Covariances

Incorporating conditioning information updates expected values:

- **Conditional Mean:** $E[X|S] = \sum_{i=1}^{n} x_i P(X_i|S)$
- **Conditional Variance:** $\text{Var}(X|S) = E[(X - E[X|S])^2|S]$
- **Conditional Covariance:**

$$\text{Cov}(X, Y|S) = E[(X - E[X|S])(Y - E[Y|S])|S] \tag{6.31}$$

Case Study: Three-Period Asset Price Evolution

Let $P_0 = 100$. Each period, the price rises by 10 with probability 0.60 or falls by 10 with probability 0.40. At Time 3, possible outcomes follow a binomial tree (Exhibit 6.12):

- $P_3 = 130$ (prob $0.6^3 = 0.216$)
- $P_3 = 110$ (prob $3 \times 0.6^2 \times 0.4 = 0.432$)
- $P_3 = 90$ (prob $3 \times 0.6 \times 0.4^2 = 0.288$)
- $P_3 = 70$ (prob $0.4^3 = 0.064$)

The unconditional price expectation at Time 3 is:

$$E[P_3] = 130(0.216) + 110(0.432) + 90(0.288) + 70(0.064) = 106$$

The unconditional variance is:

$$\text{Var}(P_3) = E[P_3^2] - (E[P_3])^2 = 11,524 - 106^2 = 288$$

Suppose at Time 1, the price is observed to be 110 ($P_1 = 110$, Exhibit 6.13). The conditional probabilities for Time 3 update to:

- $P_3 = 130$ (prob $0.6^2 = 0.36$)
- $P_3 = 110$ (prob $2 \times 0.6 \times 0.4 = 0.48$)
- $P_3 = 90$ (prob $0.4^2 = 0.16$)

The conditional expectation is:

$$E[P_3|P_1 = 110] = 130(0.36) + 110(0.48) + 90(0.16) = 114$$

The conditional variance is:

$$\text{Var}(P_3|P_1 = 110) = E[P_3^2|P_1 = 110] - E[P_3|P_1 = 110]^2 = 15,568 - 114^2 = 192$$

Suppose at Time 2, the price is 100 ($P_2 = 100$, Exhibit 6.14). The conditional statistics update to:

$$E[P_3|P_2 = 100] = 110(0.60) + 90(0.40) = 102$$

$$\text{Var}(P_3|P_2 = 100) = (110^2 \times 0.60 + 90^2 \times 0.40) - 102^2 = 96$$

Table 6.4 summarizes the updates:

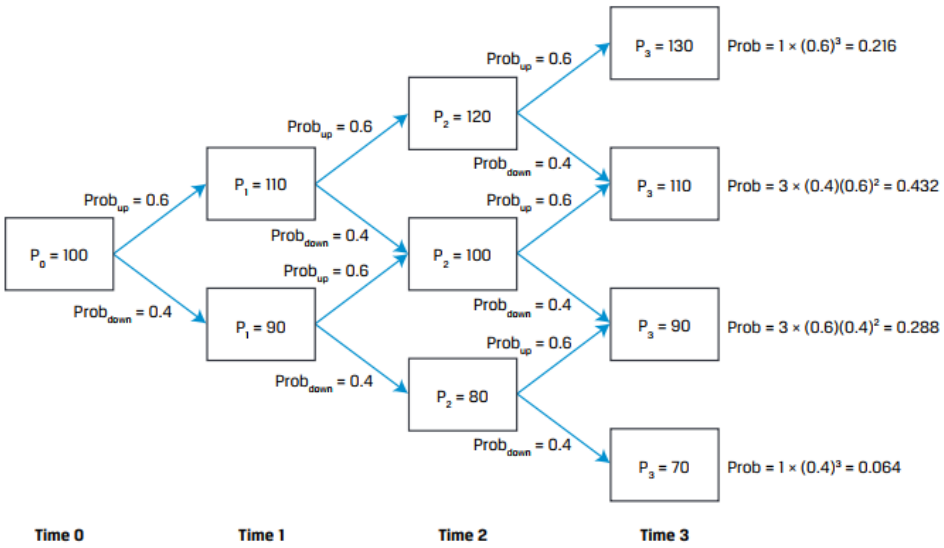


Figure 6.12: Potential Asset Price Evolution over Three Periods

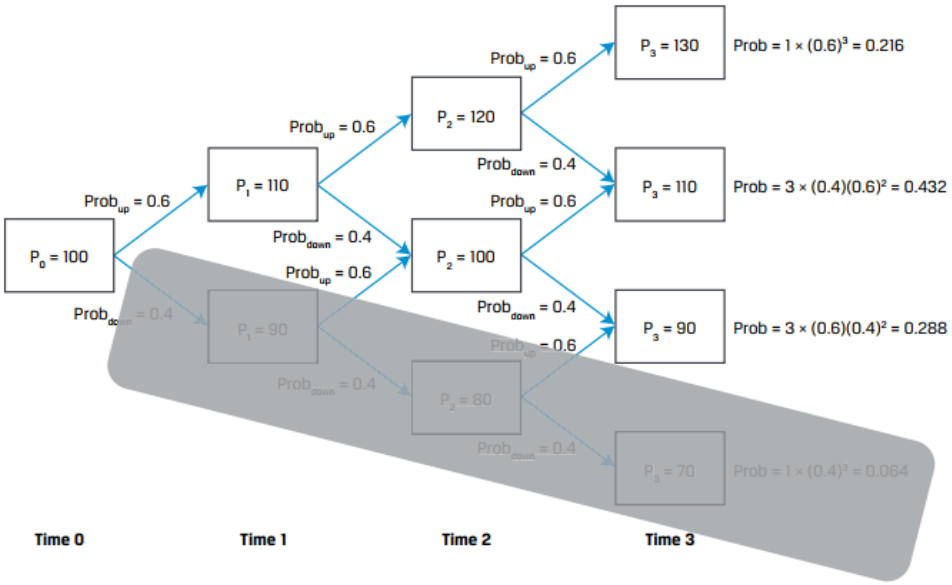


Figure 6.13: Price Evolution Given Time 1 Outcome $P_1 = 110$

Table 6.4: Expectation and Variance Under Different Conditions

Conditionality	**Expected Price**	**Variance**
Unconditional	106	288
Conditional on $P_1 = 110$	114	192
Conditional on $P_2 = 100$	102	96

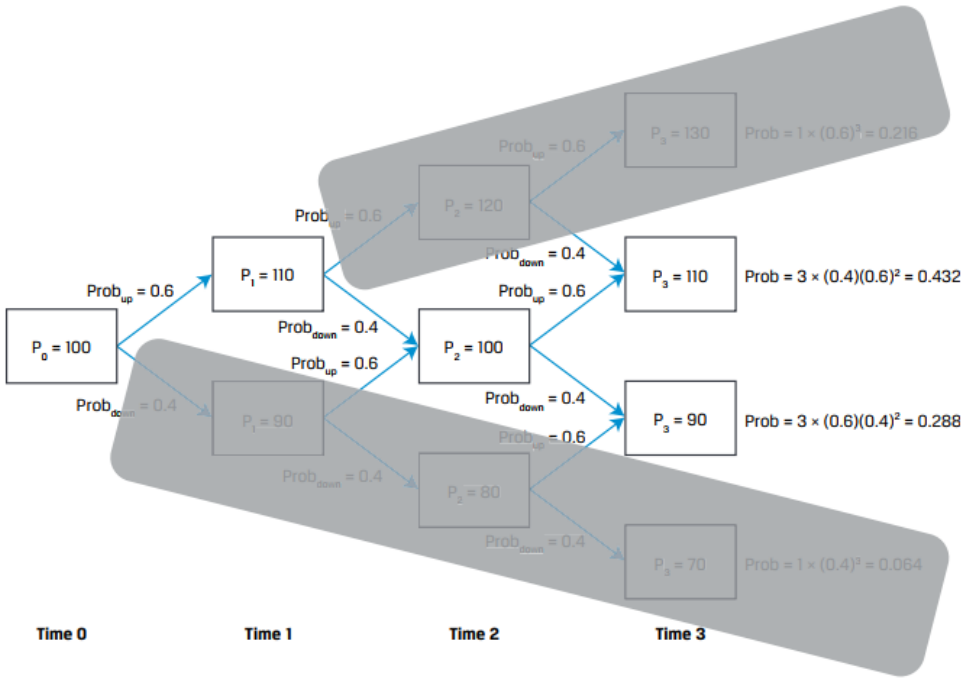


Figure 6.14: Price Evolution Given Time 2 Outcome $P_2 = 100$

6.4 Bayesian Updating

d | formulate investment problems through Bayesian updating

Bayes' theorem provides a formal framework for updating the probability of a hypothesis (prior probability) in light of additional data or evidence to form an updated probability (posterior probability).

- **A Priori Probability (Prior):** The probability of an event formulated before new data is considered.
- **A Posteriori Probability (Posterior):** The updated probability of the event after incorporating new observations.

Bayes' formula is stated as:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)} \tag{6.32}$$

where:

- $P(A|B)$ is the posterior probability of A occurring given B is true.
- $P(B|A)$ is the likelihood of observing B given A is true.
- $P(A)$ is the prior probability of A.
- $P(B)$ is the marginal probability of observing B, computed using the total probability rule:

$$P(B) = \sum_i P(B|A_i) \times P(A_i) \tag{6.33}$$

Ý nghĩa thực tế: Định lý Bayes giúp cập nhật nhận định ban đầu của chúng ta khi có thêm thông tin mới.
Quy trình 3 bước tính toán xác suất cập nhật Bayes:

Bước 1 - Xác định thông tin ban đầu (Prior & Likelihood): *Xác định xác suất tiên nghiệm của các kịch bản $P(A_i)$ và xác suất có điều kiện xảy ra thông tin mới trong từng kịch bản $P(B|A_i)$.*

Bước 2 - Tính xác suất toàn phần (Total Probability): *Tính tổng xác suất xảy ra thông tin mới $P(B) = \sum P(B|A_i)P(A_i)$. Đây chính là mẫu số trong công thức Bayes.*

Bước 3 - Tính xác suất hậu nghiệm (Posterior): *Lấy xác suất đồng thời của kịch bản mục tiêu chia cho xác suất toàn phần: $P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)}$.*

Case Study: Technology Companies and Stock Returns

Using the contingency table for technology firms in Table 6.3, we calculate:

- $P(\text{Tech}) = 0.20$ and $P(\text{Non-tech}) = 0.80$
- $P(R > 10\%|\text{Tech}) = 0.60$ and $P(R > 10\%|\text{Non-tech}) = 0.25$

The overall probability of a firm having a return $R > 10\%$ is:

$$P(R > 10\%) = (0.60 \times 0.20) + (0.25 \times 0.80) = 0.32$$

Using Bayes' formula, the probability that a firm is a technology company given its return exceeds 10% is:

$$P(\text{Tech}|R > 10\%) = \frac{P(R > 10\%|\text{Tech}) \times P(\text{Tech})}{P(R > 10\%)} = \frac{0.60 \times 0.20}{0.32} = 37.5\%$$

This represents an increase from the unconditional prior probability of 20%.

Case Study: Sustainability Investments and Earnings Expectations

EcoMed A/S will release its annual earnings next week. Prior probabilities for consensus EPS outcomes are:

- $P(\text{Exceeded Consensus}) = 0.45$
- $P(\text{Met Consensus}) = 0.30$
- $P(\text{Fell Short}) = 0.25$

Before the release, EcoMed announces expanding sustainable practices. The likelihood of this announcement under each earnings scenario is estimated as:

- $P(\text{Sustainability Announcement}|\text{Exceeded}) = 0.75$
- $P(\text{Sustainability Announcement}|\text{Met}) = 0.20$
- $P(\text{Sustainability Announcement}|\text{Fell Short}) = 0.05$

First, we find the unconditional probability of the sustainability announcement:

$$P(\text{Sustainability}) = (0.75 \times 0.45) + (0.20 \times 0.30) + (0.05 \times 0.25) = 0.41$$

We update the probabilities using Bayes' formula:

$$P(\text{Exceeded}|\text{Sustainability}) = \frac{0.75 \times 0.45}{0.41} = 82.32\%$$
$$P(\text{Met}|\text{Sustainability}) = \frac{0.20 \times 0.30}{0.41} = 14.63\%$$
$$P(\text{Fell Short}|\text{Sustainability}) = \frac{0.05 \times 0.25}{0.41} = 3.05\%$$

Note that the updated posterior probabilities sum to exactly 1.0000:

$$0.8232 + 0.1463 + 0.0305 = 1.0000$$

Kịch bản (A_i)	**Prior** $P(A_i)$	**Likelihood** $P(B|A_i)$	**Joint Probability** $P(B \cap A_i)$	**Posterior** $P(A_i|B)$
Exceeded Consensus	0.45	0.75	0.3375	82.32%
Met Consensus	0.30	0.20	0.0600	14.63%
Fell Short	0.25	0.05	0.0125	3.05%
Tổng	**1.00**		$P(\textbf{Sustainability}) = 0.4100$	**100.00%**

Module 7

Estimation and Hypothesis Testing

LEARNING OUTCOMES	
a	explain the central limit theorem and the application of confidence intervals and sampling methodologies
b	explain hypothesis testing and its components, including statistical significance, Type I and Type II errors, and the power of a test; construct appropriate hypothesis tests; and interpret the results
c	compare and contrast parametric and non-parametric tests, describe situations in which each is the more appropriate type of test, construct appropriate hypothesis tests, and interpret the results

7.1 The Central Limit Theorem, Confidence Intervals, and Sampling

a	explain the central limit theorem and the application of confidence intervals and sampling methodologies

7.1.1 Sampling Methodologies

To extract information about a population, two sampling approaches can be used:

1. **Probability Sampling:** Every member of the population has an equal chance of selection.
   - **Simple random sampling**: each member of population has same probability of being included in sample.
   - **Systematic sampling**: every k^{th} member in the population list is selected.
   - **Stratification**: the process of grouping members of the population into relatively homogeneous subgroups, or strata, before drawing samples. Strata should be *mutually exclusive and collectively exhaustive*.
   - **Cluster sampling**: requires the division or classification of the population into subpopulation groups, called clusters. A major difference between cluster and stratified random samples is that in cluster sampling, a whole cluster is regarded as a sampling unit and only sampled clusters are included.
2. **Non-Probability Sampling:** depends on factors other than probability considerations, such as a sampler's judgment or the convenience to access data.
   - **Convenience Sampling**: an element is selected from the population based on whether or not it is accessible to a researcher or on how easy it is for a researcher to access the element.
   - **Judgmental Sampling**: This sampling process involves selectively handpicking elements from the population based on a researcher's knowledge and professional judgment. Ex: when auditing financial statements, seasoned auditors can apply their sound judgment to select accounts or transactions that can provide sufficient audit coverage.

Sampling Error Sampling error of the mean = Sample mean − Population mean = $\overline{x} - \mu$

Sampling Distribution The probability distribution of a given sample statistic under repeated sampling of the population.

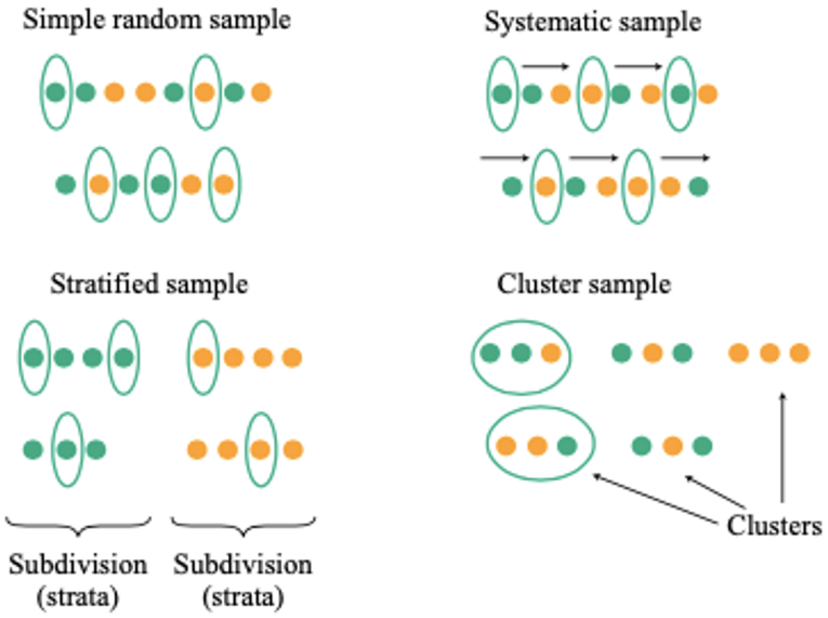


Figure 7.1: Probability Sampling

7.1.2 The Central Limit Theorem (CLT)

[1] The law of large numbers provides context for using the sample mean, $\bar{X}$, to estimate the population mean, μ. The central limit theorem (CLT) is a fundamental concept in statistics. It states that if random samples are taken from any distribution with a finite mean μ and finite variance σ^2, then the sample means, $\bar{X}$, will converge to a normal distribution as the sample size n increases. Specifically:

- The expected value of the sample mean is equal to the population mean:

$$E[\bar{X}] = \mu \tag{7.1}$$

- The variance of the sampling distribution is:

$$\sigma^2_{\bar{X}} = \frac{\sigma^2}{n} \tag{7.2}$$

- The standard error of the sample mean (the standard deviation of the sample mean) is:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad s_{\bar{X}} = \frac{s}{\sqrt{n}} \tag{7.3}$$

The standardized format of the sample mean converges in distribution to the standard normal distribution $N(0,1)$ as sample size $n \to \infty$:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1) \quad \text{as } n \to \infty \tag{7.4}$$

For symmetric and unimodal distributions, a sample size as small as 4 or 5 might provide an adequate normal approximation. For skewed distributions, a sample size of at least 25–30 is typically necessary, whereas extremely skewed distributions may require even larger sample sizes. An estimator $\hat{\theta}$ is:

- **Unbiased** if $E[\hat{\theta}] = \theta$. The sample mean is an unbiased estimator of the population mean ($E[\bar{X}] = \mu$).
- **Efficient** if it has the smallest possible standard error among all unbiased estimators. A minimum-variance unbiased estimator (MVUE) is both unbiased and efficient. The best linear unbiased estimator (BLUE) is both unbiased and maximally efficient among all linear estimators.
- **Consistent** if it converges to the true parameter value as the sample size increases. The sample mean $\bar{X}$ is a consistent estimator.
- **Robust** if it provides accurate parameter estimates even when the data include outliers or violate distribution assumptions (e.g., using trimmed means).

[1] Youtube: But what is the Central Limit Theorem?

7.1.3 Confidence Intervals

Confidence intervals are probabilistic ranges that are likely to contain the true parameter value with a specified level of confidence $(1 - \alpha)$. For large sample sizes, the confidence interval for the population mean μ is:

$$\bar{X} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \tag{7.5}$$

Where $Z_{\alpha/2}$ is the critical Z-score from the normal distribution, and $\sigma/\sqrt{n}$ is the standard error. The margin of error, $Z_{\alpha/2} \times \sigma/\sqrt{n}$, defines the range within which the true population parameter is expected, with a certain level of confidence, to fall on each side of the sample mean. Exhibit 7.2 illustrates the normal distribution curve and confidence intervals. The margin of error, $Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$, is the half-width of the confidence interval. The two

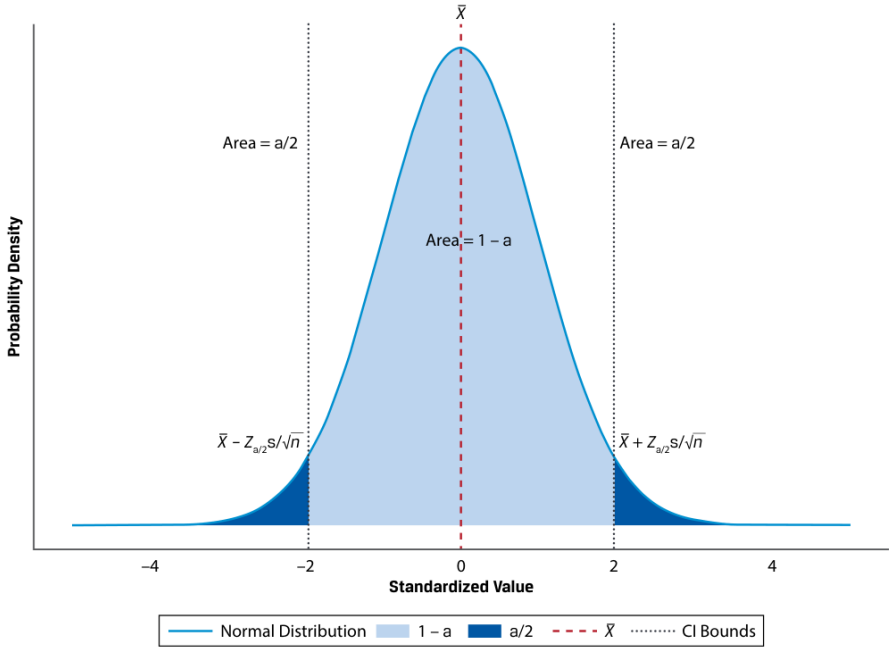


Figure 7.2: The Normal Distribution Curve and Confidence Intervals

most frequently used confidence intervals are:

- **90% Confidence Interval:** $Z_{0.005} = 1.645$, leaving 0.5% in each tail.
- **95% Confidence Interval:** $Z_{0.025} = 1.9600$, leaving 2.5% in each tail.
- **99% Confidence Interval:** $Z_{0.005} = 2.5760$, leaving 0.5% in each tail.

Exam Tip

Value at Risk (VaR) as a One-Sided Confidence Interval
VaR provides a statistical measure of potential losses in a portfolio over a specific time horizon at a given confidence level. It represents the lower bound of a one-sided confidence interval.

- $P(\text{Loss} > \text{VaR}) = \alpha$
- $P(\text{Loss} \leq \text{VaR}) = 1 - \alpha$

Assuming normally distributed asset returns and a mean return of zero, the parametric VaR is:

$$\text{VaR} = Z_\alpha \times \sigma \times P \tag{7.6}$$

Where Z_α corresponds to the one-sided percentile (e.g., 1.65 for 95%, 2.33 for 99%), σ is standard deviation, and P is the current position value.

7.2 Process of Hypothesis Testing

b	explain hypothesis testing and its components, including statistical significance, Type I and Type II errors, and the power of a test; construct appropriate hypothesis tests; and interpret the results

Hypothesis testing is the process of evaluating the accuracy of a statement regarding a population parameter (e.g., the population mean) given sample information (e.g., the sample mean).

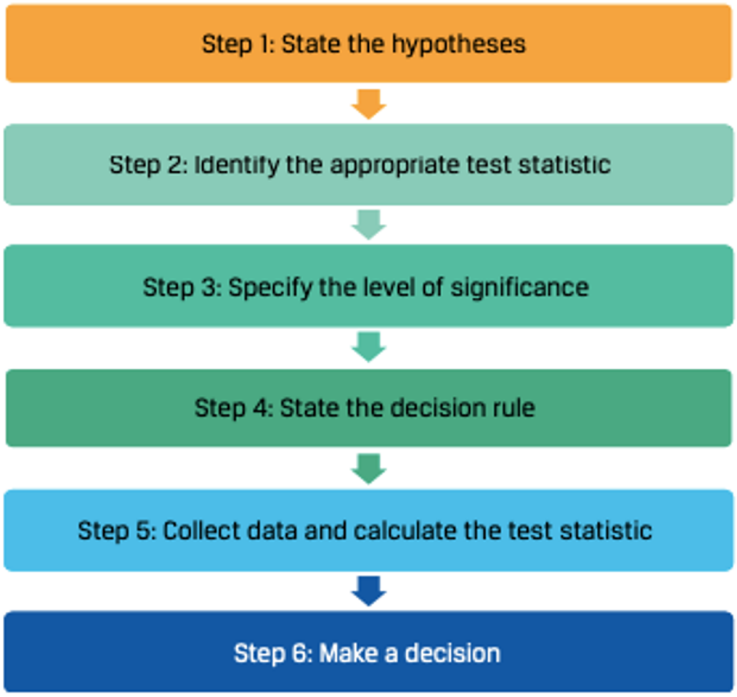


Figure 7.3: Hypothesis Testing Process

7.2.1 Stating the Hypothesis

Hypothesis is a statement about the value of a population parameter developed for the purpose of testing a theory.

1. The **null hypothesis** (H_o) generally represents the status quo and is the hypothesis that we are interested in rejecting. The null hypothesis will always include an "=" sign.
2. **Alternate hypothesis** (H_a) is essentially the statement whose validity we are trying to evaluate. The alternate hypothesis is the statement that will only be accepted if the sample data provides convincing evidence of its truth. It is the conclusion of the test if the null hypothesis is rejected.

Example:

Hypothesis statements	
$H_o : \mu = \mu_0$	$H_a : \mu \neq \mu_0$
$H_o : \mu \leq \mu_0$	$H_a : \mu > \mu_0$
$H_o : \mu \geq \mu_0$	$H_a : \mu < \mu_0$

μ = population mean
μ_0 = hypothesized value of the population mean

7.2.2 Identify Appropriate Test Statistic and Distribution

Essentially, a hypothesis test involves the comparison of a sample's test statistic to a critical value.

Test statistic is calculated as:

$$\text{Test statistic} = \frac{\text{Sample statistic} - \text{Hypothesized value}}{\text{Standard error of sample statistic}}$$

Critical value depends on the relevent distribution, sample size and level of significance (α) used to test.

Table 7.1: Test Statistics and Their Distribution

What we want to Test	Test Statistic	Probability Distribution	Degree of Freedom
Test of a single mean	$t = \frac{\overline{X} - \mu_0}{s/\sqrt{n}}$	t-Distributed	$n-1$
Test of the difference in means	$t = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}}$	t-Distributed	$n_1 + n_2 - 2$
Test of the mean of differences	$t = \frac{\overline{d} - \mu_{d0}}{s_{\overline{d}}}$	t-Distributed	$n-1$
Test of a single variance	$\chi^2 = \frac{s^2(n-1)}{\sigma_0^2}$	Chi-square distributed	$n-1$
Test of the difference in variances	$F = \frac{s_1^2}{s_2^2}$	F-distributed	$n_1 - 1, n_2 - 1$
Test of a correlation	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	t-Distributed	$n-2$
Test of independence (categorical data)	$\chi^2 = \sum_{i=1}^{m} \frac{(O_{ij} - E_{ij})^2}{E_{iji}}$	Chi-square distributed	$(r-1)(c-1)$

Note *μ_0, μ_{d0} and σ_0^2 denote hypothesized values of the mean, mean difference, and variance, respectively. The $\overline{X}$, $\overline{d}$, s^2, s, and r denote for a sample mean, mean of the differences, variance, standard deviation, and correlation, respectively, with subscripts indicating the sample, if appropriate. The sample size is indicated as n, and the subscript indicates the sampple, if appropriate. O_{ij}, E_{ij} are observed and expected frequencies, respectively, with r indicating number of rows and c indicating number of columns in the contingency table.*

7.2.3 Specify the Level of Significance

Level of significance (α) represents the chance that we are willing to take that the conclusion from the test might be wrong.

Type I and Type II Errors

- **Type I error**: Rejecting the null hypothesis when it is actually true.
- **Type II error**: Failing to reject the null hypothesis when it is actually false.

Decision	**H_o is True**	**H_o is False**
Do not reject H_o	Correct decision	Incorrect decision **Type II error**
Reject H_o	Incorrect decision **Type I error** Significance level = P(Type I error)	Correct decision Power of the test = 1 − P(Type II error)

The significance level (α) represents the probability of making a Type I error.
If we were to fail to reject the null hypothesis given the lack of overwhelming evidence in favor of the alternate, we risk a Type II error - *failing to reject the null hypothesis when it is false.*
The power of a test is the probability of correctly rejecting the null hypothesis when it is false.

- The *higher* the power of the test, the *better* it is for purposes of hypothesis testing.
- The only way to *decrease* the probability of a Type II error given the significance level (probability of Type I error) is to *increase* the sample size.

7.2.4 State the Decision Rule

One-sided versus Two-sided Tests

- Two-sided Tests

  H_o: $\mu = \mu_0$ versus H_a: $\mu \neq \mu_0$
  Two-tailed hypotheses tests have 2 rejection regions.
  Rejection Rules for Two-Tailed Hypothesis Test:

  Reject H_o when: Test statistic $<$ Lower critical value or Test statistic $>$ Upper critical value.
  Fail to reject Ho when: Lower critical value $\leq$ Test statistic $\leq$ Upper critical value.

- One-sided Tests

  1. H_o: $\mu \leq \mu_0$ versus H_a: $\mu > \mu_0$
     Reject H_o when: Test statistic $>$ Upper Critical Value
     Fail to reject H_o when: Test statistic $\leq$ Upper Critical value.
  2. H_o: $\mu \geq \mu_0$ versus H_a: $\mu < \mu_0$
     Reject H_o when: Test statistic $<$ Lower Critical Value
     Fail to reject H_o when: Test statistic $\geq$ Lower Critical value.

Decision Rule

- The action we take is based on comparing the calculated **test statistic** with a specified value or values, which we refer to as **critical values**.
- The critical value or values we choose are based on the level of significance and the probability distribution associated with the test statistic.
- If we find that the calculated value of the test statistic is more extreme than the critical value or values, then we reject the null hypothesis; we say the result is **statistically significant**.

Relation Between Confidence Intervals and Hypothesis Tests

$$\left[\begin{pmatrix} sample \\ statistic \end{pmatrix} - \begin{pmatrix} critical \\ value \end{pmatrix}\begin{pmatrix} standard \\ error \end{pmatrix}\right] \leq \begin{pmatrix} population \\ parameter \end{pmatrix} \leq \left[\begin{pmatrix} sample \\ statistic \end{pmatrix} + \begin{pmatrix} critical \\ value \end{pmatrix}\begin{pmatrix} standard \\ error \end{pmatrix}\right]$$

$$\overline{X} - Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \leq \mu_0 \leq \overline{X} + Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Confidence intervals and hypothesis test are linked by critical values:

- In a confidence interval, we aim to determine whether the hypothesized value of the population mean (μ_0), lies within a computed interval with a particular degree of confidence ($1-\alpha$). Here the interval represents the "fail-to-reject-the-null region" and is based around, or centered on the sample mean, $\overline{X}$
- In a hypothesis test, we examine whether the sample mean, $\overline{X}$ lies in the rejection region (i.e., outside the interval) or in the fail-to-reject-the-null region (i.e., within the interval) at a particular level of significance (α). Here the interval is based around or centered on the hypothesized value of the population mean (μ_0).

P-value

The **p-value** is the smallest level of significance at which the null hypothesis can be rejected.

- If the p-value is *lower* than the required level of significance, we *reject* the null hypothesis.
- If the p-value is *greater* than the required level of significance, we *fail to reject* the null hypothesis.

P-Hacking and Data Dredging **P-hacking** (also known as *data dredging, data mining*, or *specification searching*) refers to the practice of manipulating data analysis, testing multiple models, or trying various sample specifications until a statistically significant result (p-value $< \alpha$) is obtained, and reporting only those favorable results.

Cập nhật chương trình 2027

P-Hacking là một chủ đề mới và rất được nhấn mạnh trong chương trình CFA 2027 nhằm nâng cao tính trung thực trong phân tích định lượng. Học viên cần phân biệt p-hacking với phân tích khám phá chính đáng, đồng thời nắm rõ hậu quả (thổi phồng sai lầm Loại I) và các biện pháp giảm thiểu.

Key Mechanisms of P-Hacking:

- **Specification Searching:** Running multiple model variations (altering variables, functional forms, or lag structures) and reporting only the model yielding $p < 0.05$.
- **Selective Sample Trimming:** Arbitrarily removing or retaining outliers and sub-sample periods to force statistical significance.
- **Cherry-picking / Publication Bias:** Publishing or presenting only statistically significant findings while ignoring non-significant results (the "file-drawer problem").
- **HARKing (Hypothesizing After Results are Known):** Formulating hypotheses *after* observing the data results and presenting them as if they were pre-planned predictions.

Statistical Consequences of Multiple Testing: When conducting k independent tests at a significance level α, the probability of committing at least one Type I error (false positive) across all tests is:

$$P(\text{At least one Type I Error}) = 1 - (1 - \alpha)^k \tag{7.7}$$

For example, if an analyst tests 20 independent trading strategies at $\alpha = 0.05$:

$$P(\text{At least one false positive}) = 1 - (1 - 0.05)^{20} = 1 - (0.95)^{20} \approx 64.15\%$$

Thus, without correcting for multiple comparisons, finding a "statistically significant" strategy is highly likely due to pure randomness rather than true market predictability.

Mitigation Strategies: To ensure research integrity and avoid p-hacking:

1. **Pre-Registration:** Documenting hypotheses, sample choices, and testing methods prior to inspecting the data.
2. **Out-of-Sample Testing:** Validating the selected model on new, independent data periods.
3. **Adjusting P-Values:** Applying multiple testing corrections such as the *Bonferroni adjustment* ($\alpha_{adj} = \alpha/k$) or controlling the *False Discovery Rate (FDR)*.

Exam Tip: P-Hacking in CFA Exam

On the CFA exam, remember that p-hacking increases the probability of **Type I errors** (false positives). A trading model derived from p-hacking usually fails dramatically when deployed in real-world out-of-sample trading!

7.3 Parametric Testing

c | compare and contrast parametric and non-parametric tests, describe situations in which each is the more appropriate type of test, construct appropriate hypothesis tests, and interpret the results

7.3.1 Test of a Single Mean

To test hypotheses about a single population mean, the hypotheses are structured as follows:

$$H_o : \mu = \mu_0$$
$$H_a : \mu \neq \mu_0$$

where:

μ = true population mean

μ_0 = hypothesized value of the population mean

When the population standard deviation (σ) is unknown, the t-test statistic with $n-1$ degrees of freedom is computed as:

$$t = \frac{\bar{X} - \mu_0}{s_{\bar{X}}} = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

where:

$\bar{X}$ = sample mean $= \frac{1}{n}\sum_{i=1}^{n} X_i$

μ_0 = hypothesized population mean

s = sample standard deviation $= \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$

$s_{\bar{X}}$ = standard error of the sample mean $= \frac{s}{\sqrt{n}}$

n = the sample size

7.3.2 Test of Paired Differences (Dependent Samples)

When the samples of the two populations whose means we are comparing are dependent, the paired comparisons test is used. Dependence can result from events that affect both populations. Remember that to conduct a paired comparisons test, the populations must be normally distributed.
The hypotheses are structured as follows:

$H_o : \mu_d = \mu_{dz}$

$H_a : \mu_d \neq \mu_{dz}$

where:

μ_d = mean of the population of paired differnces

μ_{dz} = hypothesized mean of paired differences, which is usually zero

For the paired comparisons test, the t-stat with $n-1$ degrees of freedom is computed as:

$$t = \frac{\overline{d} - \mu_{dz}}{s_{\overline{d}}}$$

where:

$\overline{d}$ = sample mean difference $= \frac{1}{n}\sum_{i=1}^{n} d_i$

d_i = difference between the ith pair of observations

$s_{\overline{d}}$ = standard error of the sample mean difference $= \frac{s_d}{\sqrt{n}}$

s_d = sample standrad deviation $= \sqrt{\frac{\sum_{i=1}^{n}(d_i - \overline{d})^2}{n-1}}$

n = the number of paired observations

7.3.3 Test of Equality of Two Unknown Means (Independent Samples)

When testing whether the means of two independent populations are equal, the hypotheses are structured as follows:

$H_o : \mu_1 - \mu_2 = \Delta_0$

$H_a : \mu_1 - \mu_2 \neq \Delta_0$

where:

μ_1, μ_2 = means of Population 1 and Population 2

Δ_0 = hypothesized difference between population means (usually zero)

Assuming equal population variances ($\sigma_1^2 = \sigma_2^2$), the pooled t-test statistic with $n_1 + n_2 - 2$ degrees of freedom is computed as:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}}$$

where:

$\bar{X}_1, \bar{X}_2$ = sample means of Population 1 and Population 2

Δ_0 = hypothesized difference between the population means

s_p^2 = pooled sample variance = $\dfrac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$

s_1^2, s_2^2 = sample variances of Population 1 and Population 2

n_1, n_2 = sample sizes of Population 1 and Population 2

7.3.4 Test of a Single Variance

To test hypotheses concerning the variance of a single normally distributed population, the hypotheses are structured as follows:

$$H_o : \sigma^2 = \sigma_0^2$$
$$H_a : \sigma^2 \neq \sigma_0^2$$

where:

σ^2 = true population variance

σ_0^2 = hypothesized value of the population variance

The Chi-square (χ^2) test statistic with $n - 1$ degrees of freedom is computed as:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

where:

s^2 = sample variance = $\dfrac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$

σ_0^2 = hypothesized population variance

n = sample size

$n - 1$ = degrees of freedom for the Chi-square test

7.3.5 Test Concerning the Equality of Two Variances

Hypotheses related to the equality of the variance of two populations are tested with an F-test. This test is used under the assumptions that:

- The populations from which samples are drawn are normally distributed.
- The samples are independent.

Hypothesis tests concerning the variance of two populations can be one-tailed or two-tailed tests:

One-tailed tests

$$H_o : \sigma_1^2 \leq \sigma_2^2 \text{ versus } H_a : \sigma_1^2 > \sigma_2^2$$
$$H_o : \sigma_1^2 \geq \sigma_2^2 \text{ versus } H_a : \sigma_1^2 < \sigma_2^2$$

Two-tailed tests

$$H_o : \sigma_1^2 = \sigma_2^2 \text{ versus } H_a : \sigma_1^2 = \sigma_2^2$$

where:

σ_1^2 = variance of Population 1

σ_2^2 = variance of Population 2

The test statistic for the F-test is given by:

$$F = \frac{s_1^2}{s_2^2}$$

where:

s_1^2 = Variance of sample drawn from Population 1

s_2^2 = Variance of sample drawn from Population 2

Note: Always put the larger variance in the numerator when calculating the F-test statistic. Also note that the F-stat is simply the ratio of the two samples' variances.
Features of the F-distribution:

- It is skewed to the right.
- It is bounded by zero on the left.
- It is defined by two separate degrees of freedom.

The rejection region for an F-test, whether it be one-tailed or two-tailed, always lies in the right tail.

7.3.6 Parametric Test of a Correlation

The parametric pairwise correlation coefficient is often referred to as **Pearson correlation**, the **bivariate correlation**, or simply the correlation.

$$H_o : \rho = 0$$
$$H_a : \rho \neq 0$$

with:

$$\text{Test-stat} = t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Note: This would be a two-tailed t-test with n - 2 degrees of freedom.
The decision rule for the test is that we reject H_o if $t - stat > +t_{critical}$ or if $t - stat < -t_{critical}$.

7.4 Non-Parametric Test

7.4.1 Non-Parametric Test of Correlation

Spearman rank correlation coefficient is a non-parametric test that is calculated based on the ranks of two variables within their respective data sets. It lies between -1 and +1 where -1 (+1) denotes a perfectly inverse (positive) relationship between the ranks of the two variables, and 0 represents no correlation.
With n as the sample size, the Spearman rank correlation is given by

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$$

Often the results of parametric and non-parametric tests are both presented. This allows the reader to gauge how sensitive the conclusions are to the assumptions underlying the parametric tests. If the assumptions made under the parametric test are met, the use of parametric tests is preferred because they usually offer more crisp and reliable conclusions.
Spearman rank correlation đo lường mối tương quan giữa hạng của X và hạng của Y.

$$r_s = \sum_{i=1}^{n} \frac{(x_i - \overline{x})(y_i - \overline{y})}{\sigma_x \sigma_y}$$

Trong đó:

x_i, y_i *lần lượt là hạng của* X_i *và* Y_i *trong bộ dữ liệu X và Y*

$\overline{x}$, $\overline{y}$ *lần lượt là trung bình của các bộ* x_i *và* y_i

σ_x, σ_y *lần lượt là độ lệch chuẩn của các bộ* x_i *và* y_i

Ta lần lượt sử dụng các đẳng thức sau đây:

$$\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} i = \frac{n(n+1)}{2} = \sum_{i=1}^{n} y_i$$

$$\overline{x} = \frac{\sum_{i=1}^{n} i}{n} = \frac{n(n+1)/2}{n} = \frac{n+1}{2} = \overline{y}$$

$$\sigma_x = \sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2} = \sqrt{\sum_{i=1}^{n}(i - \frac{n+1}{2})^2} = \sigma_y$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

Suy ra:

$$r_s = 1 - \frac{6\sum_{i=1}^{n}(x_i - y_i)^2}{n(n^2-1)} = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$$

7.4.2 Tests of Independence Using Contingency Table Data

When evaluating whether two categorical variables are independent using contingency table data, the hypotheses are structured as follows:

H_o : The two categorical variables are independent

H_a : The two categorical variables are dependent (related)

where:

$H_o \implies$ No association between row and column classifications

$H_a \implies$ Significant association between row and column classifications

Example: If we want to test whether there is a relationship between the size and investment type, the Chi-square

Table 7.2: Size and Investment Type Classifications of 1,594 ETFs

	Size based on Market Capitalization			
Investment Type	**Small**	**Medium**	**Large**	**Total**
Value	50	110	343	503
Growth	42	122	202	366
Blend	56	149	520	725
Total	148	381	1,065	1,594

(χ^2) test statistic for independence with $df = (r-1)(c-1)$ degrees of freedom is computed as:

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where:

O_{ij} = observed frequency in cell of row i and column j

E_{ij} = expected frequency assuming independence = $\dfrac{(\text{Row } i \text{ Total}) \times (\text{Column } j \text{ Total})}{N}$

N = total number of observations in the table

r, c = number of rows and number of columns in the contingency table

$df = (r-1)(c-1)$ = degrees of freedom for the test

7.5 Parametric and Nonparametric tests

parametric test has at least one of the following two characteristics:

- It is concerned with parameters or defining features of a distribution.
- It makes a definite set of assumptions.

non-parametric test is not concerned with a parameter and makes only a minimal set of assumptions regarding the population. Non-parametric tests are used when:

- The researcher is concerned about quantities other than the parameters of the distribution.
- The assumptions made by parametric tests cannot be supported.
- When the data available is ranked (ordinal measurement scale). For example, non-parametric methods are widely used for studying populations such as movie reviews, which receive one to five stars based on people's preferences.

Nonparametric procedures are primarily used in four situations:

1. when the data do not meet distributional assumptions,
2. when there are outliers,
3. when the data are given in ranks or use an ordinal scale, or
4. when the relevant hypotheses do not concern a parameter.

Example:

Table 7.3: Nonparametric Alternatives to Parametric Tests

	Parametric	**Nonparametric**
Tests concerning a single mean	t-test z-test	Wilcoxon signed-rank test
Tests concerning differences between means	t-test approximate t-test	Mann-Whitney U test
Tests concerning mean differences (paired comparisons tests)	t-test	Wilcoxon signed-rank test Sign test

Module 8

The Return and Risk of a Financial Portfolio

LEARNING OUTCOMES	
a	calculate, interpret, and evaluate the expected return, variance, standard deviation, covariance, and correlation of portfolio returns
b	describe, calculate, and interpret the minimum-variance portfolio and portfolios that lie on the efficient frontier
c	explain the selection of an optimal portfolio, given an investor's risk aversion and the capital allocation line, and how this extends to the market portfolio and the capital market line

8.1 Calculating Portfolio Statistics

a	calculate, interpret, and evaluate the expected return, variance, standard deviation, covariance, and correlation of portfolio returns

Portfolio statistics, such as expected returns, standard deviations, and correlations, are fundamental for effective portfolio management and asset allocation. These statistics can be historical (factoring in past returns) or forward-looking (using model-based forecasts).

8.1.1 Returns of a Portfolio

The return of a portfolio is the weighted average of the returns of its individual assets. Historical portfolio returns are calculated by summing the products of the beginning-of-period asset weights and their corresponding realized returns:

$$R_{P,t} = \sum_{i=1}^{N} w_{i,t-1} R_{i,t} \tag{8.1}$$

where N is the number of assets, $w_{i,t-1}$ is the weight of asset i at the end of period $t-1$ (beginning of period t), and $R_{i,t}$ is the realized return of asset i in period t, defined as:

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1} + \text{Div}_{i,t}}{P_{i,t-1}} \tag{8.2}$$

To evaluate returns over multiple periods from $t = 1$ to T, if the weights are held constant at their initial values $w_{i,0}$ (which implies continuous rebalancing), the aggregate portfolio return is:

$$R_{P,T} = \sum_{i=1}^{N} w_{i,0} R_{i,T} \tag{8.3}$$

where $R_{i,T}$ is the cumulative return of asset i over the entire period T. The arithmetic average return of the portfolio across these periods is:

$$\bar{R}_P = \frac{1}{T} \sum_{t=1}^{T} R_{P,t} \tag{8.4}$$

When compounding returns over multiple periods without rebalancing, the asset weights will naturally shift as a result of varying asset performance. This phenomenon is known as **portfolio drift**. The cumulative return over T periods with drifting weights is:

$$R_{P,T} = \prod_{t=1}^{T} \left(1 + \sum_{i=1}^{N} w_{i,t-1} R_{i,t}\right) - 1 \tag{8.5}$$

The geometric average return, or geometric return, is defined as:

$$\bar{R}_{g,P} = \left[\prod_{t=1}^{T}(1 + R_{P,t})\right]^{\frac{1}{T}} - 1 = (1 + R_{P,T})^{\frac{1}{T}} - 1 \tag{8.6}$$

Style drift: where the portfolio deviates from its original investment strategy. Style drift can occur even without portfolio drift when asset characteristics change over time. For instance, a small-capitalization company may grow into a large-capitalization company. In some cases, portfolio drift may offset style drift, helping maintain the intended strategy.

Portfolio drift (lệch tỷ trọng mục tiêu) do tài sản tăng trưởng không đều. Nếu không tái cân bằng (rebalancing) định kỳ, tỷ trọng sẽ dồn về tài sản tăng trưởng mạnh nhất, khiến rủi ro danh mục tiệm cận tài sản đó.

Example 1: Portfolio Drift over Three Periods

An investor holds a portfolio of two assets, A and B, starting with EUR 1 invested in each ($w_{A,0} = w_{B,0} = 0.5$, total portfolio value = EUR 2). The portfolio is left untouched (no rebalancing) over three periods. The returns are:

Period	Asset A Return ($R_{A,t}$)	Asset B Return ($R_{B,t}$)
1	+20%	+5%
2	−10%	+15%
3	+25%	−5%

Calculate the portfolio returns and weights at each step, and determine the cumulative arithmetic and geometric returns. **Solution:**

- **Period 1:**
  - Portfolio return: $R_{P,1} = 0.5(20\%) + 0.5(5\%) = 12.5\%$.
  - Ending value: Asset A = EUR 1.20; Asset B = EUR 1.05. Total value = EUR 2.25.
  - Ending weights for Period 2: $w_{A,1} = \frac{1.20}{2.25} \approx 53.33\%$; $w_{B,1} = \frac{1.05}{2.25} \approx 46.67\%$.
- **Period 2:**
  - Portfolio return: $R_{P,2} = 0.5333(-10\%) + 0.4667(15\%) = 1.67\%$.
  - Ending value: Asset A = EUR 1.08; Asset B = EUR 1.2075. Total value = EUR 2.2875.
  - Ending weights for Period 3: $w_{A,2} = \frac{1.08}{2.2875} \approx 47.21\%$; $w_{B,2} = \frac{1.2075}{2.2875} \approx 52.79\%$.
- **Period 3:**
  - Portfolio return: $R_{P,3} = 0.4721(25\%) + 0.5279(-5\%) = 9.16\%$.
  - Ending value: Asset A = EUR 1.35; Asset B = EUR 1.1471. Total value = EUR 2.4971.
  - Ending weights: $w_{A,3} = \frac{1.35}{2.4971} \approx 54.06\%$; $w_{B,3} = \frac{1.1471}{2.4971} \approx 45.94\%$.
- **Cumulative Returns:**
  - Cumulative arithmetic return: $R_{P,T} = R_{P,1} + R_{P,2} + R_{P,3} = 12.5\% + 1.67\% + 9.16\% = 23.33\%$.
  - Arithmetic average return: $\bar{R}_P = \frac{23.33\%}{3} \approx 7.78\%$.
  - Cumulative geometric return: $R_{P,T} = (1.125 \times 1.0167 \times 1.0916) - 1 \approx 24.86\%$.
  - Geometric average return: $\bar{R}_{g,P} = (1.125 \times 1.0167 \times 1.0916)^{\frac{1}{3}} - 1 \approx 7.68\%$.

Table 8.1: Portfolio Drift Characteristics by Weighting Scheme

Weighting Method	Impact of Drift	Resulting Portfolio Behavior
Equal-Weighted	Weights drift as top performers grow larger	Potential bias toward high growth and higher-volatility assets
Non-Equally Weighted	Large-allocation assets drift and dominate portfolio performance	Momentum-driven returns with increased concentration risk

8.1.2 Variance and Standard Deviation of Portfolio Returns

The risk of a portfolio is measured by its variance and standard deviation. The variance of portfolio returns (σ_P^2) incorporates both individual asset variances and the covariances between all asset pairs:

$$\sigma_P^2 = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j \sigma_{i,j} \tag{8.7}$$

where $\sigma_{i,j}$ represents the covariance between asset i and asset j. Note that $\sigma_{i,i}$ is the variance of asset i (σ_i^2). Standard deviation (σ_P) is the square root of variance. If covariances are modeled as random variables, the expected variance of portfolio returns is:

$$E[\sigma_P^2] = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j E[\sigma_{i,j}] \tag{8.8}$$

Covariance is linked to the correlation coefficient ($\rho_{i,j}$) by: $\sigma_{i,j} = \rho_{i,j}\sigma_i\sigma_j$

Due to the non-linear interaction of volatility and correlation, the expected covariance $E[\sigma_{i,j}] \neq E[\rho_{i,j}] \times E[\sigma_i] \times E[\sigma_j]$ in the presence of joint dependencies. For a two-asset portfolio, the portfolio variance formula is:

$$\sigma_P^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_{1,2} = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{1,2}\sigma_1\sigma_2 \tag{8.9}$$

Example 2: Portfolio Allocation (Two-Asset)

A portfolio allocates 80% to the S&P 500 Index ($w_A = 0.8$) and 20% to the MSCI Emerging Markets Index ($w_B = 0.2$). Expected returns are 9.93% and 18.20%, and standard deviations are 16.21% and 33.11%, respectively. The covariance between the two is 0.0050. Calculate the expected portfolio return, variance, and standard deviation. **Solution:**

1. **Expected Portfolio Return:** $E[R_P] = 0.8(9.93\%) + 0.2(18.20\%) = 11.58\%$.
2. **Portfolio Variance:**

$$E[\sigma_P^2] = (0.8^2 \times 0.1621^2) + (0.2^2 \times 0.3311^2) + (2 \times 0.8 \times 0.2 \times 0.0050)$$

$$E[\sigma_P^2] = 0.01684 + 0.00439 + 0.00160 \approx 0.02283$$

3. **Portfolio Standard Deviation:** $\sigma_P = \sqrt{0.02283} \approx 15.11\%$.

Tương quan thấp giữa S&P 500 và MSCI EM tạo đồ thị cong lõm (concave), giúp danh mục có rủi ro (14.59%) thấp hơn cả S&P 500 (16.21%). Nếu tương quan bằng +1.0, đa dạng hóa sẽ hoàn toàn không có tác dụng giảm rủi ro.

Example 3: Varying Correlations

Asset 1 has expected return 7.0% and standard deviation 12.0%. Asset 2 has expected return 15.0% and standard deviation 25.0%. Compare the portfolio risk (σ_P) for different asset weights under correlations (ρ) of +1.0, +0.5, +0.2, and −1.0. **Solution:** The calculated portfolio standard deviation (σ_P) for each scenario is summarized in Table 8.3.

Table 8.2: Portfolio Return and Risk under Different Weights (S&P 500 vs. MSCI EM)

Portfolio	**S&P 500 Weight (%)**	**MSCI EM Weight (%)**	**Expected Return (%)**	**Risk / SD (%)**
1	0	100	18.20	33.11
2	10	90	17.37	29.85
3	20	80	16.55	26.70
4	30	70	15.72	23.71
5	40	60	14.89	20.93
6	50	50	14.07	18.47
7	60	40	13.24	16.47
8	70	30	12.41	15.12
9	**80**	**20**	**11.58**	**14.59**
10	90	10	10.76	14.98
11	100	0	9.93	16.21

Table 8.3: Portfolio Return and Risk Under Different Correlations

Weight of Asset 1 (%)	**Portfolio Return (%)**	**Portfolio Standard Deviation (%) with ρ of:** +1.0	+0.5	+0.2	−1.0
0	15.0	25.0	25.0	25.0	25.0
20	13.4	22.4	21.3	20.6	17.6
40	11.8	19.8	17.9	16.6	10.2
50	11.0	18.5	16.3	14.9	6.5
60	10.2	17.2	15.0	13.4	2.8
80	8.6	14.6	12.9	11.7	4.6
100	7.0	12.0	12.0	12.0	12.0

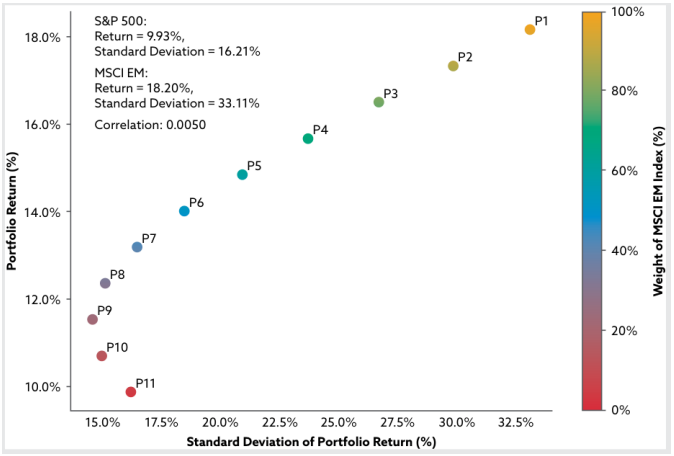


Figure 8.1: Feasible Set of Portfolios from S&P 500 and MSCI EM (Covariance = 0.0050)

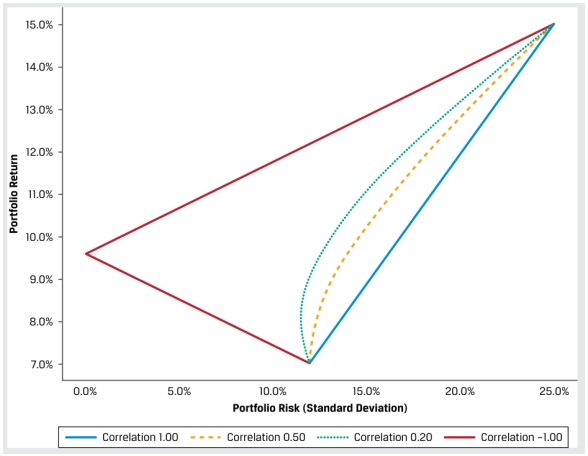


Figure 8.2: Portfolio Opportunity Frontier for Different Correlation Coefficients

As correlation declines, the frontier curves further to the left, which indicates a greater reduction in portfolio risk for a given expected return. When $\rho = -1$, the portfolio risk can be reduced to zero.

Matrix Notation of Portfolio Risk

For a portfolio with N assets, the number of unique covariances that must be computed is:

$$\text{Unique Covariances} = \frac{N(N-1)}{2} \tag{8.10}$$

Along with the N individual variances, the total parameters needed for an N-asset portfolio is $\frac{N(N+1)}{2}$. For example, a 100-security portfolio requires 4,950 unique covariances. This structure is organized into the symmetric

variance-covariance matrix V:

$$\mathbf{V} = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} & \cdots & \sigma_{1,N} \\ \sigma_{2,1} & \sigma_2^2 & \sigma_{2,3} & \cdots & \sigma_{2,N} \\ \sigma_{3,1} & \sigma_{3,2} & \sigma_3^2 & \cdots & \sigma_{3,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{N,1} & \sigma_{N,2} & \sigma_{N,3} & \cdots & \sigma_N^2 \end{bmatrix} \tag{8.11}$$

The matching normalized matrix is the **correlation matrix $\boldsymbol{\rho}$**:

$$\boldsymbol{\rho} = \begin{bmatrix} 1 & \rho_{1,2} & \rho_{1,3} & \cdots & \rho_{1,N} \\ \rho_{2,1} & 1 & \rho_{2,3} & \cdots & \rho_{2,N} \\ \rho_{3,1} & \rho_{3,2} & 1 & \cdots & \rho_{3,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho_{N,1} & \rho_{N,2} & \rho_{N,3} & \cdots & 1 \end{bmatrix} \tag{8.12}$$

Thực tế: Bloomberg thường nhân covariance matrix với 10^4 (đơn vị $\%^2$). Khi quy đổi về số thập phân để tính toán, bắt buộc phải chia cho 10^4 để tránh sai số hàng ngàn lần.

Example 4: A Portfolio with Three Assets (Covariance Matrix)

A portfolio contains three assets. The expected returns, standard deviations, and correlation matrix are given below:

- Expected Returns: $E[R_1] = 10\%$, $E[R_2] = 12\%$, $E[R_3] = 8\%$
- Volatilities: $\sigma_1 = 15\%$, $\sigma_2 = 20\%$, $\sigma_3 = 10\%$
- Correlation Matrix $\boldsymbol{\rho}$:

$$\boldsymbol{\rho} = \begin{bmatrix} 1 & 0.2 & -0.3 \\ 0.2 & 1 & 0.4 \\ -0.3 & 0.4 & 1 \end{bmatrix}$$

Construct the variance-covariance matrix $\mathbf{V}$. **Solution:** Using the relationship between covariance and correlation, $\sigma_{i,j} = \sigma_i \sigma_j \rho_{i,j}$:

1. **Asset Variances (main diagonal of the covariance matrix):**
   - $\sigma_1^2 = 0.15^2 = 0.0225$
   - $\sigma_2^2 = 0.20^2 = 0.0400$
   - $\sigma_3^2 = 0.10^2 = 0.0100$
2. **Asset Covariances (off-diagonal elements):**
   - $\sigma_{1,2} = \sigma_{2,1} = 0.15 \times 0.20 \times 0.2 = 0.0060$
   - $\sigma_{1,3} = \sigma_{3,1} = 0.15 \times 0.10 \times (-0.3) = -0.0045$
   - $\sigma_{2,3} = \sigma_{3,2} = 0.20 \times 0.10 \times 0.4 = 0.0080$

Therefore, the variance-covariance matrix $\mathbf{V}$ is:

$$\mathbf{V} = \begin{bmatrix} 0.0225 & 0.0060 & -0.0045 \\ 0.0060 & 0.0400 & 0.0080 \\ -0.0045 & 0.0080 & 0.0100 \end{bmatrix}$$

Example 5: Return and Risk of a Three-Asset Portfolio

Using the data from Example 4, assume the portfolio weights are $w_1 = 0.4$, $w_2 = 0.3$, $w_3 = 0.3$. Calculate the expected portfolio return and standard deviation. **Solution:**

1. **Expected Portfolio Return:**

$$E[R_P] = (0.4 \times 10\%) + (0.3 \times 12\%) + (0.3 \times 8\%) = 10\%$$

2. **Expected Portfolio Variance:**

$$\sigma_P^2 = \sum_{i=1}^{3}\sum_{j=1}^{3} w_i w_j \sigma_{i,j}$$

$$\sigma_P^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + w_3^2\sigma_3^2 + 2w_1w_2\sigma_{1,2} + 2w_1w_3\sigma_{1,3} + 2w_2w_3\sigma_{2,3}$$

$$\sigma_P^2 = (0.4^2 \times 0.0225) + (0.3^2 \times 0.04) + (0.3^2 \times 0.01) + 2(0.4)(0.3)(0.006) + 2(0.4)(0.3)(-0.0045) + 2(0.3)(0.3)(0.008)$$

$$\sigma_P^2 = 0.0036 + 0.0036 + 0.0009 + 0.00144 - 0.00108 + 0.00144 = 0.0099$$

3. **Portfolio Standard Deviation:** $\sigma_P = \sqrt{0.0099} \approx 9.95\%$.

8.1.3 Expected Return and Variance under Economic Scenarios

Expected returns and covariances can be derived from joint probabilities over M economic scenarios. Under scenario M occurring with probability p_M, the portfolio return is:

$$R_{P,M} = \sum_{i=1}^{N} w_i R_{i,M} \tag{8.13}$$

The expected portfolio return over all scenarios is:

$$E[R_P] = \sum_{M=1}^{M} p_M R_{P,M} = \sum_{M=1}^{M} p_M \left(\sum_{i=1}^{N} w_i R_{i,M} \right) \tag{8.14}$$

The portfolio variance is the weighted sum of squared deviations from the expected return:

$$\sigma_P^2 = \sum_{M=1}^{M} p_M \left(R_{P,M} - E[R_P] \right)^2 \tag{8.15}$$

Example 6: Expected Portfolio Returns and Scenarios

Consider three assets, A, B, and C, with weights $w_A = 0.4$, $w_B = 0.3$, $w_C = 0.3$. Their scenario returns are:

Scenario	Probability	Asset A (%)	Asset B (%)	Asset C (%)
1 (Expansion)	30%	15	10	5
2 (Steady)	50%	10	12	8
3 (Recession)	20%	-5	20	12

Calculate the expected portfolio return and standard deviation. **Solution:**

1. **Scenario Returns:**
   - Scenario 1 (Expansion): $R_{P,1} = 0.4(15) + 0.3(10) + 0.3(5) = 10.5\%$.
   - Scenario 2 (Steady): $R_{P,2} = 0.4(10) + 0.3(12) + 0.3(8) = 10.0\%$.
   - Scenario 3 (Recession): $R_{P,3} = 0.4(-5) + 0.3(20) + 0.3(12) = 7.6\%$.
2. **Expected Portfolio Return:**

$$E[R_P] = 0.3(10.5\%) + 0.5(10.0\%) + 0.2(7.6\%) = 9.67\%$$

3. **Expected Portfolio Variance:**
   - Scenario 1 contribution: $0.3 \times (10.5\% - 9.67\%)^2 = 0.3 \times (0.83\%)^2 \approx 0.0000207$
   - Scenario 2 contribution: $0.5 \times (10.0\% - 9.67\%)^2 = 0.5 \times (0.33\%)^2 \approx 0.0000054$
   - Scenario 3 contribution: $0.2 \times (7.6\% - 9.67\%)^2 = 0.2 \times (-2.07\%)^2 \approx 0.0000857$

$$\sigma_P^2 = 0.0000207 + 0.0000054 + 0.0000857 = 0.0001118$$

4. **Portfolio Standard Deviation:** $\sigma_P = \sqrt{0.0001118} \approx 1.06\%$.

Summation vs. Matrix Notation Comparison

For portfolios with a large number of assets, matrix notation provides a compact, computationally efficient mathematical representation.

Table 8.4: Comparison of Summation and Matrix Notation

Concept	Summation Notation	Matrix Notation
Asset Weights Vector	w_i	$\mathbf{w} = [w_1, w_2, \ldots, w_N]^T$
Expected Returns Vector	$E[R_i]$	$\boldsymbol{\mu} = [E[R_1], E[R_2], \ldots, E[R_N]]^T$
Portfolio Return	$E[R_P] = \sum_{i=1}^{N} w_i E[R_i]$	$E[R_P] = \mathbf{w}^T \boldsymbol{\mu}$
Covariance Matrix	$\sigma_{i,j} = \rho_{i,j}\sigma_i\sigma_j$	$\mathbf{V}$ (symmetric $N \times N$ matrix)
Portfolio Variance	$\sigma_P^2 = \sum_{i=1}^{N}\sum_{j=1}^{N} w_i w_j \sigma_{i,j}$	$\sigma_P^2 = \mathbf{w}^T\mathbf{V}\mathbf{w}$

Example 7: The Three-Asset Portfolio Revisited (Matrix Form)

Using the data from Example 5, confirm the return and variance using matrix multiplication.

$$\mathbf{w} = \begin{bmatrix} 0.4 \\ 0.3 \\ 0.3 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} 0.10 \\ 0.12 \\ 0.08 \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} 0.0225 & 0.0060 & -0.0045 \\ 0.0060 & 0.0400 & 0.0080 \\ -0.0045 & 0.0080 & 0.0100 \end{bmatrix}$$

Solution:

1. **Expected Return:**

$$E[R_P] = \mathbf{w}^T\boldsymbol{\mu} = \begin{bmatrix} 0.4 & 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.10 \\ 0.12 \\ 0.08 \end{bmatrix} = 0.4(0.10) + 0.3(0.12) + 0.3(0.08) = 10\%$$

2. **Portfolio Variance ($\mathbf{w}^T\mathbf{V}\mathbf{w}$):** First, compute the matrix-vector product $\mathbf{V}\mathbf{w}$:

$$\mathbf{V}\mathbf{w} = \begin{bmatrix} 0.0225(0.4) + 0.006(0.3) - 0.0045(0.3) \\ 0.006(0.4) + 0.040(0.3) + 0.008(0.3) \\ -0.0045(0.4) + 0.008(0.3) + 0.010(0.3) \end{bmatrix} = \begin{bmatrix} 0.00945 \\ 0.01680 \\ 0.00360 \end{bmatrix}$$

Next, multiply the row vector $\mathbf{w}^T$ by the column vector ($\mathbf{V}\mathbf{w}$):

$$\sigma_P^2 = 0.4(0.00945) + 0.3(0.0168) + 0.3(0.0036) = 0.00378 + 0.00504 + 0.00108 = 0.0099$$

This yields $\sigma_P^2 = 0.0099$, which perfectly matches the result from the summation formula in Example 5.

8.1.4 The Benefits of Diversification

The power of diversification lies in combining assets that are not perfectly positively correlated to reduce overall portfolio risk. Consider an equally weighted portfolio ($w_i = 1/N$) where all assets have the same variance σ^2 and the average correlation coefficient is $\bar{\rho}$. The portfolio variance is:

$$\sigma_P^2 = \frac{\sigma^2}{N} + \left(1 - \frac{1}{N}\right)\bar{\rho}\sigma^2 \tag{8.16}$$

As the number of assets $N \to \infty$:

- The contribution of the individual asset variance, $\frac{\sigma^2}{N}$, converges to 0.
- The multiplier $\left(1 - \frac{1}{N}\right)$ approaches 1.

This leaves only the average covariance term to determine the portfolio's risk:

$$\lim_{N\to\infty} \sigma_P^2 = \bar{\rho}\sigma^2 = \text{Average Covariance} \tag{8.17}$$

Nhận xét: Danh mục đa dạng hóa tốt chỉ còn chịu rủi ro hệ thống (covariance trung bình) do rủi ro phi hệ thống đã bị triệt tiêu. Thị trường chỉ bù đắp lợi nhuận cho rủi ro không thể đa dạng hóa (rủi ro hệ thống).

Example 8: Approximating Standard Deviation of a Portfolio with Many Assets

An equally weighted portfolio has 50 assets ($N = 50$). The average asset variance is $\sigma^2 = 0.0400$ and the average correlation coefficient is $\bar{\rho} = 0.3$. Calculate the exact portfolio standard deviation and compare it to the large-N approximation. **Solution:**

1. **Exact Portfolio Variance using Equation 8.16:**

$$\sigma_P^2 = \frac{0.04}{50} + \left(1 - \frac{1}{50}\right)(0.3)(0.04) = 0.0008 + 0.98(0.012) = 0.01256$$

   Exact standard deviation: $\sigma_P = \sqrt{0.01256} \approx 11.21\%$.

2. **Approximated Variance using Equation 8.17:**

$$\sigma_P^2 \approx \bar{\rho}\sigma^2 = 0.3(0.04) = 0.0120$$

   Approximated standard deviation: $\sigma_P \approx \sqrt{0.0120} \approx 10.95\%$.

The approximation is very close to the exact value, demonstrating that for a sufficiently large portfolio, the portfolio variance is dominated by the average covariance between the assets.

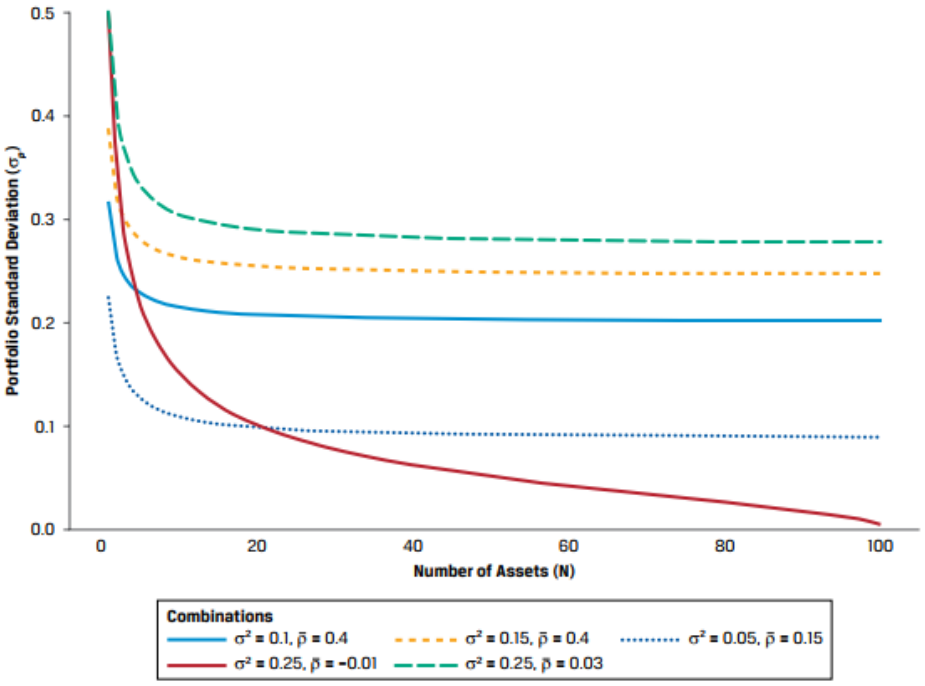


Figure 8.3: Portfolio Risk Reduction as the Number of Assets (N) Increases

Example 9: Portfolio Size and Portfolio Variance (GBM Simulation)

A simulation generates 250 daily returns for 100 randomly selected assets using geometric Brownian motion (GBM). Portfolios ranging from $N = 2$ to 50 assets are constructed. As shown in Exhibit 8.4, the standard deviation of portfolio returns decreases sharply as N increases from 2 to 20, but the marginal benefit of adding more assets diminishes beyond $N = 30$. **Key Takeaway:** A portfolio of 20 to 30 assets captures most of the benefits of diversification, provided the assets have low correlations. Adding assets beyond this threshold yields negligible reduction in portfolio risk, as the remaining risk is systematic risk, which cannot be diversified away.

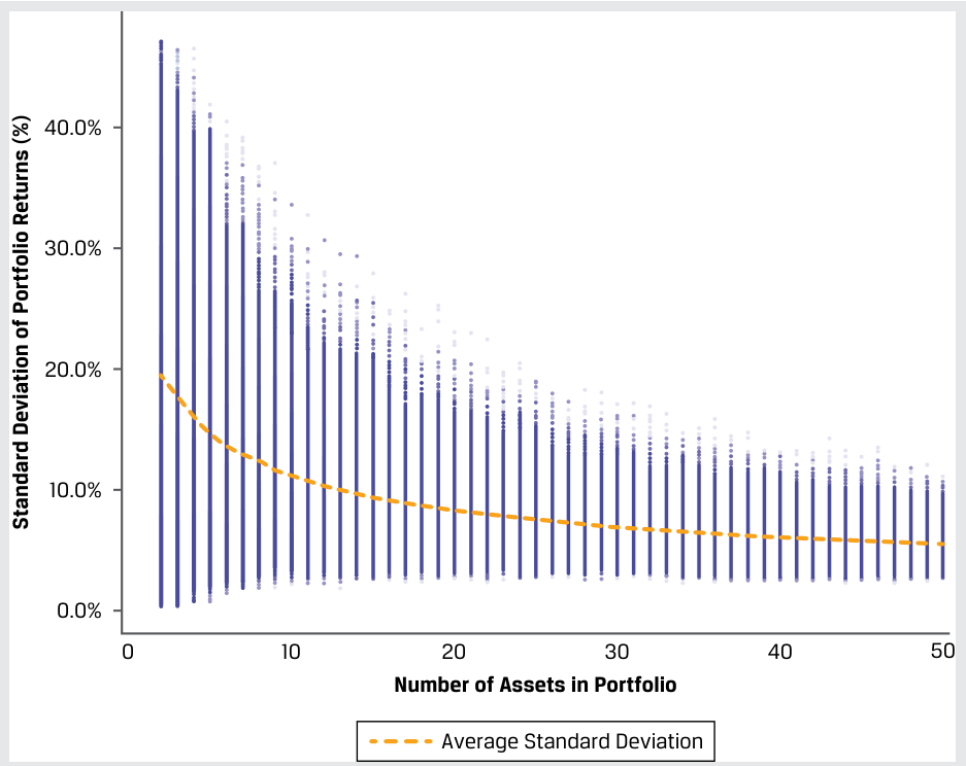


Figure 8.4: GBM Simulation of Portfolio Standard Deviation vs. Number of Assets

8.2 The Minimum-Variance Portfolio and the Efficient Frontier

b	describe, calculate, and interpret the minimum-variance portfolio and portfolios that lie on the efficient frontier

Portfolio optimization involves finding the optimal weights of assets to minimize risk for a given target return, or maximize expected return for a given risk level.

8.2.1 The Feasible Set and Efficient Frontier

The **feasible set** (or investment opportunity set) represents all possible portfolios that can be constructed from a given group of assets. When there are multiple risky assets, the feasible set forms a cloud of portfolios in risk-return space. The boundary of this cloud on the upper-left represents the optimal trade-off between risk

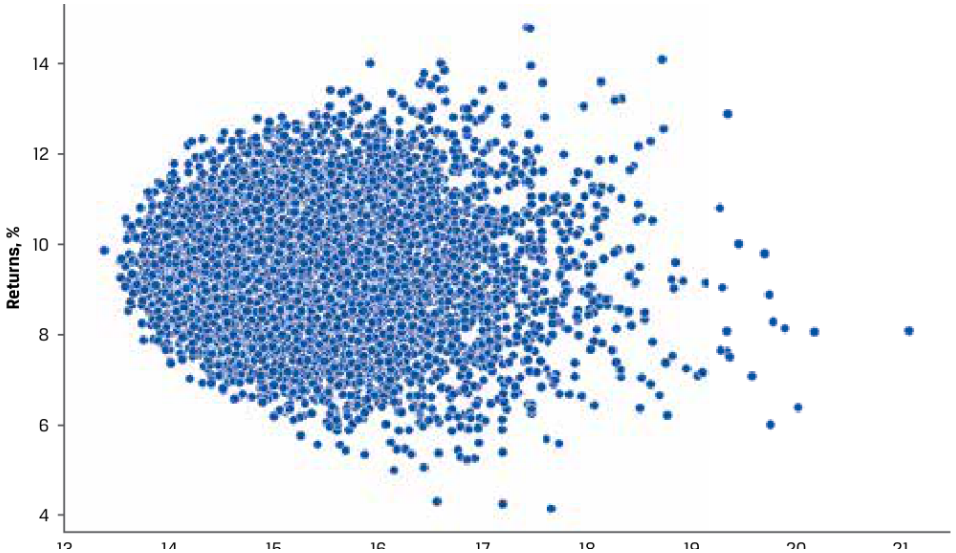


Figure 8.5: The Cloud of Feasible Portfolios in Risk-Return Space

and return:

- The **Minimum-Variance Frontier** is the boundary that minimizes variance for each level of expected return.

- The **Global Minimum-Variance Portfolio** is the single portfolio on the frontier that has the lowest possible risk.
- The **Markowitz Efficient Frontier** is the portion of the minimum-variance frontier that lies above the Global Minimum-Variance portfolio. Risk-averse, utility-maximizing investors will only choose portfolios on this segment.

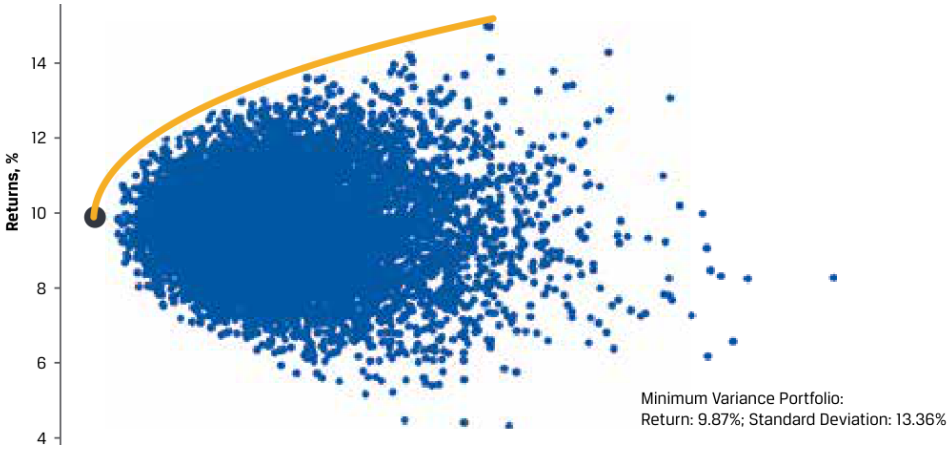


Figure 8.6: The Minimum-Variance Frontier and the Efficient Frontier

Mathematically, the optimization problem is formulated as:

$$\text{Minimize } \sigma_P^2 = \sum_{i=1}^{N}\sum_{j=1}^{N} w_i w_j \sigma_{i,j} \quad \text{subject to } \sum_{i=1}^{N} w_i = 1, \quad w_i \geq 0, \quad E[R_P] = k \tag{8.18}$$

where k is the target expected return. Under matrix form:

$$\text{Minimize } \sigma_P^2 = \mathbf{w}^T \mathbf{V} \mathbf{w} \quad \text{subject to } \mathbf{w}^T \mathbf{1} = 1, \quad \mathbf{w}^T \boldsymbol{\mu} = k, \quad \mathbf{w} \geq \mathbf{0} \tag{8.19}$$

The constraint $w_i \geq 0$ represents the **no-short-selling** constraint.

8.2.2 Derivation of the Minimum-Variance Portfolio Weights

For a two-asset portfolio, we can find the weights that minimize variance without complex optimization. Let w_A be the weight of Asset A and $w_B = 1 - w_A$ be the weight of Asset B.

$$\sigma_P^2 = w_A^2 \sigma_A^2 + (1 - w_A)^2 \sigma_B^2 + 2 w_A (1 - w_A) \sigma_{A,B}$$

Taking the first derivative of σ_P^2 with respect to w_A and setting it equal to zero:

$$\frac{d\sigma_P^2}{dw_A} = 2 w_A \sigma_A^2 - 2(1 - w_A)\sigma_B^2 + 2(1 - 2w_A)\sigma_{A,B} = 0$$

$$w_A(\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}) - \sigma_B^2 + \sigma_{A,B} = 0$$

Solving for the optimal weight w_A^* gives the global minimum-variance weight:

$$w_A^* = \frac{\sigma_B^2 - \sigma_{A,B}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}} \tag{8.20}$$

Correspondingly, the weight for Asset B is:

$$w_B^* = 1 - w_A^* = \frac{\sigma_A^2 - \sigma_{A,B}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}} \tag{8.21}$$

Example 10: Calculating Minimum-Variance Weights

We revisit Example 2 where Asset A (S&P 500) has standard deviation $\sigma_A = 16.21\%$ ($\sigma_A^2 = 0.0263$), Asset B (MSCI EM) has standard deviation $\sigma_B = 33.11\%$ ($\sigma_B^2 = 0.1096$), and the covariance is $\sigma_{A,B} = 0.0107$ (corresponding to a 20% correlation). Calculate the weights, return, and standard deviation of the global minimum-variance portfolio. **Solution:**

1. **Optimal Weights using Equation 8.20:**

$$w_A^* = \frac{0.1096 - 0.0107}{0.0263 + 0.1096 - 2(0.0107)} = \frac{0.0989}{0.1145} \approx 86.4\%$$

Weight of Asset B:

$$w_B^* = 1 - 0.864 = 13.6\%$$

2. **Expected Portfolio Return:** $E[R_P] = 0.864(9.93\%) + 0.136(18.20\%) = 11.05\%$.

3. **Portfolio Standard Deviation (Risk):**

$$\sigma_P = \sqrt{(0.864^2 \times 0.0263) + (0.136^2 \times 0.1096) + 2(0.864)(0.136)(0.0107)}$$

$$\sigma_P = \sqrt{0.01963 + 0.00203 + 0.00251} = \sqrt{0.02417} \approx 15.55\%$$

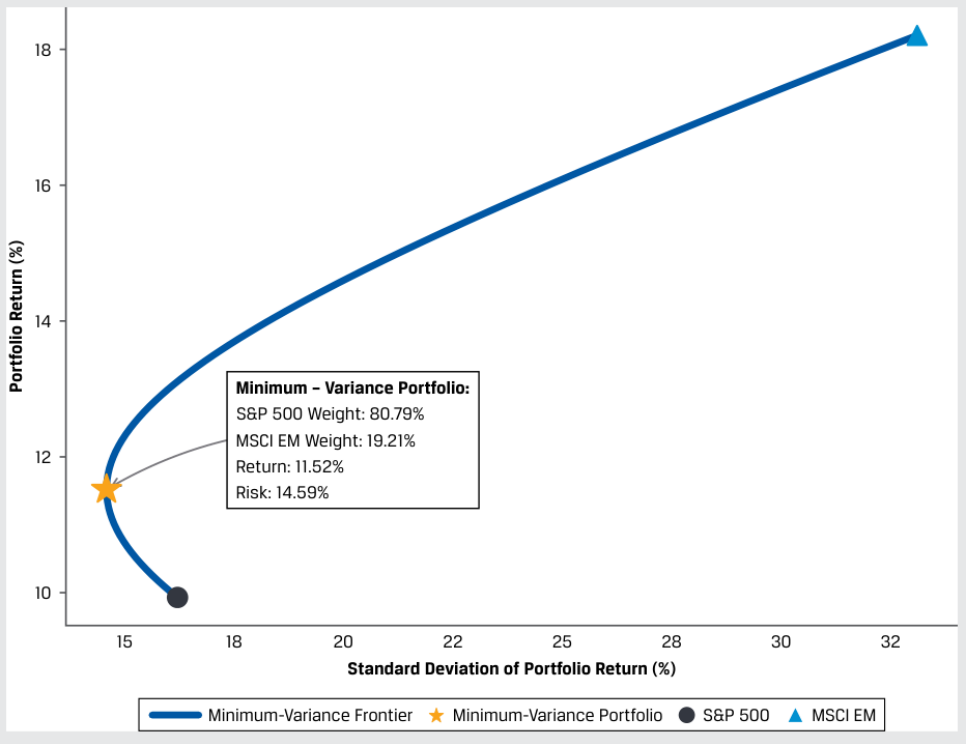


Figure 8.7: The Minimum-Variance Frontier and the S&P 500 / MSCI EM Indices

Example 11: Selected Frontier Portfolios

We examine selected portfolios on the efficient frontier established between the S&P 500 and MSCI EM under the covariance assumption of $\sigma_{A,B} = 0.0050$ (as shown in Exhibit 8.8).

- Portfolio P6 (Global MVP): 71.8% S&P 500, 28.2% MSCI EM. Expected return: 12.26%, Risk: 14.96%.
- Portfolio P4: 68.5% S&P 500, 31.5% MSCI EM. Expected return: 12.53%, Risk: 15.27%.
- Portfolio P5: 49.8% S&P 500, 50.2% MSCI EM. Expected return: 14.08%, Risk: 18.52%.
- Portfolio P1: 16.3% S&P 500, 83.7% MSCI EM. Expected return: 16.85%, Risk: 27.85%.

Portfolios lying below P6 (offering lower expected returns for the same level of risk) are inefficient portfolios.

MVP phân chia đường biên hiệu quả làm hai phần. Nhà đầu tư duy lý chỉ chọn danh mục ở nửa trên MVP (đường biên hiệu quả) vì mang lại tỷ suất sinh lời cao nhất cho cùng mức rủi ro.

Example 12: Capital Allocation Weights

An analyst creates a target portfolio return of 9.0% by combining the MSCI World Index (expected return 10.4%, standard deviation 14.80%) and a risk-free asset returning 1.0% ($\sigma_f = 0.0\%$). Calculate the weights and portfolio standard deviation. **Solution:**

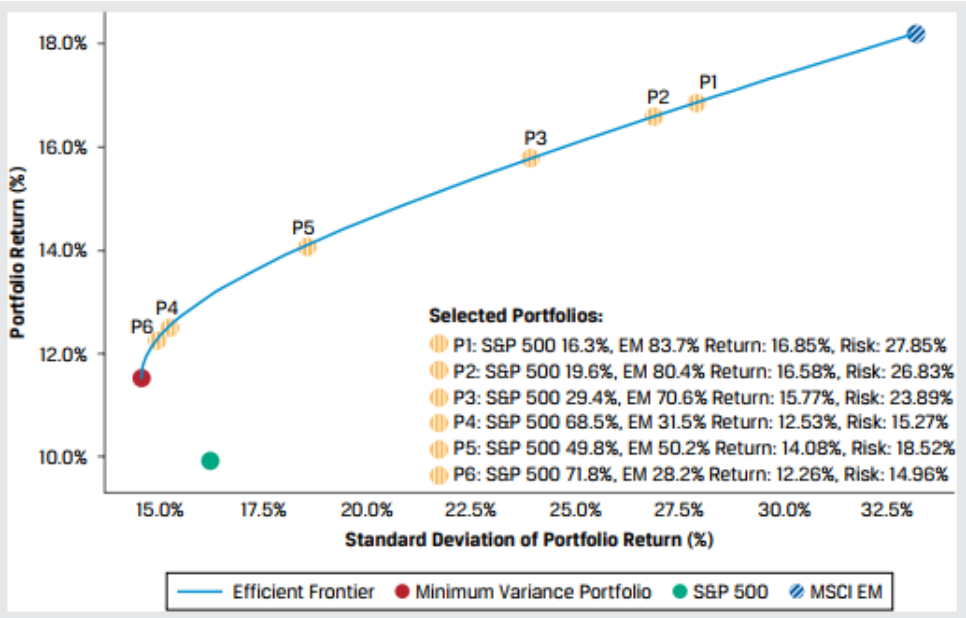


Figure 8.8: The Efficient Frontier and Selected Opportunity Portfolios

1. **Weights:** Let w_M be the weight allocated to the MSCI World Index:

$$R_P = w_M R_M + (1 - w_M) R_f \implies 9\% = 10.4\% w_M + 1\%(1 - w_M)$$

$$8\% = 9.4\% w_M \implies w_M = \frac{8\%}{9.4\%} \approx 85.1\%$$

   Weight allocated to the risk-free asset: $w_f = 1 - 0.851 = 14.9\%$.

2. **Portfolio Standard Deviation:** Since the risk-free asset has zero variance and covariance:

$$\sigma_P = w_M \sigma_M = 0.851 \times 14.80\% \approx 12.59\%$$

8.2.3 Portfolio Optimization

1. **Return-constrained optimization** by minimizing portfolio risk, σ_p, for a given level of return.
2. **Risk-constrained optimization** by finding the highest expected return $E[r_p]$, for a given level of risk.
3. **Risk-adjusted return optimization** by maximizing the Sharpe ratio, $\frac{E[r_p]-r_f}{\sigma_p}$, where r_f represents the risk-free rate.

The portfolio optimization process incorporates several key inputs:

1. Expected returns, $E[r_i]$, for each individual asset.
2. Volatility, σ_i, for each individual asset.
3. Asset correlations, $\rho_{i,j}$, or the off-diagonal covariances from the variance-covariance matrix $\mathbf{V}$.

The optimization process must also account for investor-specific factors, such as **risk tolerance**, the capacity and willingness to absorb losses, investment horizon, cash return requirements, and other investment objectives.

Example 13: Portfolio Optimization with Several Assets

An analyst aims to optimize a traditional 60/40 stock/bond portfolio ($R_{60/40} = 10\%$, $\sigma_{60/40} = 13\%$) using five assets: Global Developed Equities (12%, 20%), Emerging Market Equities (16%, 33%), Long-Term Bonds (5%, 8%), International Bonds (7%, 15%), and Short-Term Bills (3%, 1%). **Solution:**

1. **Return-Constrained Optimization:** Minimize portfolio risk for a target expected return of 11%.
2. **Risk-Constrained Optimization:** Maximize expected return for a target portfolio risk of 12%.

The resulting optimal asset weights are detailed in Table 8.5.

Table 8.5: Optimal Portfolio Weights vs. Traditional 60/40 Portfolio

Asset / Metric	Return-Constrained	Risk-Constrained
Global Developed Equities	53.45%	48.64%
Emerging Market Equities	14.75%	5.33%
Long-Term Government Bonds	0.00%	6.61%
International Government Bonds	31.80%	39.42%
Short-Term Government Bills	0.00%	0.00%
Expected Portfolio Return	**11.00%**	**9.78%**
Expected Portfolio Risk	**14.81%**	**12.00%**

Both portfolios lie on the multi-asset efficient frontier (Exhibit 8.9), outperforming the traditional 60/40 allocation.

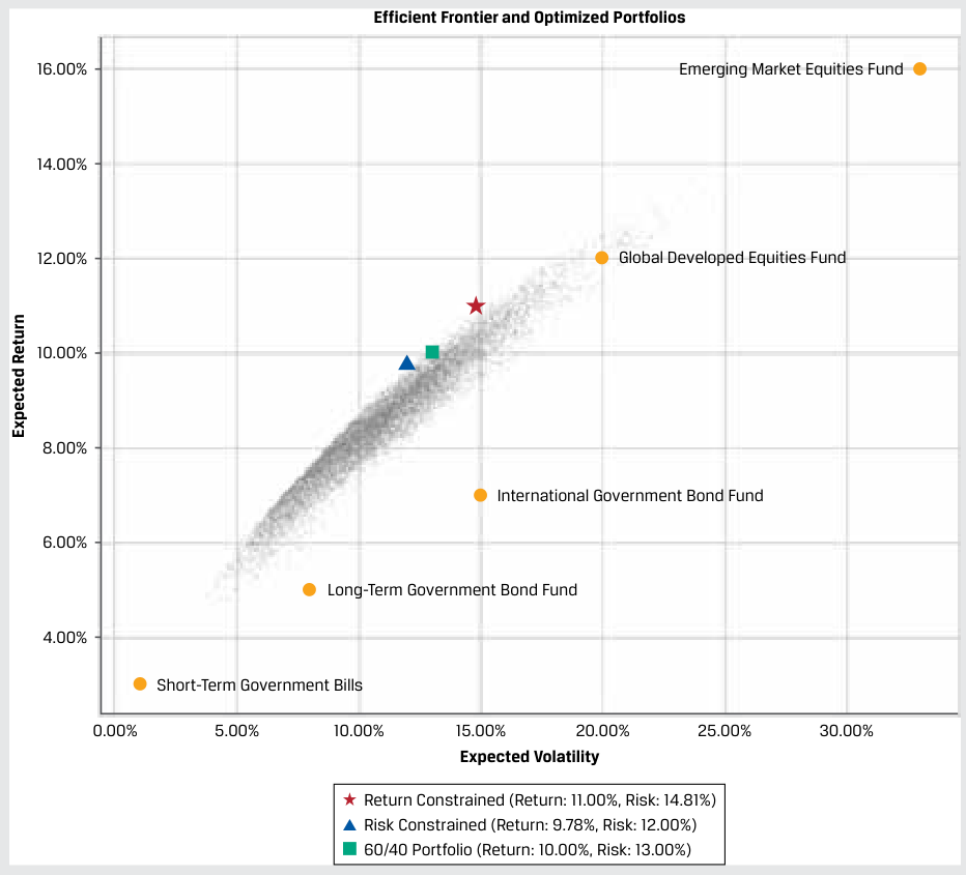


Figure 8.9: Efficient Frontier and Optimized Constrained Portfolios

8.3 Optimal Portfolio Selection

c | explain the selection of an optimal portfolio, given an investor's risk aversion and the capital allocation line, and how this extends to the market portfolio and the capital market line

Optimal portfolio selection balances expected returns with risk, matching the opportunity set against an investor's personal preferences.

Cập nhật chương trình 2027

Cấu phần "Optimal Portfolio Selection" bao gồm các khái niệm về Đường phân bổ vốn (CAL), Đường thị trường vốn (CML), Mô hình định giá tài sản vốn (CAPM) và Beta đã được chuyển từ phần Portfolio Management sang tích hợp trực tiếp vào Quantitative Methods năm 2027. Đây là phần lý thuyết cốt lõi và rất quan trọng cho kỳ thi.

8.3.1 Investor Utility and Indifference Curves

The utility function represents an investor's satisfaction or ranking of portfolios based on risk and expected return. For a risk-averse investor, utility is formulated as:

$$U = E[R] - \frac{1}{2}A\sigma^2 \tag{8.22}$$

where $E[R]$ is expected return, σ^2 is variance of returns, and A is the **coefficient of risk aversion**.

- $A > 0$: Risk-averse investor. Utility decreases with higher risk.
- $A = 0$: Risk-neutral investor. Utility depends solely on expected return.
- $A < 0$: Risk-seeking investor. Utility increases with risk.

An **indifference curve** maps the combinations of risk and expected return that yield the same level of utility. Steeper indifference curves represent higher risk aversion (A), meaning the investor requires a much higher

Table 8.6: Risk Preferences and Indifference Curves

Preference	Aversion (A)	Indifference Slope	Utility Curve Shape
Risk-seeking	$A < 0$	Downward/Concave	Convex utility in wealth
Risk-neutral	$A = 0$	Flat/Horizontal	Linear utility in expected return
Risk-averse	$A > 0$	Upward/Convex	Concave utility in wealth

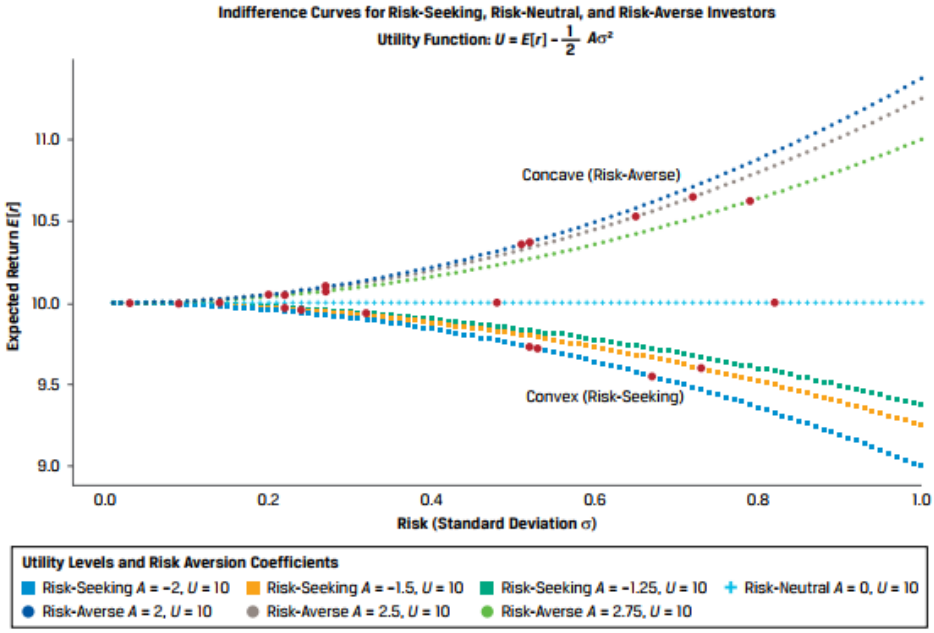


Figure 8.10: Indifference Curves for Risk-Averse, Risk-Neutral, and Risk-Seeking Investors

expected return to accept a small increase in risk.

Knowledge Check: Utility calculations

An investor evaluates three investments:

Investment	Expected Return $E[R]$ (%)	Standard Deviation σ (%)
A	12	30
B	15	35
C	21	40

Determine the optimal choice for:

1. An investor with $A = 4$.

2. An investor with $A = 2$.
3. A risk-seeking investor with $A = -4$.

Solution: Using $U = E[R] - \frac{1}{2}A\sigma^2$:

1. **Case 1 ($A = 4$):**
   - $U_A = 0.12 - 0.5(4)(0.09) = -0.060$
   - $U_B = 0.15 - 0.5(4)(0.1225) = -0.095$
   - $U_C = 0.21 - 0.5(4)(0.16) = -0.110$

   The investor chooses **Investment A** (highest utility / lowest negative utility).
2. **Case 2 ($A = 2$):**
   - $U_A = 0.12 - 0.5(2)(0.09) = 0.0300$
   - $U_B = 0.15 - 0.5(2)(0.1225) = 0.0275$
   - $U_C = 0.21 - 0.5(2)(0.16) = 0.0500$

   The investor chooses **Investment C**. Despite higher risk, the expected return compensation is sufficient.
3. **Case 3 ($A = -4$):**
   - $U_A = 0.12 - 0.5(-4)(0.09) = 0.300$
   - $U_B = 0.15 - 0.5(-4)(0.1225) = 0.395$
   - $U_C = 0.21 - 0.5(-4)(0.16) = 0.530$

   The investor chooses **Investment C**. Utility increases with both return and risk.

8.3.2 The Capital Allocation Line (CAL) and Sharpe Ratio

When a risk-free asset (with return R_f and standard deviation $\sigma_f = 0$) is introduced, investors can combine it with a risky portfolio P. The expected return of the combined portfolio is:

$$E[R_c] = w_f R_f + (1 - w_f)E[R_P] \tag{8.23}$$

The standard deviation is:

$$\sigma_c = (1 - w_f)\sigma_P \implies (1 - w_f) = \frac{\sigma_c}{\sigma_P} \tag{8.24}$$

Substituting $(1 - w_f)$ into the expected return equation yields the **Capital Allocation Line (CAL)**:

$$E[R_c] = R_f + \left(\frac{E[R_P] - R_f}{\sigma_P}\right)\sigma_c \tag{8.25}$$

The slope of the CAL is the **Sharpe ratio (SR)** of the risky portfolio:

$$\text{Sharpe Ratio} = \frac{E[R_P] - R_f}{\sigma_P} \tag{8.26}$$

The optimal risky portfolio (tangency portfolio) is the point on the efficient frontier that maximizes the Sharpe ratio, creating the steepest possible CAL. The optimal portfolio for a specific investor lies at the point of tangency between their highest attainable indifference curve and the CAL:

- More risk-averse investors select portfolios closer to the risk-free asset (left on the CAL).
- Less risk-averse investors select portfolios closer to or beyond the tangency portfolio P, utilizing leverage (borrowing at the risk-free rate) to buy more of the risky portfolio.

Example 14: Calculating the Sharpe Ratio

Assuming a risk-free rate of 1.0%, compare:

- Portfolio A: expected return 9.0%, weight in MSCI World 85.1%, standard deviation 12.59%.
- Portfolio B: expected return 12.0%, weight in MSCI World 117.1%, standard deviation 17.32%.

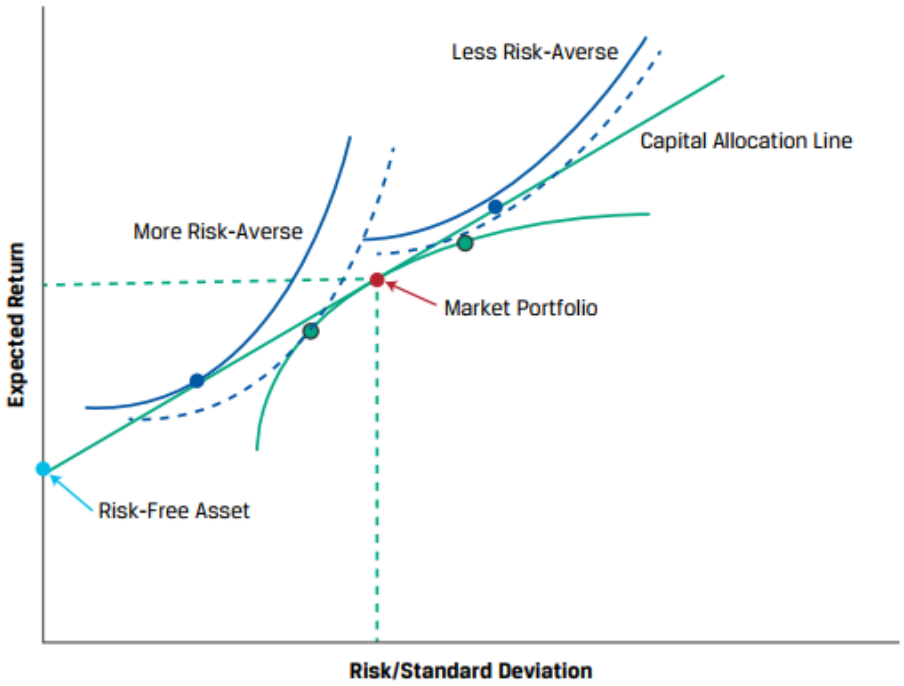


Figure 8.11: Optimal Portfolio Selection via Tangency between Indifference Curves and CAL

Solution:

- $\text{SR}_A = \frac{9\%-1\%}{12.59\%} \approx 0.635$
- $\text{SR}_B = \frac{12\%-1\%}{17.32\%} \approx 0.635$
- Both portfolios lie on the same Capital Allocation Line (CAL) and have the same Sharpe ratio. Portfolio B is a leveraged portfolio, borrowing 17.1% at the risk-free rate to increase investment in the risky asset portfolio.

8.3.3 The Capital Market Line (CML)

If all investors share identical assumptions and hold mean-variance efficient portfolios, the optimal risky portfolio is the **Market Portfolio** (M). The Market Portfolio theoretically contains all risky assets in the universe, weighted by their market values. The Capital Allocation Line using the Market Portfolio as the risky portfolio is called the **Capital Market Line (CML)**:

$$E[R_P] = R_f + \left(\frac{E[R_M] - R_f}{\sigma_M}\right)\sigma_P \tag{8.27}$$

where $E[R_M]$ and σ_M are the expected return and standard deviation of the market portfolio. According to the **mutual fund theorem**, all rational investors can achieve optimal asset allocation by holding only two assets: the risk-free asset and the market portfolio.

8.3.4 The Capital Asset Pricing Model (CAPM) and Beta

Total risk consists of two parts:

- **Unsystematic Risk** (diversifiable / idiosyncratic risk): Unique to a specific company or sector. Can be eliminated in a well-diversified portfolio.
- **Systematic Risk** (undiversifiable / market risk): Driven by macroeconomic factors that affect all assets. Cannot be diversified away.

Since investors can eliminate unsystematic risk at no cost through diversification, they are not compensated for bearing it. The expected return of an asset depends solely on its systematic risk. Systematic risk is measured by **Beta** (β), which quantifies the sensitivity of an asset's returns to movements in the market portfolio:

$$\beta_i = \frac{\sigma_{i,M}}{\sigma_M^2} = \frac{\rho_{i,M}\sigma_i\sigma_M}{\sigma_M^2} = \rho_{i,M}\frac{\sigma_i}{\sigma_M} \tag{8.28}$$

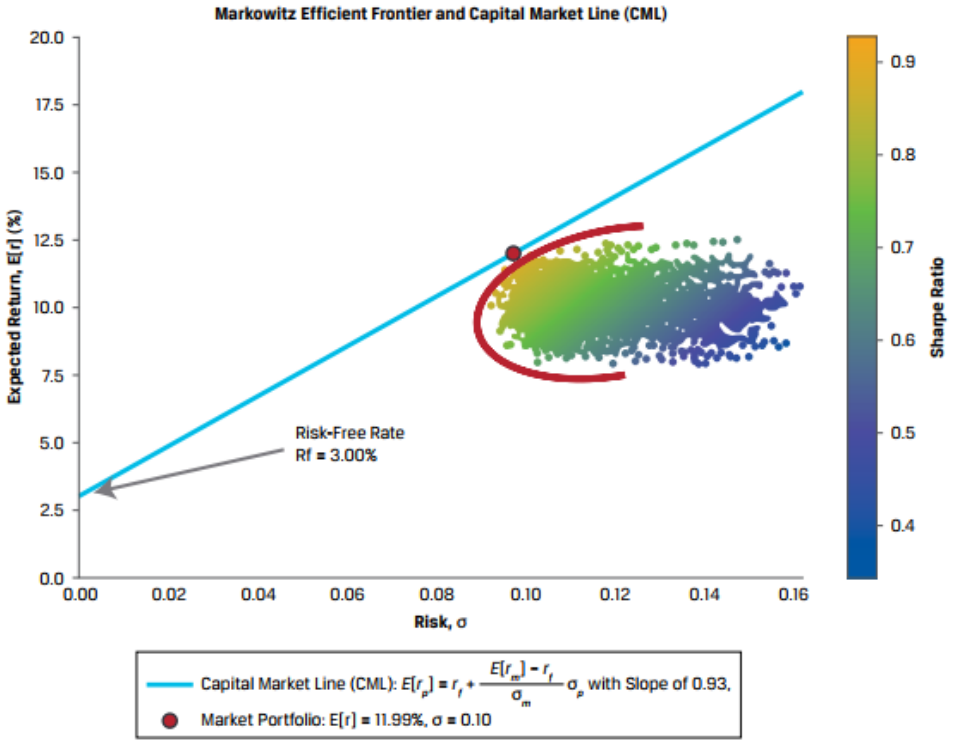


Figure 8.12: The Capital Market Line (CML) and the Markowitz Efficient Frontier

where $\sigma_{i,M}$ is the covariance between asset i and the market, and σ_M^2 is the variance of the market portfolio.

- $\beta = 1.0$: Asset has the same systematic risk as the market.
- $\beta > 1.0$: Asset is more volatile than the market (aggressive asset).
- $\beta < 1.0$: Asset is less volatile than the market (defensive asset).

The **Capital Asset Pricing Model (CAPM)** states that the expected return of asset i is:

$$E[R_i] = R_f + \beta_i \left(E[R_M] - R_f\right) \tag{8.29}$$

where $(E[R_M] - R_f)$ is the **market risk premium (MRP)**. The graphical representation of CAPM is the **Security Market Line (SML)**, which plots expected return against Beta. In contrast to the CML (which uses total risk / standard deviation and applies only to efficient portfolios), the SML uses systematic risk / Beta and applies to **all** assets, whether efficient or individual.

$$E[R_i] = R_f + \left(\frac{E[R_M] - R_f}{\sigma_M^2}\right) \sigma_{i,M} \tag{8.30}$$

So sánh: CML dùng tổng rủi ro (σ) áp dụng cho danh mục hiệu quả. SML dùng rủi ro hệ thống (β) áp dụng cho mọi tài sản đơn lẻ lẫn danh mục. Cổ phiếu đơn lẻ không nằm trên CML nhưng vẫn nằm trên SML.

Knowledge Check: The Capital Asset Pricing Model

Consider: $R_f = 2.0\%$, $E[R_M] = 8.0\%$, $\sigma_M = 15.0\%$, and an investor's optimal portfolio has standard deviation $\sigma_P = 10.0\%$.

1. Calculate the expected return of the portfolio using CML.
2. Calculate the Beta of the portfolio.
3. Calculate the expected return using CAPM and confirm it matches the CML.
4. Calculate the Sharpe ratio of the market portfolio.

Solution:

1. **CML expected return:**

$$E[R_P] = R_f + \left(\frac{E[R_M] - R_f}{\sigma_M}\right) \sigma_P = 2\% + \left(\frac{8\% - 2\%}{15\%}\right) (10\%) = 2\% + 0.4(10\%) = 6\%$$

2. **Beta:** For an efficient portfolio on the CML, the correlation with the market is $\rho = 1.0$:

$$\beta_P = \frac{\sigma_P}{\sigma_M} = \frac{10\%}{15\%} \approx 0.6667$$

3. **CAPM return:**

$$E[R_P] = R_f + \beta_P(E[R_M] - R_f) = 2\% + 0.6667(8\% - 2\%) = 2\% + 4\% = 6\%$$

The results match exactly, confirming the consistency between CML and CAPM for efficient portfolios.

4. **Sharpe ratio:** $\text{SR}_M = \frac{8\%-2\%}{15\%} = 0.40$.

Estimating Beta using Linear Regression

In practice, Beta is estimated using ordinary least squares (OLS) linear regression of the asset's excess returns against the market's excess returns. This is known as the **Market Model**:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i \left(R_{M,t} - R_{f,t}\right) + \epsilon_{i,t} \tag{8.31}$$

where α_i is the intercept, β_i is the slope, and $\epsilon_{i,t}$ is the error term. Under CAPM, the expected alpha (α_i) should be exactly 0 in equilibrium.

Example 15: Estimating LVMH's Beta via Bloomberg

An analyst regresses monthly returns of LVMH Moët Hennessy Louis Vuitton against the CAC 40 index over 3 years (36 data points). The Bloomberg regression output shows:

- Regression line: $y = 0.939x + 1.318$ ($\beta = 0.939$, $\alpha = 1.318\%$)
- Correlation coefficient $R = 0.850$ ($R^2 = 0.722$)
- Standard error of Beta: 0.101

The expected market return is 9.577% and the risk-free rate is 2.560%. Calculate LVMH's expected return.
Solution:

1. **Beta Interpretation:** A beta of 0.939 indicates that LVMH has slightly lower systematic risk than the French equity market average (CAC 40 Index).

2. **Expected Return (CAPM):**

$$E[R_{\text{LVMH}}] = R_f + \beta(E[R_M] - R_f) = 2.56\% + 0.939(9.577\% - 2.56\%)$$

$$E[R_{\text{LVMH}}] = 2.56\% + 0.939(7.017\%) = 2.56\% + 6.589\% \approx 9.15\%$$

8.3.5 Limitations of Portfolio Theory and CAPM

Several assumptions underlying Markowitz optimization and CAPM do not hold in real-world markets:

- **Distributional Assumptions:** Financial returns are rarely normally distributed, exhibiting fat tails (high kurtosis) and skewness.
- **Time-Varying Volatility and Correlation:** Variances and correlations are not constant but fluctuate, especially rising during systemic market crises.
- **Market Inefficiencies:** Information asymmetries exist, leading to mispricings and non-zero alphas.
- **Estimation Errors:** Small errors in the expected return and covariance inputs can result in dramatically different and highly concentrated optimal weights (input sensitivity).
- **Historical limits:** Historical relationships may not persist in the future.
- **Transaction Costs and Taxes:** Real-world investors must pay brokerage fees, spreads, and taxes. Capital distributions (dividends) and capital gains face differing tax structures, which reduces the after-tax efficient frontier.

Module 9

Simulation of Financial Asset Prices and Returns

LEARNING OUTCOMES	
a	describe historical simulation and explain how it can be used in investment applications
b	describe bootstrap resampling, and explain how it can be used in investment applications
c	describe Monte Carlo simulation and explain how it can be used in investment applications

9.1 Historical Simulation

a | describe historical simulation and explain how it can be used in investment applications

Mô phỏng lịch sử là phương pháp phi tham số đơn giản: áp dụng trực tiếp biến động quá khứ lên danh mục hiện tại. Ưu điểm: bảo toàn các hiện tượng thực tế như đuôi béo (fat tails) hay tương quan mà không cần giả định phân phối phức tạp.

9.1.1 Overview of Simulation Methods

Simulation predicts potential outcomes for a system based on theoretical scenarios by defining objectives, selecting variables, assigning values, evaluating scenarios, and interpreting results. Figure 9.1 outlines the key steps.



Figure 9.1: Steps in the Simulation Process

The simulation process comprises:

- **Step 1: Setup/parameterize.** Define the target output variable (e.g., risk-adjusted return) and identify the necessary inputs and mathematical relationships.
- **Step 2: Scenario generation.** Generate input values (scenarios) representing different market conditions.
- **Step 3: Scenario evaluation/calculation engine.** Process scenarios through a mathematical model to calculate outputs.
- **Step 4: Result compilation/analysis.** Aggregate outputs into a probability distribution to assess risks.

Three common simulation approaches differ primarily in how they generate scenarios (Step 2):

- **Historical simulation** applies actual past market movements directly to current portfolio factors.

- **Bootstrap simulation** resamples observed historical data with replacement to generate synthetic datasets.
- **Monte Carlo simulation** generates random scenarios based on assumed probability distributions.

Table 9.1 compares these approaches.

Table 9.1: Comparing the Steps for Different Simulation Approaches

Simulation Approach	Step 1: Setup/ Parameterize	Step 2: Scenario Generation	Step 3: Scenario Evaluation	Step 4: Result Analysis
Historical Simulation	Use historical data to initialize model parameters.	Incorporate actual past observations for scenario inputs.	Run the model using each historical scenario.	Examine performance under real-world conditions encountered in the past.
Bootstrap Simulation	Adopt historical data to define parameters (with replacement).	Formulate synthetic scenarios by resampling the observed dataset.	Conduct calculations for each resampled scenario.	Assess variability while preserving the original data structure.
Monte Carlo Simulation	Specify probability distributions for model inputs.	Draw random samples from these distributions to shape scenario inputs.	Execute calculations using each random input vector.	Evaluate the resulting distribution of outputs to determine risks and probabilities.

The three simulation approaches differ primarily in Step 2 (scenario generation); the other steps remain functionally identical.

9.1.2 Historical Simulation Overview

Historical simulation uses historical prices and returns to model future outcomes under the assumption that past patterns approximate future risks. This non-parametric approach captures empirical features like fat tails, kurtosis, and volatility clustering without distributional assumptions. However, it relies on the assumption that past patterns represent the future, which fails during structural shifts.

9.1.3 Implementing Historical Simulation

In historical simulation, scenarios are assembled from observed historical changes in each variable, as shown in Figure 9.2.

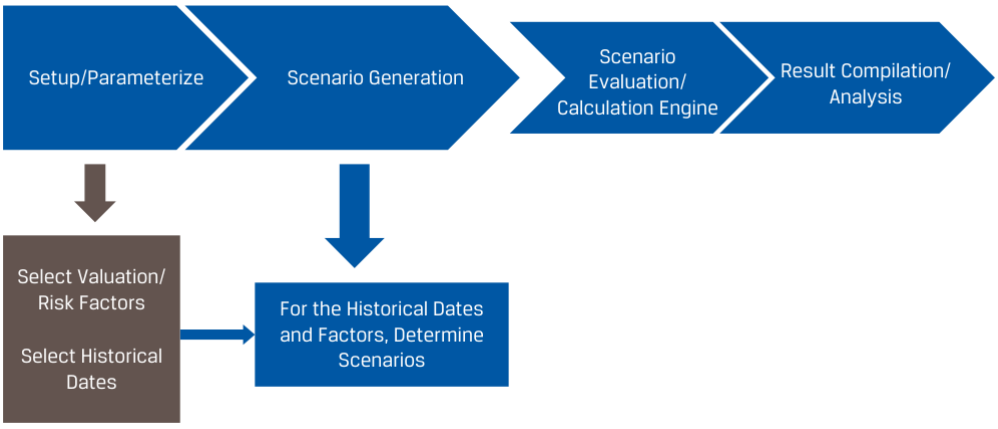


Figure 9.2: Simulation Framework Applied to Historical Simulation

To construct a robust simulation, key valuation and risk factors need to be identified, which serve as the fundamental inputs in the simulation process. For equities, the primary variables are usually continuously compounded (logarithmic) price returns:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{9.1}$$

For fixed-income instruments, the simulation models changes in the yield curve. The price of a bond is determined by discounting future cash flows using simulated yields:

$$PV_B = \sum_{t=1}^{T} \frac{CF_t}{(1+y_t)^t} \tag{9.2}$$

Historical simulation is commonly used to calculate portfolio Value at Risk (VaR). The case study below applies this method to a long-only portfolio.

Case Study: Historical Simulation of a Long-Only Portfolio

A portfolio manager wants to determine the daily Value at Risk (VaR) of a portfolio at the 95

- HSI (Hang Seng Index): HKD 55,000,000
- CSI 300 Index: HKD 40,000,000

The manager uses a one-year history of daily log returns (246 trading days).
Step 1: Calculate log returns.
Daily log returns for both indexes are calculated as: $r = \ln(P_1/P_0)$. For example, on Day 1:

$$r_{HS,1} = \ln\left(\frac{20,145.29}{19,781.41}\right) = 1.82\%$$

Returns are shown in Figure 9.3 and Table 9.2.

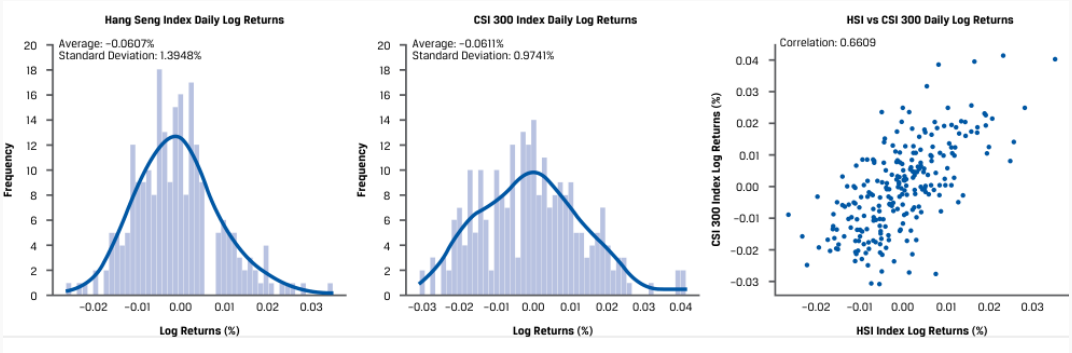


Figure 9.3: Comparison of Hang Seng Index and Shanghai Shenzhen CSI 300 Index Daily Log Returns

Table 9.2: Hang Seng Index and CSI 300 Daily Log Returns

Scenario No.	Hang Seng Index in HKD	CSI 300 in HKD	Hang Seng Index Log Return	CSI 300 Log Return
0, $t=0$	19,781.41	4,381.91	–	–
1	20,145.29	4,391.77	1.82%	0.22%
2	20,793.11	4,416.16	3.17%	0.55%
3	21,052.17	4,504.74	1.24%	1.99%
4	20,991.64	4,543.39	-0.29%	0.85%
...	...	...	...	...
244	16,624.84	3,648.65	1.73%	-0.14%
245	17,043.53	3,752.58	2.49%	2.81%
246	17,047.39	3,772.51	0.02%	0.53%

Step 2: Calculate daily profit and loss (P/L).
For each day, the P/L is computed as: $P/L = \text{Position value} \times (e^r - 1)$. For Day 1:

$$\text{Total P/L} = 55\text{M} \times (e^{1.82\%} - 1) + 40\text{M} \times (e^{0.22\%} - 1) = \text{HKD } 1,098,262$$

The P/L across all scenarios are compiled in Table 9.3.

Table 9.3: Portfolio P/L for the Long-Only Portfolio

Scenario	HSI Log Return	CSI 300 Log Return	HSI P/L (HKD)	CSI 300 P/L (HKD)	Total Portfolio P/L (HKD)
1	1.82%	0.22%	1,010,165	88,097	1,098,262
2	3.17%	0.55%	1,771,429	220,606	1,992,035
3	1.24%	1.99%	686,246	803,973	1,490,219
4	-0.29%	0.85%	-159,269	341,449	182,180
...	...	...	...	...	...
244	1.73%	-0.14%	959,778	-55,961	903,817
245	2.49%	2.81%	1,386,693	1,139,941	2,526,634
246	0.02%	0.53%	11,001	212,563	223,564

Step 3: Create the P/L distribution.
Daily P/L results generate the empirical distribution shown in Figure 9.4.

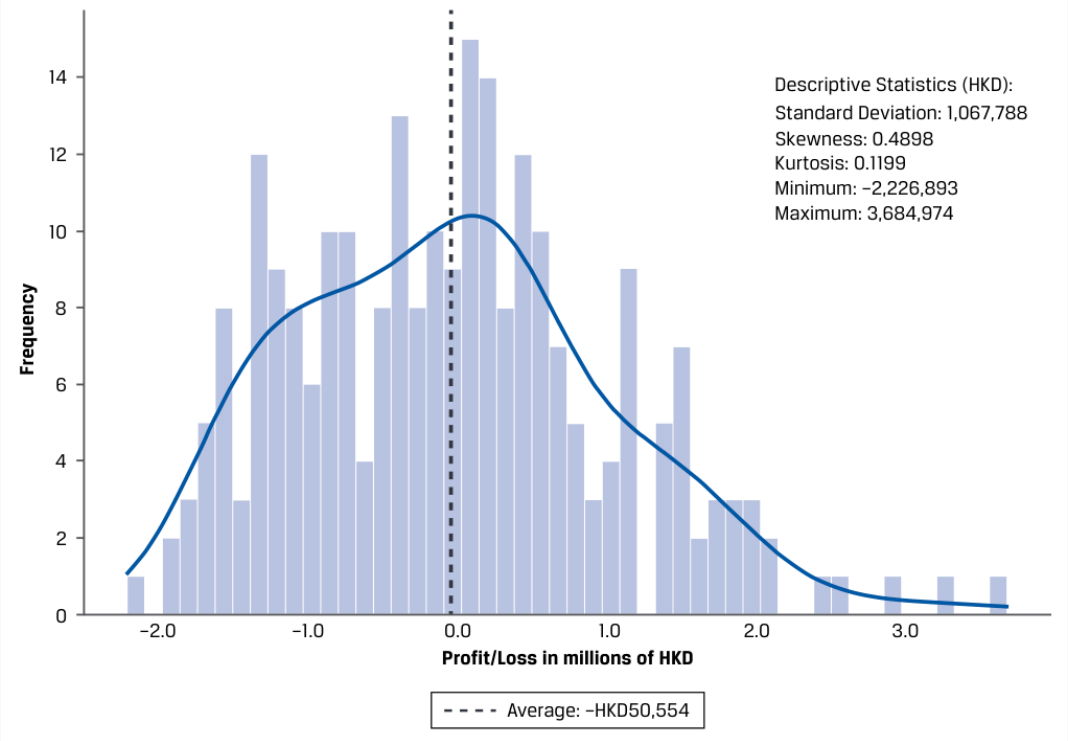


Figure 9.4: Distribution of Total Portfolio Daily Profit and Loss

Step 4: Determine the VaR.
The daily P/L estimates are ranked from lowest to highest (Table 9.4).

Table 9.4: Ranking of Total Portfolio Daily Profit and Loss

Rank	Day	Total P/L (HKD)
1	262	-2,226,893
2	157	-1,949,811
3	39	-1,888,929
...	...	...
11	49	-1,651,318
12	43	-1,624,801
13	25	-1,607,739
...	...	...
246	176	3,684,974

For 246 observations, the 5% tail corresponds to the 12.3rd rank from the bottom. Using linear interpolation between the 12th (HKD -1,624,801) and 13th (HKD -1,607,739) observations:

$$\text{VaR}_{95\%} = 0.3 \times 1,607,739 + 0.7 \times 1,624,801 = \text{HKD } 1,619,682$$

For comparison, a parametric VaR is calculated assuming a joint normal distribution with parameters in Table 9.5.

Table 9.5: Descriptive Statistics

Statistic	Hang Seng Index	CSI 300 (in HKD)
Average Log Return, μ	-0.0607%	-0.0611%
Standard Deviation, σ	1.3948%	0.9741%
Correlation, ρ	0.6609	–

Assuming zero mean returns for daily calculations, the portfolio value variance (for $D_{HS} = 55$M and $D_{SS} = 40$M) is:

$$\sigma_p^2 = D_{HS}^2\sigma_{HS}^2 + D_{SS}^2\sigma_{SS}^2 + 2D_{HS}D_{SS}\rho\sigma_{HS}\sigma_{SS} = 1.1354\,(\text{million HKD})^2 \tag{9.3}$$

The standard deviation is $\sigma_p = \text{HKD } 1.0656$ million. The 95

$$\text{VaR}_{95\%} = 1.96 \times 1.0656 = \text{HKD } 2.0885 \text{ million}$$

The difference arises because parametric VaR relies on the normality assumption, whereas historical VaR directly reflects empirical tail behavior and any inherent drift.

Lưu ý: VaR tham số ($z = 1.96$) lớn hơn VaR lịch sử vì giả định phân phối chuẩn phân bổ đều xác suất về hai đuôi, trong khi dữ liệu thực tế ít biến động cực đại hơn. VaR tham số phụ thuộc mô hình, còn VaR lịch sử phụ thuộc mẫu.

Case Study: Modeling Investment Scenarios for a Sustainable Energy Startup

A venture capital firm is evaluating an investment with a five- to seven-year horizon. It analyzes 50 past investments to estimate risks by reviewing deviations between projected and realized revenues.

Step 1: Determine the historical distribution of returns.

Figure 9.5 and Table 9.6 show the distribution of revenue forecast deviations.

$$deviation = \frac{Actual - Forecast}{Forecast}$$

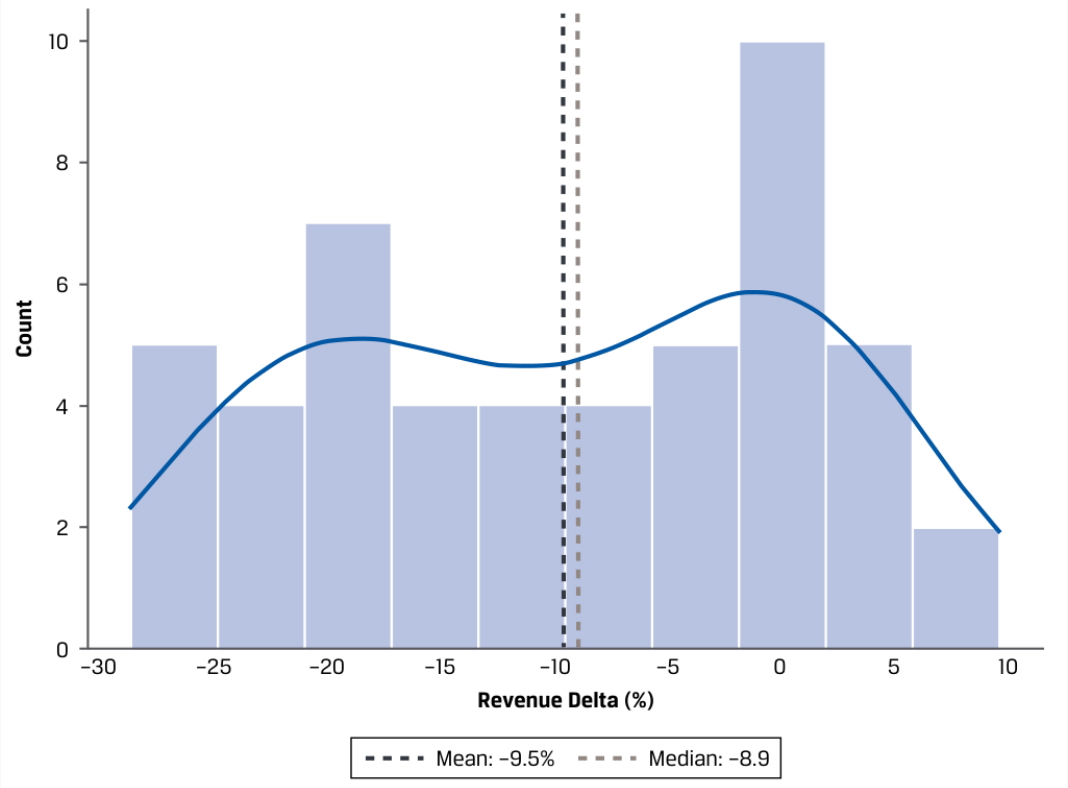


Figure 9.5: Deviations from Revenue Forecasts

Table 9.6: The Two Tail Ends of the Revenue Forecast Deviations

Investment	Return Deviation (%)
1	-28.56
2	-25.60
3	-25.56
...	...
48	5.49
49	7.30
50	9.69

The deviations range from -28.56% to 9.69%, suggesting optimistic forecasts and a heavily left-skewed, high-variance distribution.

Step 2: Establish the return deviation distribution.

To find the 2.5th and 97.5th percentiles of the return deviations, we interpolate linearly:

- **2.5th percentile** (between ranks 2 and 3):

$$x = -25.60 + \frac{2.5 - 2}{3 - 2} \times (-25.56 - (-25.60)) = -25.58\% \tag{9.4}$$

- **97.5th percentile** (between ranks 48 and 49):

$$x = 5.49 + \frac{97.5 - 96}{98 - 96} \times (7.30 - 5.49) = 6.89\%$$

Thus, the 2.5th percentile deviation is -25.58%, and the 97.5th percentile is 6.89%.

Step 3: Calculate the Net Present Value (NPV) range.

The baseline NPV is GBP 1,000,000. Every 1% deviation in revenue changes NPV by GBP 123,897.22 (β). We have $NPV = NPV_{base} + \delta NPV$, with:

$$\delta NPV = \beta \times \delta Rev(\%)$$

If Actual = Forecast, that means $\delta Rev = 0$, $NPV = GBP1,000,000$. Extreme scenario NPVs are calculated as:

- **Downside (2.5th percentile):**

$$\text{NPV} = 1,000,000 + (-25.58 \times 123,897.22) \approx -\text{GBP } 2,170,000$$

- **Upside (97.5th percentile):**

$$\text{NPV} = 1,000,000 + (6.89 \times 123,897.22) \approx \text{GBP } 1,854,000$$

The potential NPV ranges from $-\text{GBP } 2,170,000$ to GBP $1,854,000$.

Using empirical data to define realistic outcomes allows the firm to adjust expectations and mitigate overconfidence in forecast models.

9.1.4 Features of Historical Simulation

Historical simulation is a valuable tool for financial analysis, but its implementation requires attention to data quality and availability.

Missing data in global investment portfolios may be caused by:

- **Asynchronous global markets:** Different trading holidays and time zones.
- **Non-existence of securities:** A security listing after the simulation window begins.
- **Thinly traded securities:** Sparse or stale pricing due to low volume.

To compensate for missing data, analysts often use a **proxy**—a substitute financial instrument or index representing the missing security's risk factor.

For fixed-income securities, applying historical price changes directly can be misleading. Instead, simulations model changes in benchmark yields (e.g., zero-coupon curves at specific maturities) rather than the bond prices directly.

Strengths of historical simulation:

- **Simplicity:** Straightforward implementation without complex statistical models.
- **No distribution assumptions:** Preserves empirical distribution features like fat tails and kurtosis.
- **Implicit correlation:** Preserves natural historical correlations between assets.
- **Low computational cost:** Scenarios are sourced directly from past observations.

Disadvantages of historical simulation:

- **Backward looking:** Relies on the assumption that past risk distributions represent the future.
- **Limited scenario variety:** Cannot simulate extreme events that have not occurred in the historical sample.
- **Data dependency:** Results depend entirely on the quality and completeness of historical datasets.

9.2 Bootstrapping

b | describe bootstrap resampling, and explain how it can be used in investment applications

Bootstrapping (resampling with replacement) tạo ra các kịch bản tổng hợp bằng cách coi mẫu hiện tại là đại diện quần thể, khác với mô phỏng lịch sử coi dữ liệu quá khứ là cố định và duy nhất.

9.2.1 Bootstrapping Overview

Resampling repeatedly draws samples from observed data to make statistical inferences. It is particularly useful when population distributions are unknown or sample sizes are small.

Bootstrapping is a resampling method that generates a simulated distribution by resampling with replacement from the observed sample (treating the sample as a proxy for the population).

Resampling with replacement means each data point has a probability of $1/n$ (where n is sample size) of being selected in each draw, allowing the same value to be selected multiple times. For example, from sample $[2, 5, 8]$, bootstrap samples could be $[5, 5, 8]$ or $[2, 2, 5]$.

Bootstrapping differs from historical simulation:

- **Bootstrapping** resamples with replacement to create synthetic datasets, preserving the empirical distribution structure while generating new combinations.
- **Historical simulation** applies the actual chronological sequence of past market movements directly as fixed scenarios.

Table 9.7 compares these two non-parametric approaches.

Table 9.7: Comparison of Historical Simulation and Bootstrapping

Feature	**Historical Simulation**	**Bootstrapping**
Source of data	Historical dataset	Observed sample
Sampling method	Simple sampling of historical data	Resampling with replacement from the sample
Assumptions	Assumes past data reflects future	Assumes sample represents population
Primary use cases	Risk management, financial stress tests	Estimating confidence intervals, standard errors
Computational intensity	Low to moderate	Moderate

By using the observed sample as a proxy for the unknown population, bootstrapping eliminates the need for external data or theoretical models, making it extremely flexible and widely applicable in situations where we have little prior information about the population distribution.

9.2.2 Implementing Bootstrapping

Bootstrapping is non-parametric, flexible, and simple to implement. Its accuracy depends on how well the sample represents the true distribution; larger samples are always preferred.

The only functional change when using the bootstrap method is the source of scenarios (Step 2), as shown in Figure 9.6.

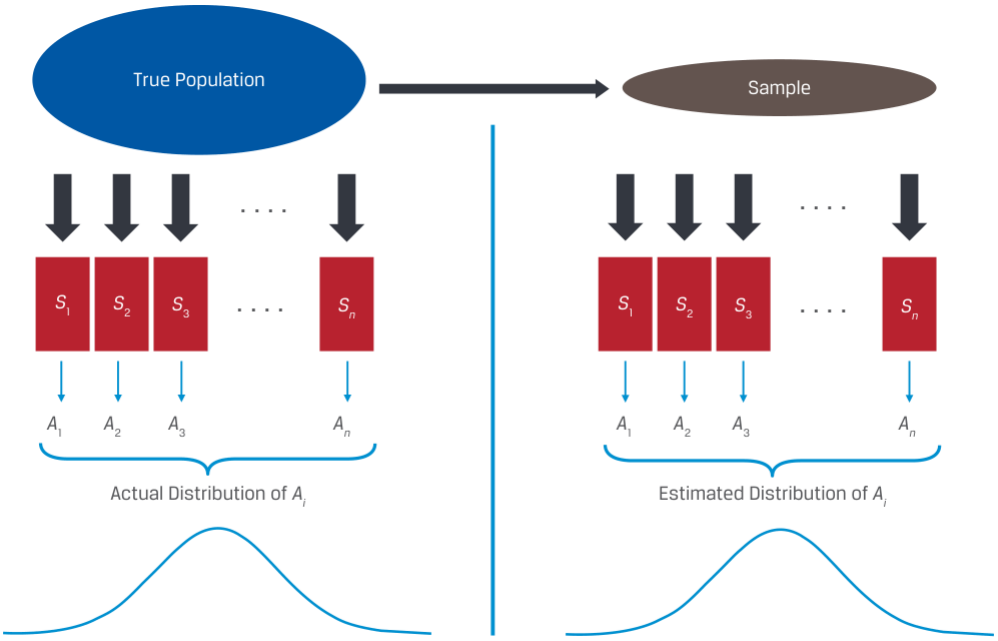


Figure 9.6: Bootstrap Resampling

Case Study: Bootstrapping Techniques to Value a European-Style Call Option on Brighton Rock plc Stock

We value a 10-week European call option on Brighton Rock plc stock using bootstrapping. The strike price is equal to the current stock price ($K = P_0 =$ GBP 141.25).

Step 1: Setup/parameterization

- Option Payoff: $V_{\text{Call option}} = \max[P_T - K, 0]$ where $K =$ GBP 141.25. - Steps: $n = 10$ weeks. - Interest rate: $r = 5.0\%$ annualized (continuously compounded). - Sample data: 25 weekly log returns ($t = 1$ to $t = 25$) calculated from 26 weeks of historical price data (Table 9.8).

Table 9.8: Weekly Closing Prices and Log Returns for Brighton Rock plc

Week	Price (GBP)	Log Return (r_i)	Week	Price (GBP)	Log Return (r_i)
0, $t = 0$	100.00	NA	13	114.07	-0.0056
1, $t = 1$	100.26	0.0026	14	115.83	0.0153
2	102.36	0.0207	15	123.01	0.0601
3	102.92	0.0055	16	128.71	0.0453
4	103.55	0.0061	17	131.07	0.0182
5	103.75	0.0019	18	131.88	0.0062
6	109.44	0.0534	19	132.15	0.0020
7	112.79	0.0302	20	129.54	-0.0199
8	109.26	-0.0318	21	133.28	0.0285
9	108.96	-0.0027	22	134.02	0.0055
10	110.69	0.0158	23	135.02	0.0074
11	111.57	0.0079	24	137.09	0.0152
12	114.71	0.0278	25, $t = 25$	141.25	0.0299
Average		**0.0138**	**Cumulative**		**0.3454**

We run 500 simulations to estimate the option value.

Step 2: Scenario generation.

We construct simulated price paths by drawing 10 random integers between 1 and 25 (with replacement) for

each scenario and assigning the corresponding historical return. Table 9.9 lists Scenario 1's details.

Table 9.9: Random Numbers and Corresponding Log Returns for a Scenario

Pull (Random Number)	**Corresponding Log Return (r_i)**
21	0.0285
25	0.0299
4	0.0061
13	-0.0056
16	0.0453
8	-0.0318
7	0.0302
8	-0.0318
15	0.0601
10	0.0158
Total return	**0.1467**

Step 3: Scenario evaluation/calculation engine.
Log returns are additive, so for Scenario 1:

$$r_{\text{Scenario 1}} = \sum_{i=1}^{10} r_i = 0.1467$$

The terminal stock price is:

$$P_T = \text{GBP } 141.25 \times e^{0.1467} = \text{GBP } 163.57$$

The option payoff at maturity is:

$$\text{Payoff} = \max(163.57 - 141.25, 0) = \text{GBP } 22.32$$

Figure 9.7 shows the price paths of 500 simulations.

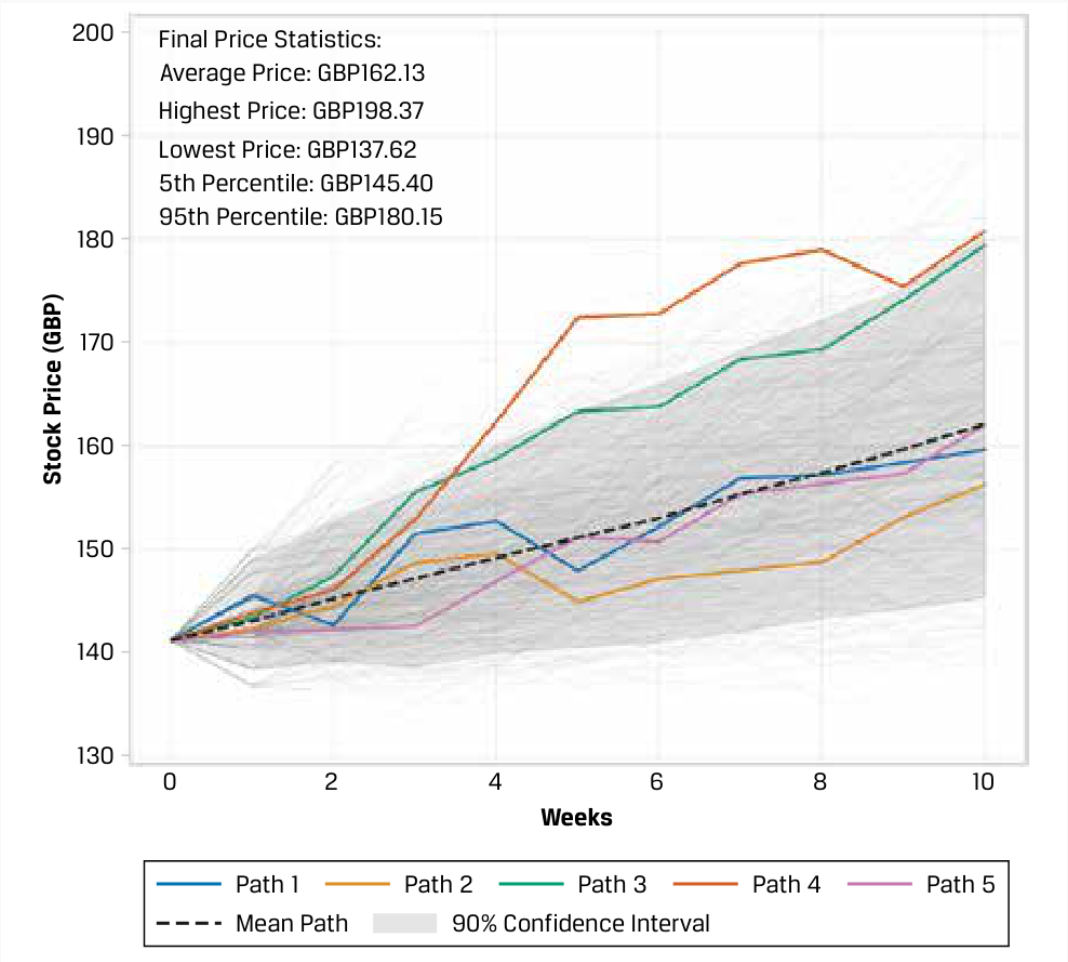


Figure 9.7: Bootstrap Share Price Path Evolution Graph

Step 4: Result compilation/analysis.
Figure 9.8 shows the resulting histograms for 500 simulations (mean option value is ≈ GBP 20.90).

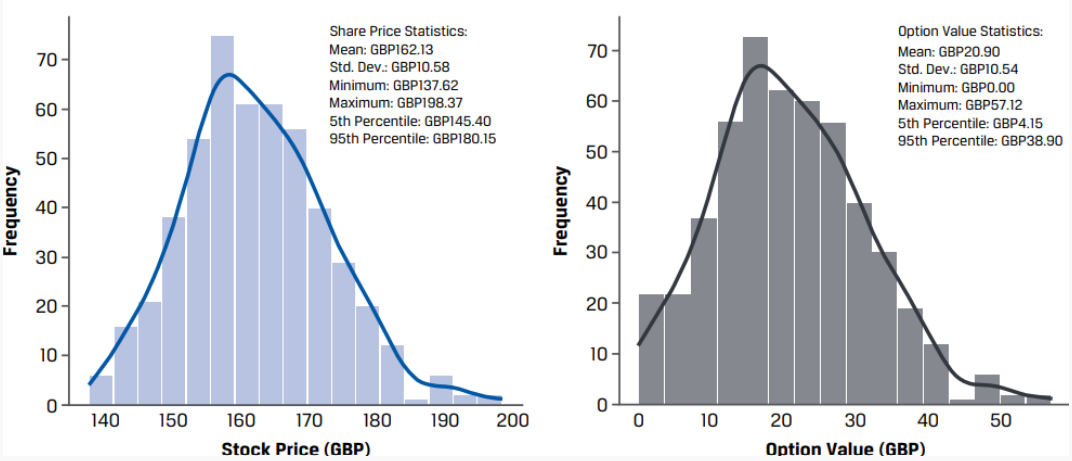


Figure 9.8: Stock Price and Option Final Value Distribution Comparison after 500 Simulations

Figure 9.9 shows the histograms for 10,000 simulations (mean option value is ≈ GBP 21.34), showing smoother tails.

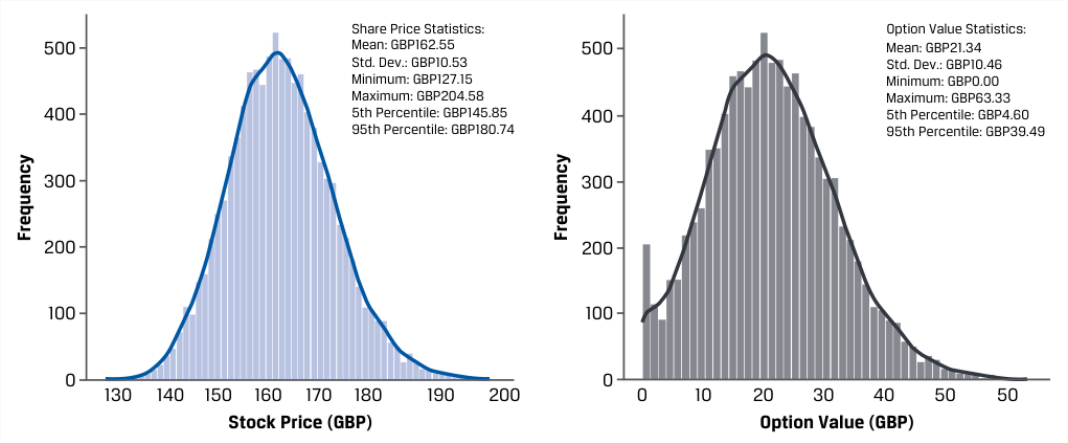


Figure 9.9: Stock Price and Option Value Distribution Comparison after 10,000 Simulations

Under the standard risk-free BSM model, this option is worth < GBP 5. The bootstrapped price (≈ GBP 21) is much higher because bootstrapping uses historical returns, incorporating the large positive drift (1.38% weekly) observed in the sample data, whereas BSM assumes a risk-neutral drift.

Lưu ý: Định giá quyền chọn bằng bootstrapping ra GBP 21 (so với BSM dưới GBP 5) vì dùng drift lịch sử (1.38%/tuần) thay vì lãi suất phi rủi ro (5%/năm) như lý thuyết bắt buộc. Đây là ví dụ về lỗi sai nguyên lý tài chính.

Case Study: Analyzing Portfolio Volatility of a Target Date Fund Using Bootstrapping

A company evaluates a target date retirement fund with only 24 months of available return data. To estimate its volatility over various horizons, bootstrapping is used to generate robust standard error estimates.
Step 1: Setup and parameterization.
Table 9.10 lists the observed monthly returns and corresponding NAVs starting from USD 100. Figure 9.10 illustrates their evolution.

Table 9.10: Monthly Returns and NAVs

Month	Monthly Return (%)	NAV (USD)	Month	Monthly Return (%)	NAV (USD)
0	0.00%	100.00	13	-0.77%	134.50
1	-4.46%	95.54	14	11.00%	149.30
2	9.16%	104.29	15	10.26%	164.62
3	-3.94%	100.18	16	7.64%	177.20
4	13.55%	113.75	17	-3.94%	170.22
5	6.02%	120.60	18	14.53%	194.95
6	-6.33%	112.97	19	-6.70%	181.89
7	-1.97%	110.74	20	-7.20%	168.79
8	8.71%	120.39	21	-7.80%	155.62
9	3.75%	124.90	22	-4.34%	148.87
10	-5.56%	117.96	23	-4.50%	142.17
11	6.09%	125.14	24	8.79%	154.67
12	8.31%	135.54			

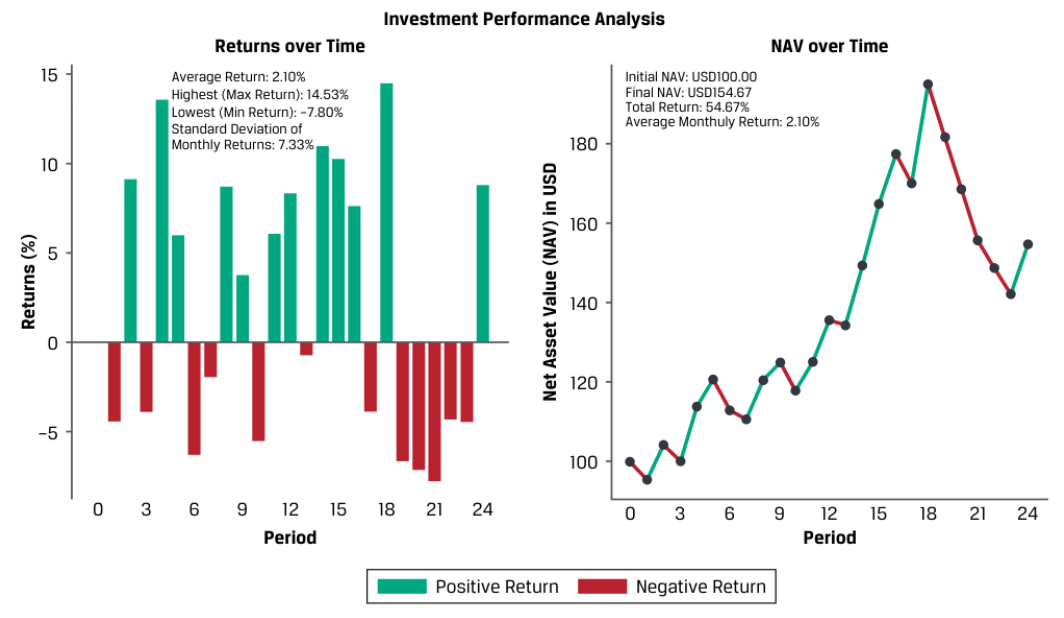


Figure 9.10: Returns and NAV of the Fund

Step 2: Scenario generation with bootstrapping.
We construct 1,000 scenarios, each simulating a 50-month path by randomly drawing 50 returns from the 24-month sample (with replacement). For example: - Random draws: $[8, 7, 22, \ldots, 18]$. - Assigned returns: $[8.71\%, -1.97\%, -4.34\%, \ldots, 14.53\%]$. The sample standard deviation is 7.61%.
Step 3: Scenario evaluation and calculation.
For each scenario, the monthly returns' standard deviation is computed. The resulting frequency distribution is shown in Figure 9.11.

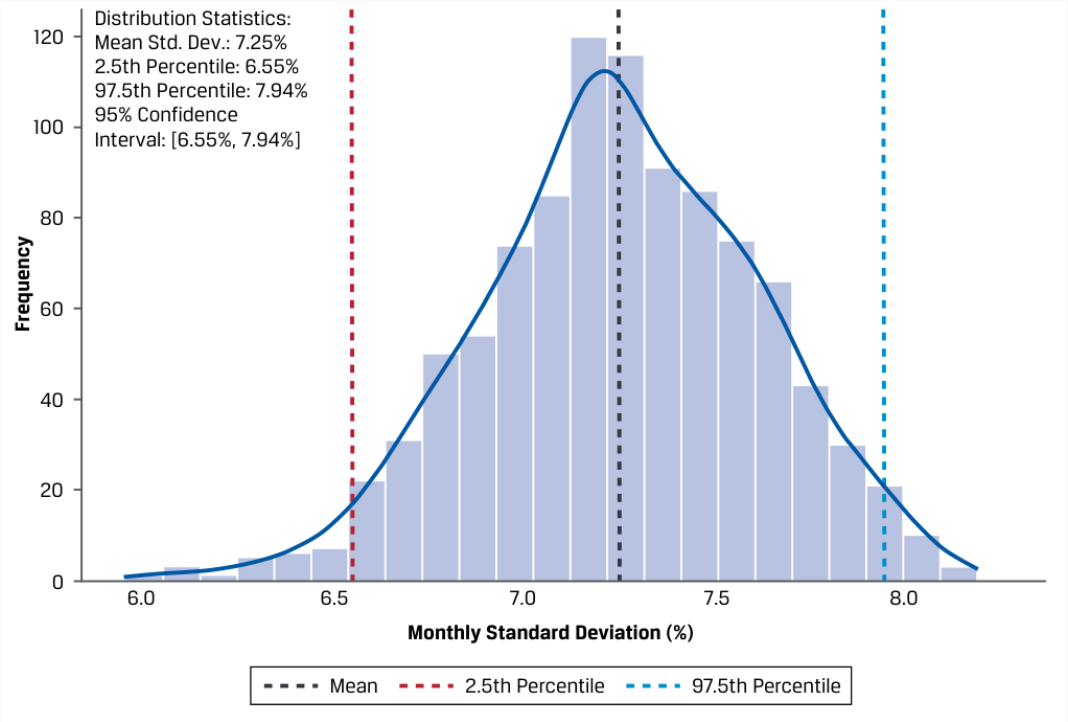


Figure 9.11: Distribution of Monthly Return Standard Deviations

Step 4: Result compilation and analysis.
The 95% confidence interval for the monthly volatility (2.5th to 97.5th percentiles) ranges from 6.55% to 7.94%. This range allows the company to compare the fund's risk profile against competitors and determine if its volatility is acceptable.

9.2.3 Features of Bootstrapping Simulations

Bootstrapping avoids parametric distribution assumptions, allows constructing confidence intervals, and assesses estimator stability. Its effectiveness depends on sample representativeness. Larger sample sizes yield more accurate results.

Strengths of bootstrapping:

- **Minimal assumptions:** Bootstrapping does not require assumptions about the underlying population distribution.
- **Statistical robustness:** Unlike many other methods, bootstrapping is particularly useful when dealing with small datasets.
- **Robust variability estimates:** Provides robust estimates of standard errors and confidence intervals for limited datasets.
- **Adaptability:** Bootstrapping can be applied to almost any statistic (e.g., mean, median, variance, correlation) in various fields.
- **No need for a "large" external dataset:** bootstrapping works directly from the sample at hand, making it ideal for situations where data are scarce.

Disadvantages of bootstrapping:

- **Computationally intensive:** Requires running many iterations to build the simulated distribution.
- **Sample dependency:** Highly sensitive to sample quality; cannot generate values outside the sample range.
- **Time-series limitations:** Simple bootstrapping breaks down autocorrelation in time-series data.
- **No analytical insight:** Bootstrapping is purely empirical and does not provide theoretical insight into the population or the underlying process.

9.3 Monte Carlo Simulation

c | describe Monte Carlo simulation and explain how it can be used in investment applications

Mô phỏng Monte Carlo là phương pháp tham số: giả định phân phối xác suất và chạy hàng chục ngàn kịch bản ngẫu nhiên. Hữu ích nhất để định giá tài sản phi tuyến hoặc phụ thuộc đường đi (Asian options, MBS).

9.3.1 Monte Carlo Simulation Overview

A Monte Carlo simulation generates a large number of random samples from a specified probability distribution or distributions to produce a simulated frequency distribution of outcomes.

Unlike historical simulation and bootstrapping, Monte Carlo simulation is a parametric method that requires explicit assumptions about the input variables' probability distributions. This allows analysts to evaluate the sensitivity of output variables to changes in parameters like mean, variance, and correlation.

In finance, Monte Carlo simulation is widely used to value complex securities and derivatives (e.g., path-dependent contingent claims) that lack analytical closed-form pricing formulas.

Case Study: European Call Option Valuation Using Monte Carlo Simulation

We evaluate a European call option using Monte Carlo simulation and compare the results to the analytic BSM formula:

- Option Type: European Call, Strike (K): CAD 90
- Current Price (P_0): CAD 85, Time (T): 1 Year
- Volatility (σ): 25%, Risk-free rate (r): 5.0%

Step 1: Setup/parameterization
- Valuation: $V_{\text{Call option}} = \max[P_T - K, 0]$ where $K =$ CAD 90. - Under risk-neutral pricing, stock price paths are assumed to follow a log-normal distribution.
Step 2: Scenario generation
The log return return is simulated using Geometric Brownian Motion over one step ($s = 1.0$):

$$r_t = \left(r_f - \frac{\sigma^2}{2}\right) \times s + \sigma\sqrt{s}Z \tag{9.5}$$

where $Z \sim N(0,1)$ and the drift is: $0.05 - (0.25^2/2) = 0.01875$. The ending stock price is $P_T = P_0 \times e^{r_t}$.
Step 3: Scenario evaluation/calculation engine
We run 5,000 simulations to generate ending prices (Figure 9.12).

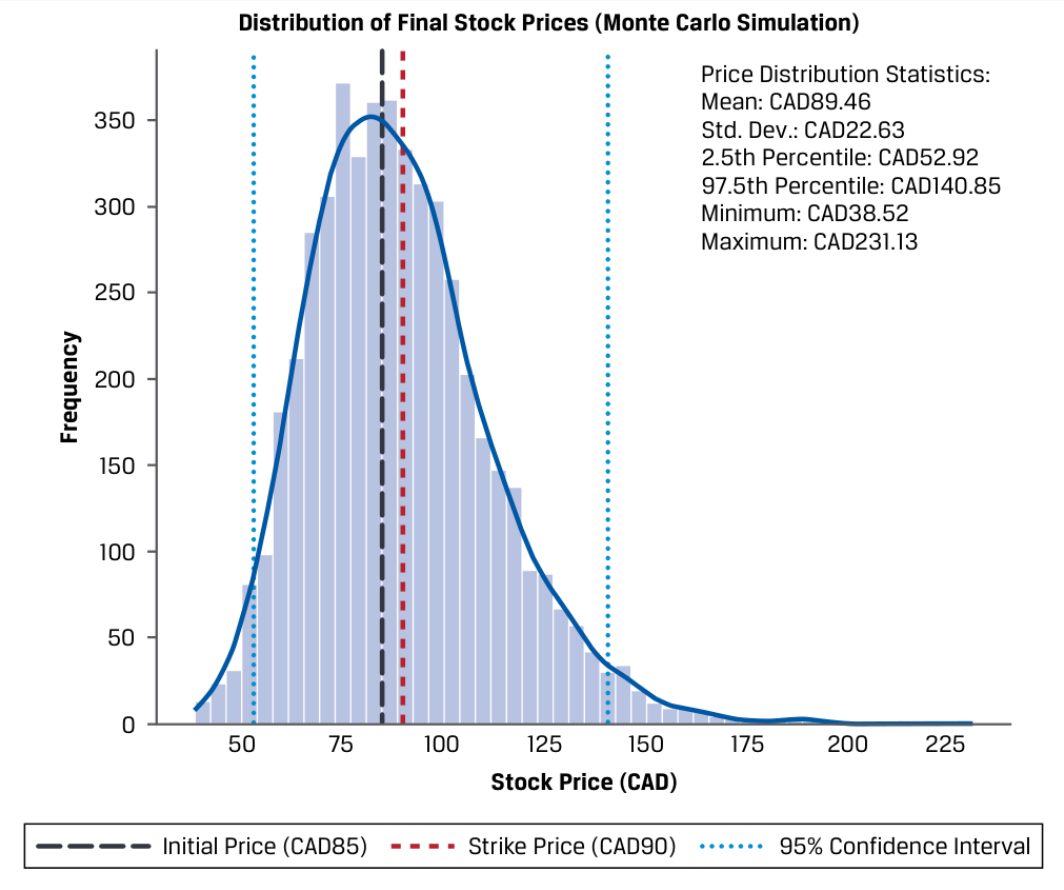


Figure 9.12: Distribution of Stock Prices, Simulation Result

Step 4: Result compilation/analysis
The resulting option value distribution is shown in Figure 9.13.

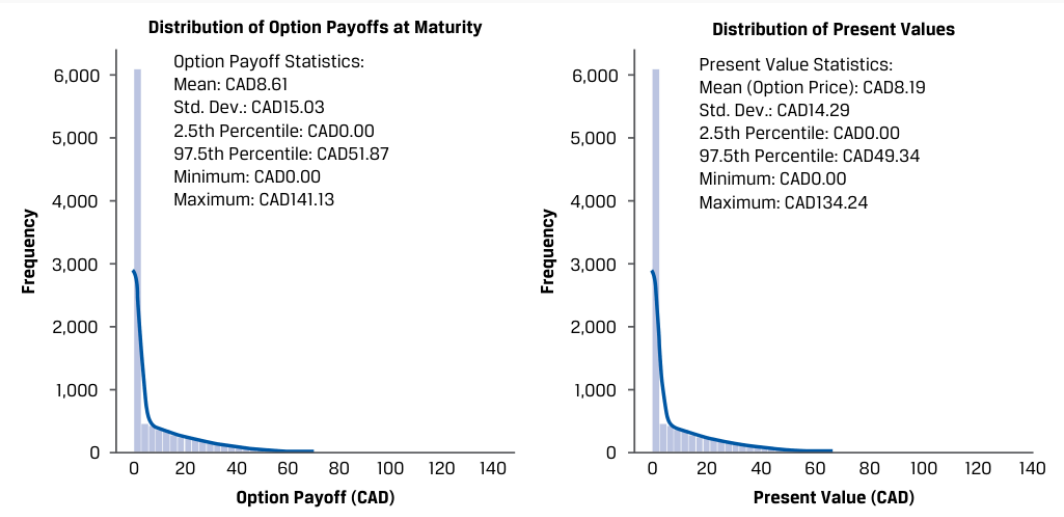


Figure 9.13: Distribution of European Call Option Values for Simulation

The mean option payoff is CAD 8.61. Discounted at the risk-free rate of 5.0

$$V_0 = 8.61 \times e^{-0.05 \times 1} = \text{CAD } 8.19$$

This closely matches the BSM formula value of CAD 8.18.

Lưu ý: Mô phỏng Monte Carlo hội tụ về giá trị BSM khi tăng số kịch bản do cùng giả định chuyển động Brown hình học. Tuy nhiên, với quyền chọn phụ thuộc đường đi (như kiểu Á), ta bắt buộc phải dùng mô phỏng.

Case Study: Asian Option Valuation Using Monte Carlo Simulation

An Asian call option's payoff depends on the average price of the underlying asset: $V_{\text{Payoff}} = \max[P_T - \text{Average price}, 0]$. We value a 21-day Asian option on a stock denominated in Swiss francs (CHF).

Step 1: Setup/Parameterization

- Objective: Estimate $V_0 = E[\max[P_{21} - \text{Average price (Days 1–21)}, 0]] \times e^{-rT}$. - Input parameters: $P_0 =$ CHF 25.00, $T = 21$ days, $\sigma = 25\%$, $r = 5.0\%$.

Step 2: Scenario Generation

Daily stock returns follow a log-normal distribution:

$$\text{Log price change}_t = \left(0.05 - \frac{0.25^2}{2}\right) \times \left(\frac{1}{252}\right) + 0.25 \times \sqrt{\frac{1}{252}} \times Z \tag{9.6}$$

where $Z \sim N(0,1)$. Prices evolve as $P_t = P_{t-1} \times e^{r_t}$. Table 9.11 shows one simulated run.

Table 9.11: Asian Option Valuation: Daily Log Returns and Prices for One Simulated Run

Day	Return (%)	Price (CHF)	Day	Return (%)	Price (CHF)
1	-2.32%	24.43	12	0.22%	24.80
2	-0.70%	24.26	13	-1.43%	24.45
3	1.45%	24.61	14	4.87%	25.67
4	0.49%	24.73	15	-2.80%	24.96
5	-1.01%	24.48	16	-1.25%	24.65
6	1.15%	24.76	17	-1.54%	24.27
7	1.16%	25.05	18	0.24%	24.33
8	0.46%	25.16	19	-4.29%	23.31
9	-0.58%	25.01	20	0.41%	23.41
10	0.12%	25.04	21	-1.28%	23.11
11	-1.17%	24.75			
Average					**24.54**

For this run, the ending price is CHF 23.11, which is below the average price of CHF 24.54, yielding a payoff of zero. Figure 9.14 shows the price path of this run.

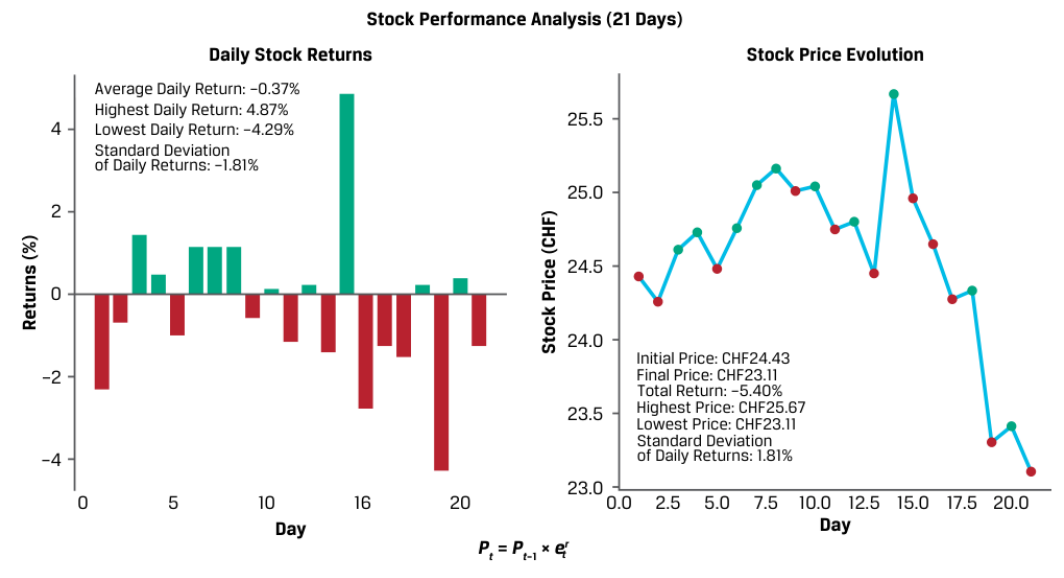


Figure 9.14: Asian Option Valuation: Daily Log Return and Prices for the Same Run

Step 3: Scenario evaluation/calculation engine.

We run 500 price path simulations (Figure 9.15).

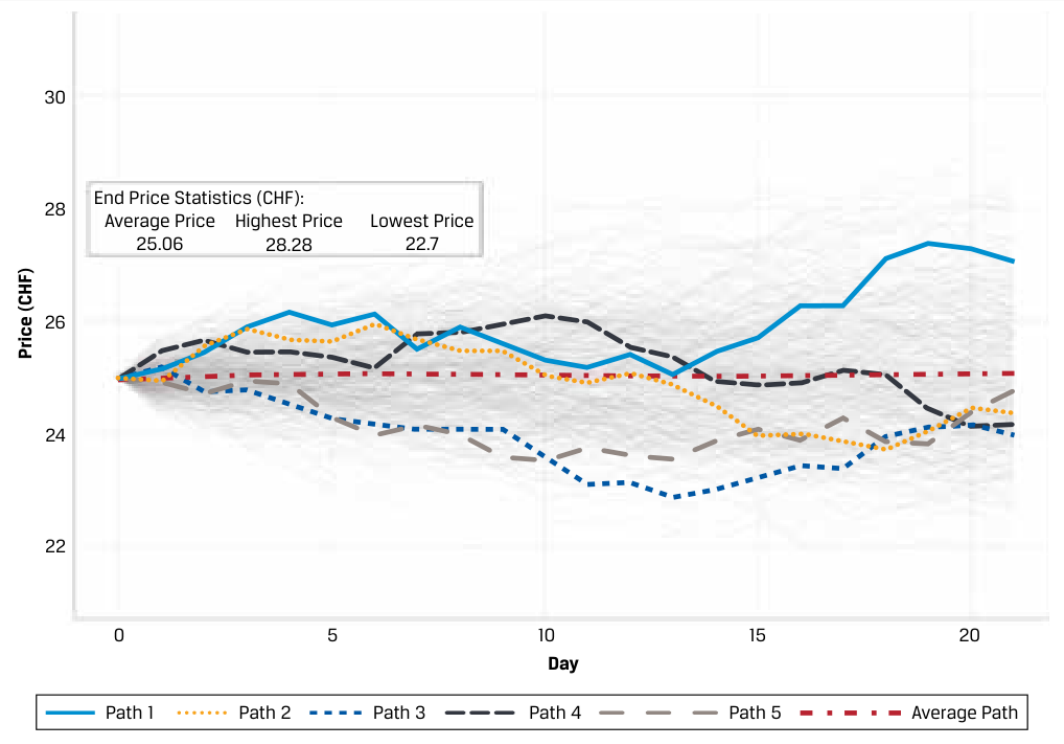


Figure 9.15: Asian Option Valuation: 500 Simulated Stock Price Paths

Step 4: Result compilation/analysis.
The distribution of payoffs is shown in Figure 9.16.

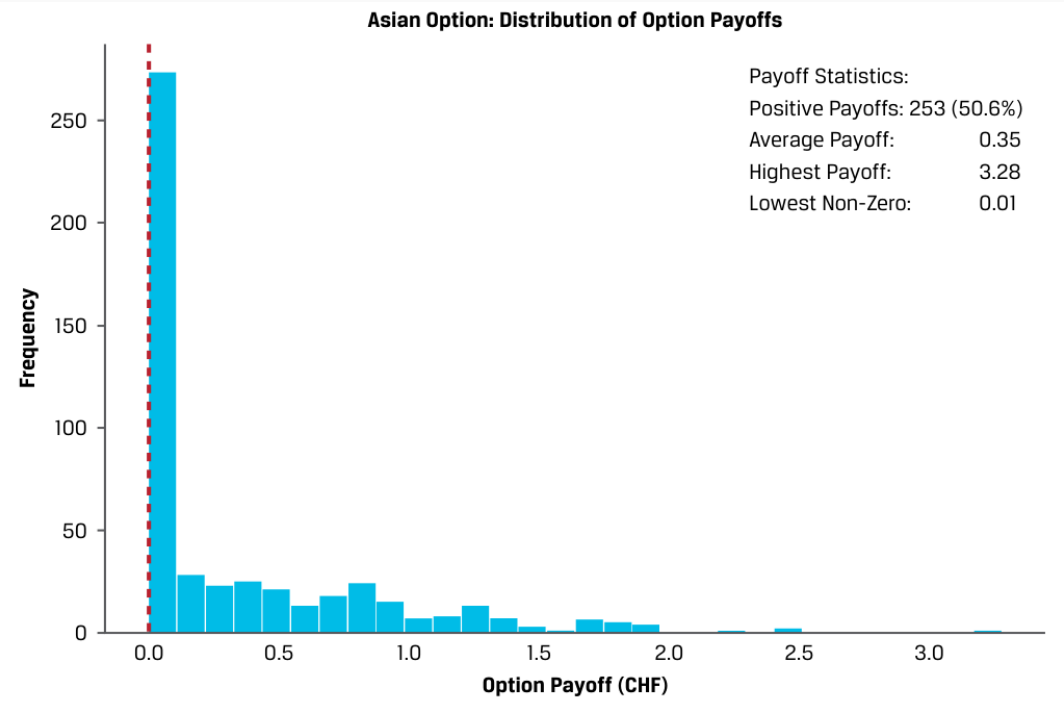


Figure 9.16: Asian Option Valuation: Distribution of Option Payoffs

The average payoff across all 500 simulations was CHF 0.35. Discounting using the risk-free rate yields:

$$V_0 = 0.35 \times e^{-0.05 \times \frac{21}{252}} = 0.35 \times 0.9958 = \text{CHF } 0.3485$$

9.3.2 Multivariate Monte Carlo Simulation

Simulating multiple variables simultaneously is necessary to capture complex correlations and dependencies in financial portfolios or multi-asset derivatives.

Correlated variables are generated by drawing independent standard normal variables and transforming them using the Cholesky decomposition—a matrix factorization technique that decomposes a covariance or correlation matrix. The framework is shown in Figure 9.17.

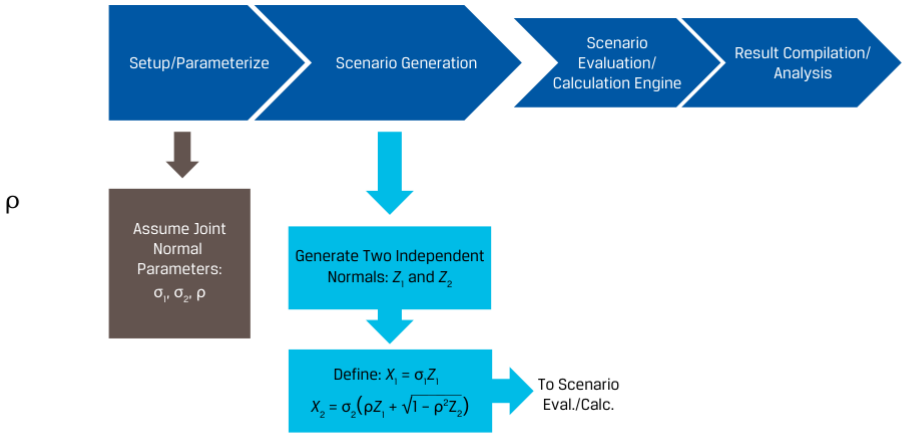


Figure 9.17: Simulation Framework Applied to Multivariate Monte Carlo

Step 1: Setup/parameterize

We assume a joint normal distribution for the two variables with standard deviations σ_1 and σ_2 and correlation ρ.

Step 2: Scenario generation

The Cholesky decomposition transforms independent standard normal variables Z_1 and Z_2 into correlated variables X_1 and X_2:

$$X_1 = \sigma_1 Z_1 \tag{9.7}$$

$$X_2 = \sigma_2 \left(\rho Z_1 + \sqrt{1-\rho^2} Z_2\right) \tag{9.8}$$

This approach ensures that the final variables, X_1 and X_2, adhere to our calibration:

$$E[X_1] = \sigma_1 E[Z_1] = 0 \tag{9.9}$$

$$E[X_2] = \sigma_2 E[\rho Z_1 + \sqrt{1-\rho^2} Z_2] = 0 \tag{9.10}$$

The variances are:

$$E[X_1^2] = \sigma_1^2; \quad E[X_2^2] = \sigma_2^2 \tag{9.11}$$

Since Z_1 and Z_2 are independent, the covariance is:

$$\sigma_{X_1,X_2} = E[X_1 X_2] = \rho \sigma_1 \sigma_2 \tag{9.12}$$

Xem thêm về mô phỏng 2 biến có tương quan, sử dụng Cholesky decomposition tại phần (C.1)

Knowledge Check

An analyst is reviewing intermediate outputs from a Monte Carlo simulation designed to model log price changes. The simulation involves two normally distributed variables, both of which have a mean of 0 and standard deviations of 1.5% and 2.0%, respectively. The analyst has specified a correlation of 30% between the two variables, and the simulations are conducted using the Cholesky decomposition method.

In a specific scenario, the random draws from independent standard normal distributions are $Z_1 = 0.4$ and $Z_2 = -1.2$. The first variable, x_1, is calculated using:

$$x_1 = \sigma_1 \times Z_1 = 1.5\% \times 0.4 = 0.60\%$$

The value of the second variable, x_2, is closest to:

A. -2.40%.

B. -2.05%.

C. -1.80%.

Solution:
B là phương án đúng. Áp dụng công thức liên kết hai biến số tương quan bằng Cholesky decomposition:

$$x_2 = \sigma_2 \left(\rho Z_1 + \sqrt{1-\rho^2} Z_2\right)$$

$$x_2 = 2.0\% \times \left(0.3 \times 0.4 + \sqrt{1-0.3^2} \times (-1.2)\right)$$

$$x_2 = 2.0\% \times (0.12 + 0.9539 \times (-1.2)) = 2.0\% \times (0.12 - 1.1447) = 2.0\% \times (-1.0247) \approx -2.05\%$$

Phương án A tính sai khi bỏ qua yếu tố tương quan và nhân trực tiếp: $2.0\% \times Z_2 = -2.40\%$.
Phương án C sai vì dùng sai tỷ lệ của x_1.

Lưu ý: Biến đổi Cholesky duy trì cấu trúc hiệp phương sai trong Monte Carlo đa biến. Nếu thiếu, các biến ngẫu nhiên sẽ độc lập, làm mất đi tính tương quan và hiệu ứng đa dạng hóa danh mục.

Case Study: Analyzing the Potential Distribution of a Long–Short Strategy

A manager implements a daily long–short strategy, investing EUR 10 million (long) in the DAX (Germany) and shorting EUR 10 million in the MADX (Spain) (Figure 9.18). We simulate the portfolio's P/L distribution.

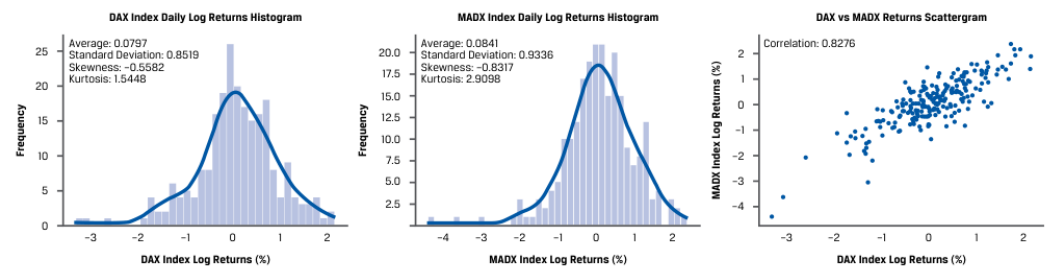


Figure 9.18: The Distribution of the Daily Log Returns for the DAX and MADX over One Year

We make the following parameters: - Model: Bivariate normal distribution on log returns. - Standard deviations: DAX $\sigma_1 = 0.82\%$, MADX $\sigma_2 = 0.86\%$. - Correlation coefficient: $\rho = 0.8276$. - Expected returns: Assumed to be zero.
We generate independent standard normals Z_1, Z_2 and apply Cholesky decomposition:

$$X_1 = \sigma_{\text{DAX}} \times Z_1 = 0.0082 Z_1$$

$$X_1 = \sigma_{\text{MADX}} \times (\rho Z_1 + \sqrt{1-\rho^2} Z_2)$$

$$X_2 = 0.0086 \times \left(0.8276 Z_1 + \sqrt{1-0.8276^2} Z_2\right) = 0.0071 Z_1 + 0.0048 Z_2$$

The portfolio return is $R_P = X_1 - X_2$ (due to shorting MADX). Using 250 simulations, the resulting P/L distribution is shown in Figure 9.19.

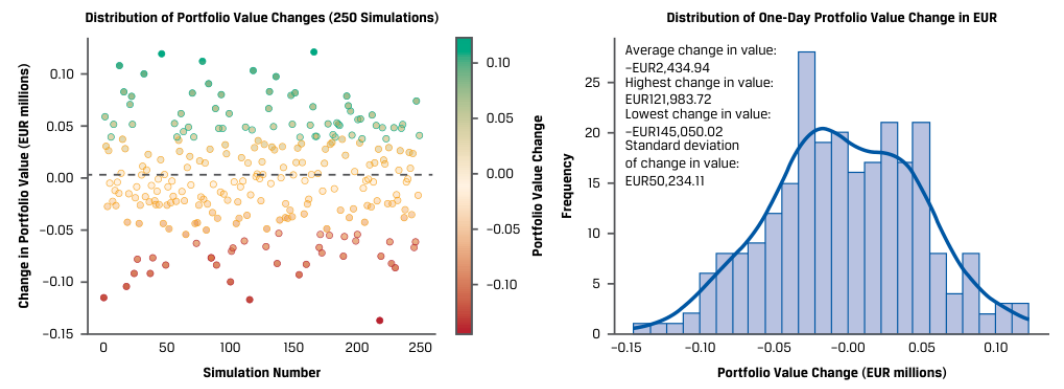


Figure 9.19: Portfolio Value Change Distributions: Simulation Results

The simulation results show an average value change of -EUR 2,434.94 (theoretically zero) and standard deviation of EUR 50,234.11. Increasing the number of simulations would reduce simulation noise and move the mean closer to the theoretical zero.

9.3.3 Features of Monte Carlo Simulation

Monte Carlo simulation is highly flexible because the modeler controls the probability distributions, parameters, and relationships of the inputs.

Strengths of Monte Carlo simulation:

- **Wide applicability:** Can simulate complex, non-linear, and path-dependent cash flows.
- **Correlation handling:** Monte Carlo simulation effectively models the correlations between assets.
- **Forward looking:** Allows testing portfolio behavior under different distributional parameters and hypothetical future scenarios.
- **No historical data constraint:** Can be applied to new instruments by defining pricing parameters.
- **Flexibility for non-normal distributions:** Monte Carlo simulation is adaptable to non-normal return distributions, accounting for skewness and kurtosis.

Disadvantages of Monte Carlo simulation:

- **Computational intensity:** Requires simulating thousands of trials, which can be computationally intensive.
- **Model dependency:** Results depend entirely on input distribution and correlation assumptions ("garbage in, garbage out").
- **Complexity:** Harder to design, calibrate, and explain compared to historical simulation.
- **Data requirements:** Monte Carlo simulation often relies on extensive historical data to estimate such parameters as volatility and correlation.

Table 9.12 compares the three simulation approaches.

Table 9.12: Comparison of Simulation Approaches

Feature	Historical Simulation	Bootstrapping	Monte Carlo Simulation
Source of data	Historical dataset	Observed sample	Assumed distribution
Sampling method	Sampling from historical data	Resampling with replacement	Random sampling from predefined distribution
Assumptions	Assumes past reflects future	Assumes sample represents population	Assumes generation process can be modeled statistically
Flexibility	Limited to historical patterns	Limited by sample size	Highly flexible
Computational intensity	Low to moderate	Moderate	High

9.3.4 Practical Implementation of Simulation Approaches Using Excel and Other Tools

Excel and Google Sheets provide functions to implement these techniques. Table 9.13 summarizes these functions.

Implementing Historical Simulation in Excel:

- Input historical price or return data in Column A.
- Calculate returns in Column B.
- Sort returns in Column C: `=SORT(B:B, 1, TRUE)`.
- Calculate VaR using percentile functions: `=PERCENTILE.EXC(B:B, 0.05)`.

Implementing Bootstrapping in Excel:

- Input historical data in Column A and calculate returns in Column B.

Table 9.13: Functions Used for Simulation

Technique	Excel Functions	Google Sheets	Description
Monte Carlo	RAND(), NORM.INV(), LOGNORM.INV(), RANDBETWEEN(), IF(), SUM(), AVERAGE(), STDEV(), CONFIDENCE.NORM()	RAND(), NORMINV(), LOGNORMINV(), RANDBETWEEN(), IF(), SUM(), AVERAGE(), STDEV(), CONFIDENCE()	Generate random values from normal, lognormal, or uniform distributions.
Historical	INDEX(), SORT(), PERCENTILE.EXC(), PERCENTILE.INC(), LARGE(), SMALL(), AVERAGE(), STDEV(), IFERROR()	INDEX(), SORT(), PERCENTILE.EXC(), PERCENTILE.INC(), LARGE(), SMALL(), AVERAGE(), STDEV(), IFERROR()	Sort, rank, and calculate percentiles of historical return series.
Bootstrapping	RAND(), INDEX(), MATCH(), COUNTIF(), SUM(), AVERAGE(), STDEV(), LARGE(), SMALL(), CONFIDENCE.NORM()	RAND(), INDEX(), MATCH(), COUNTIF(), SUM(), AVERAGE(), STDEV(), LARGE(), SMALL(), CONFIDENCE()	Resample with replacement using RAND() and INDEX().

- Resample returns with replacement in Column C: `=INDEX(B:B, RANDBETWEEN(2, N))`, which randomly selects a row.
- Replicate the formula across multiple rows and columns to generate paths.

Implementing Monte Carlo in Excel:

- Define initial stock price, volatility, risk-free rate, and time steps.
- Generate standard normal variables: `=NORM.INV(RAND(), 0, 1)`.
- Simulate price paths using geometric Brownian motion:

$$P_{t+\Delta t} = P_t \times e^{(r-0.5\sigma^2)\Delta t + \sigma\sqrt{\Delta t}Z} \tag{9.13}$$

  In Excel, this is implemented as: `=P_t * EXP((r - 0.5 * o^2) * dt + o * SQRT(dt) * NORM.INV(RAND(), 0, 1))` (where `o` is volatility σ).

Module 10

Applications of Simple Linear Regression in Finance

LEARNING OUTCOMES	
a	describe, interpret, and explain simple linear regression, including coefficient estimation using the least squares criterion
b	describe and compare the assumptions of simple linear regression, identify violations through analyzing residuals, evaluate the estimated model's goodness-of-fit and regression coefficients, and results of ANOVA estimates
c	calculate and interpret predicted values, the standard error of the estimate, and prediction intervals for the dependent variable in a simple linear regression model and describe different functional forms
d	calculate and interpret the variable estimates of the capital asset pricing model (CAPM)

10.1 Linear Regression Estimation

a	describe, interpret, and explain simple linear regression, including coefficient estimation using the least squares criterion

10.1.1 Linear Regression Analysis

Regression analysis estimates the relationship between a dependent variable (Y), whose variation we want to explain, and an independent variable (X), which is hypothesized to influence or explain changes in Y.

In simple linear regression, the population model is expressed as:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, \ldots, n \tag{10.1}$$

where:

- Y_i is the dependent variable (or response variable) for observation i.
- X_i is the independent variable (or explanatory variable) for observation i.
- β_0 is the intercept coefficient, representing the expected value of Y when $X = 0$.
- β_1 is the slope coefficient, representing the change in the expected value of Y for a one-unit change in X.
- ϵ_i is the error term (or disturbance), capturing random noise and variables omitted from the model.

The estimated regression line is:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i \tag{10.2}$$

where $\hat{Y}_i$ is the predicted value of the dependent variable for observation i, and $\hat{\beta}_0$ and $\hat{\beta}_1$ are the sample estimates of β_0 and β_1.

10.1.2 Coefficient Estimation of Linear Regression

The Ordinary Least Squares (OLS) estimation criterion determines the coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ by minimizing the Sum of Squared Errors (SSE), which is the sum of the squared differences between the observed values Y_i and the predicted values $\hat{Y}_i$:

$$\text{SSE} = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n} e_i^2 \tag{10.3}$$

The OLS estimators for the slope and intercept are calculated as:

$$\hat{\beta}_1 = \frac{\text{Cov}(X,Y)}{\text{Var}(X)} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} \tag{10.4}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \tag{10.5}$$

where $\bar{X}$ and $\bar{Y}$ are the sample means of the independent and dependent variables, respectively.

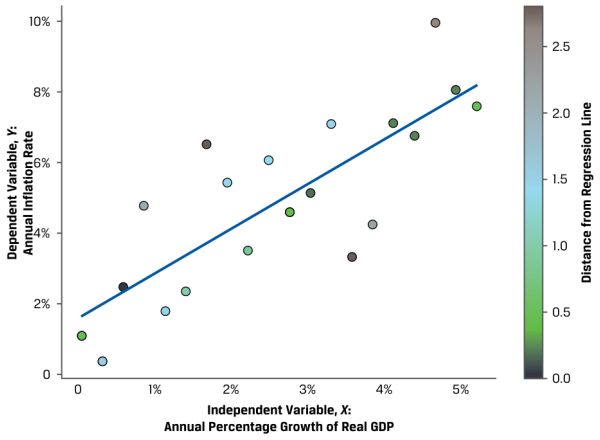
Nhận xét: OLS đảm bảo tổng sai số bằng 0 ($\sum e_i = 0$) và đường hồi quy đi qua điểm trung bình $(\bar{X}, \bar{Y})$. Tuy nhiên, việc bình phương sai số (e_i^2) khiến OLS rất nhạy cảm với các điểm dị biệt (outliers).

10.1.3 Estimation Residuals

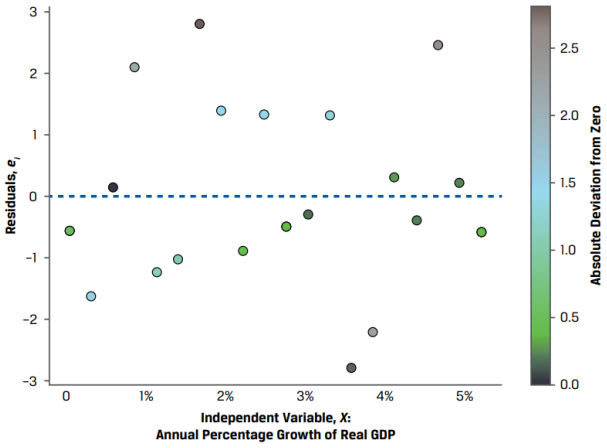
The difference between the observed value Y_i and the estimated/predicted value $\hat{Y}_i$ is the residual (e_i):

$$e_i = Y_i - \hat{Y}_i \tag{10.6}$$

The residuals are the sample counterparts to the population error terms (ϵ_i). Figure 10.1b illustrates the residuals plotted around a horizontal line at $e_i = 0$, reflecting the vertical distance of each observation from the regression line.



(a) OLS Regression: Inflation vs Real GDP



(b) Residuals of Estimated OLS Equation

Figure 10.1: OLS Regression Fit and Residual Analysis

10.1.4 Data Types

Financial regression models typically utilize three types of data structures:

- **Cross-sectional data:** Observations on multiple entities (e.g., different companies, sectors, or countries) at a single point in time. It is indexed by $i = 1, \ldots, n$.
- **Time-series data:** Repeated observations of a single entity (e.g., a country's GDP growth or a company's stock return) across multiple time periods. It is indexed by $t = 1, \ldots, T$.
- **Panel data:** A combination of cross-sectional and time-series data, observing multiple entities repeatedly over time.

Example: Estimating Inflation using Real GDP Growth

An analyst estimates the linear relationship between annual inflation (Y, dependent variable) and annual real GDP growth (X, independent variable) for a country over 24 periods. The estimated OLS model yields

the following equation:

$$\hat{Y}_i = 1.59\% + 1.27X_i$$

Interpretation:

- **Intercept ($\hat{\beta}_0 = 1.59\%$):** If the real GDP growth rate is 0%, the expected annual inflation rate is 1.59%.
- **Slope ($\hat{\beta}_1 = 1.27$):** For every 1% increase in the annual real GDP growth rate, the expected annual inflation rate increases by 1.27%.

If the real GDP growth is projected to be 3%, the predicted inflation rate is:

$$\hat{Y} = 1.59\% + 1.27 \times 3\% = 5.40\%$$

10.2 Limitations of Linear Regression Models

b	describe and compare the assumptions of simple linear regression, identify violations through analyzing residuals, evaluate the estimated model's goodness-of-fit and regression coefficients, and results of ANOVA estimates

Lưu ý: Mô hình OLS chỉ đáng tin cậy khi thỏa mãn các giả định. Nếu vi phạm (như phương sai thay đổi - Heteroskedasticity hay tự tương quan - Autocorrelation), ước lượng sẽ lệch lạc. Đồ thị phần dư (Residual Plots) là công cụ trực quan tốt nhất để phát hiện vi phạm.

10.2.1 Assumptions of the Simple Linear Regression Model

To ensure that the OLS estimators are Blue (Best Linear Unbiased Estimators) and that statistical inferences are valid, 4 key assumptions must hold:

1. **Linearity:** The relationship between the dependent variable Y and the independent variable X is linear in parameters.
2. **Homoskedasticity:** The variance of the error terms (ϵ_i) is constant across all observations: $\text{Var}(\epsilon_i) = \sigma^2$ for all i.
3. **Independence:** The error terms are independent of one another, implying they are uncorrelated across observations: $\text{Cov}(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$.
4. **Normality:** The error terms (ϵ_i) are normally distributed. This is crucial for small sample hypothesis testing.

10.2.2 Residual Analysis

Residual plots (plotting residuals e_i against X_i or time t) are crucial diagnostics to identify violations of OLS assumptions:

- **Violation of Linearity:** Residuals show a distinct curved (e.g., U-shaped) pattern when plotted against X, suggesting that a linear form is inappropriate. Figure 10.2 illustrates this violation.
- **Heteroskedasticity:** The spread of the residuals increases or decreases systematically with X (creating a funnel shape). Non-constant variance invalidates standard error estimates, making t-statistics unreliable. Figure 10.3 shows a comparison of regressions on data with two variance regimes.
- **Autocorrelation (Serial Correlation):** Residuals exhibit a systematic trend or cyclical pattern over time (e.g., positive residuals followed by positive residuals). Common in time-series data, this violates the independence assumption. Figure 10.4 compares independent and autocorrelated residuals.

10.2.3 Evaluating Goodness-of-Fit and Regression Coefficients

Goodness-of-fit measures how well the independent variable explains variation in the dependent variable. Total variation is decomposed into:

$$\text{SST} = \text{SSR} + \text{SSE} \tag{10.7}$$

where:

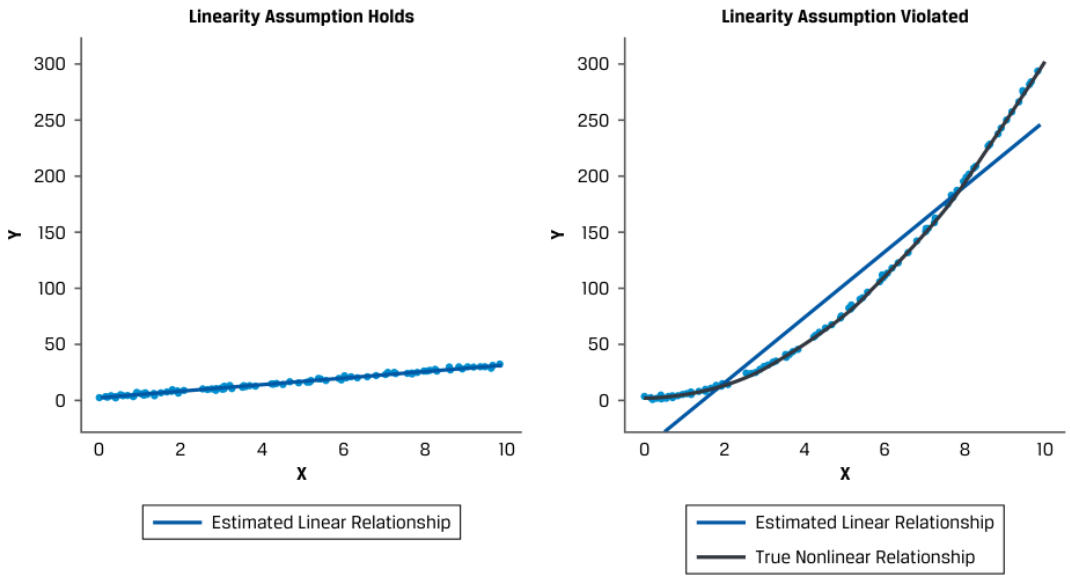


Figure 10.2: Fitting a Linear and a Non-Linear Model to Non-Linear Data

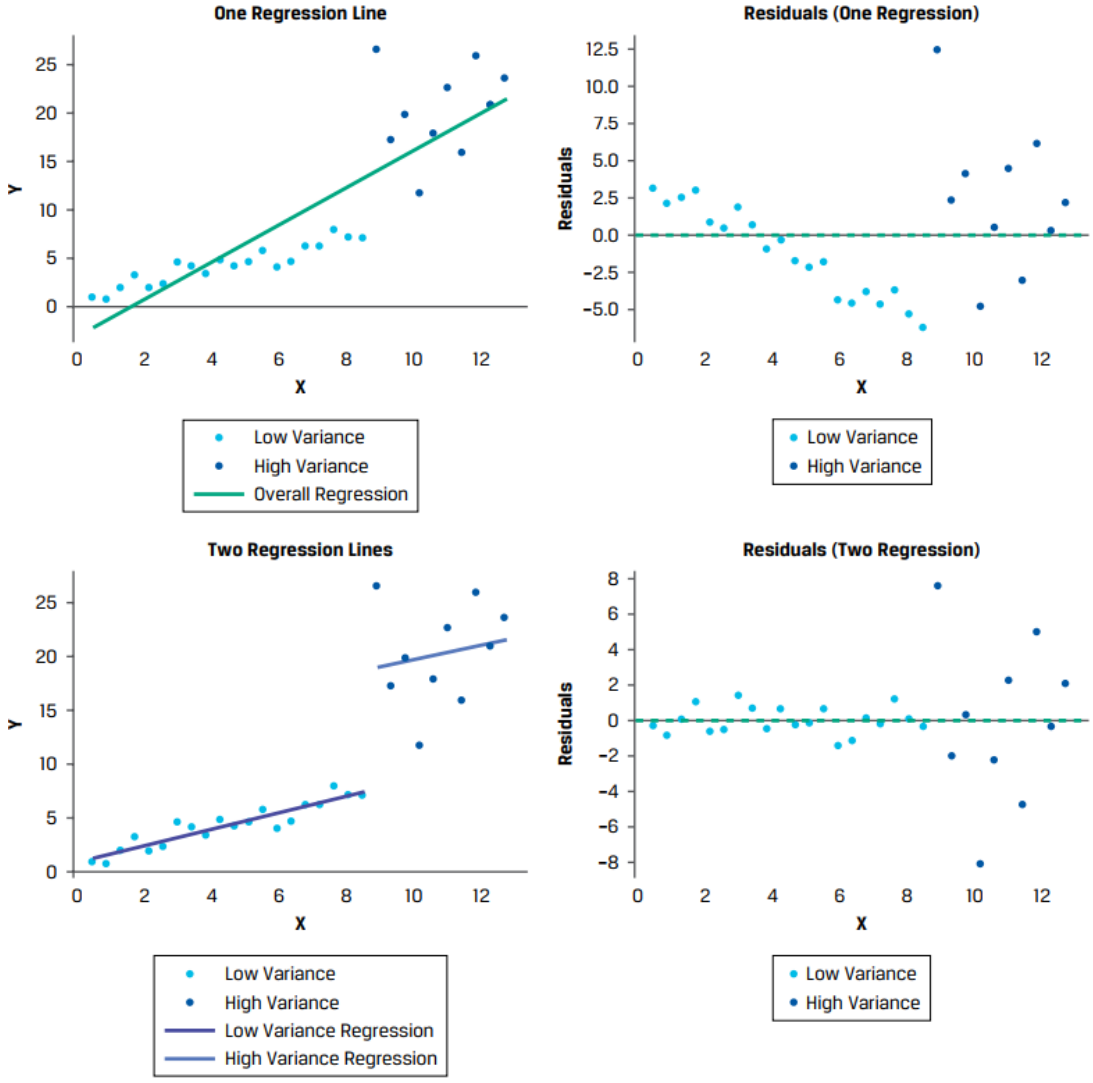


Figure 10.3: Regressions on Data with Two Different Regimes and Residuals

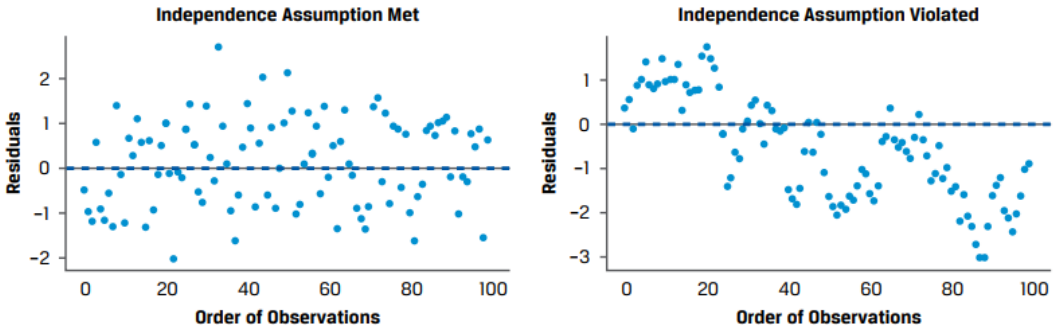


Figure 10.4: Comparison of Residual Plots: Independence (Left) and Autocorrelation (Right)

- **Total Sum of Squares (SST):** Total variation of the dependent variable around its mean: SST $= \sum(Y_i - \bar{Y})^2$.
- **Regression Sum of Squares (SSR):** Variation explained by the regression line: SSR $= \sum(\hat{Y}_i - \bar{Y})^2$.
- **Sum of Squared Errors (SSE):** Unexplained variation (residuals): SSE $= \sum(Y_i - \hat{Y}_i)^2$.

The **Coefficient of Determination (R^2)** measures the proportion of total variation in Y explained by the regression model:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} \tag{10.8}$$

In simple linear regression, the correlation coefficient r is related to R^2 by:

$$r = \text{sign}(\hat{\beta}_1) \times \sqrt{R^2} \tag{10.9}$$

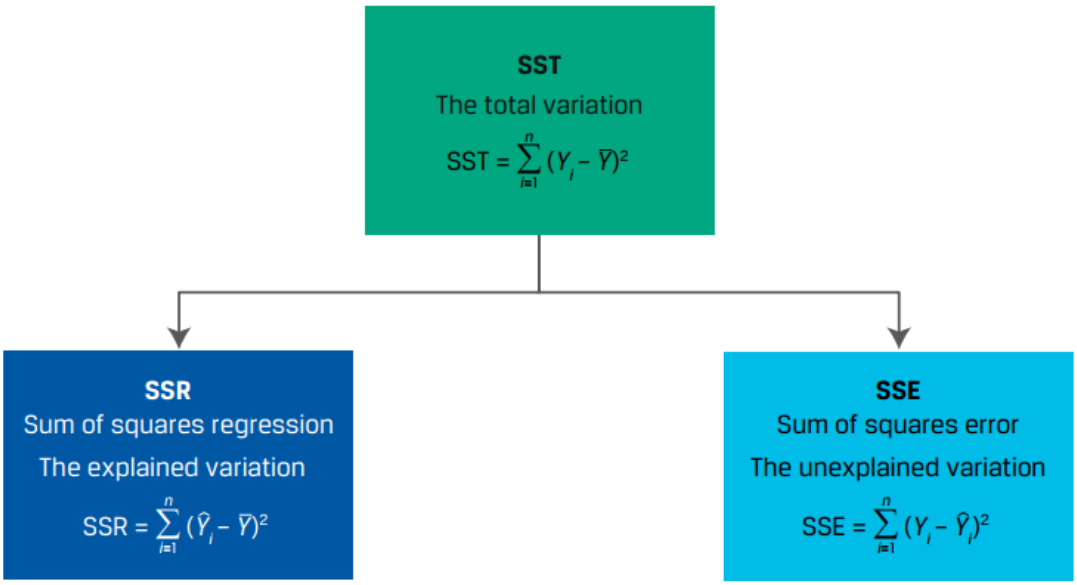


Figure 10.5: The Components of Total Variation in Regression Analysis

10.2.4 Hypothesis Testing and ANOVA Estimates

Analysis of Variance (ANOVA) summarizes the regression outputs to test the overall model significance. Table 10.1 shows the standard ANOVA layout.

The **Standard Error of the Estimate (SEE)** measures the dispersion of the observed values around the regression line:

$$s_e = \sqrt{\text{MSE}} = \sqrt{\frac{\text{SSE}}{n-2}} \tag{10.10}$$

To test whether the independent variable has a significant linear relationship with the dependent variable, we conduct a hypothesis test on the slope coefficient:

$$H_0 : \beta_1 = 0 \quad \text{vs} \quad H_a : \beta_1 \neq 0$$

Table 10.1: ANOVA Table for Simple Linear Regression

Source	df	Sum of Squares	Mean Square	F-statistic
Regression	1	SSR	MSR = SSR/1	$F = \text{MSR}/\text{MSE}$
Residual	$n-2$	SSE	$\text{MSE} = \text{SSE}/(n-2)$	–
Total	$n-1$	SST	–	–

The test statistic follows a t-distribution with $n-2$ degrees of freedom:

$$t = \frac{\hat{\beta}_1 - 0}{s_{\hat{\beta}_1}} \tag{10.11}$$

where the standard error of the slope estimator $s_{\hat{\beta}_1}$ is:

$$s_{\hat{\beta}_1} = \frac{s_e}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}} \tag{10.12}$$

Example: Evaluating FIRN AG Stock Regression

An analyst regresses the excess returns of FIRN AG stock (Y) on the market risk premium (X) using 24 monthly observations. The regression output is shown below:

Source	df	SS	MS	F-statistic
Regression	1	0.029	0.029	63.81
Residual	22	0.010	0.000455	–
Total	23	0.039	–	–

Variable	Coefficient	Standard Error	t-statistic
Intercept	0.0077	0.0007	11.328
Market Premium	0.8260	0.1030	7.988

Calculations & Analysis:

1. **Goodness-of-Fit (R^2):**
$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{0.029}{0.039} = 0.7436\,(74.36\%)$$
Interpretation: 74.36% of the monthly excess returns variation of FIRN AG is explained by the market risk premium.

2. **Correlation Coefficient (r):**
$$r = \sqrt{0.7436} = 0.8623$$
Since the slope coefficient is positive (0.8260), the correlation between FIRN AG and the market is +0.8623.

3. **Standard Error of the Estimate (SEE):**
$$s_e = \sqrt{\text{MSE}} = \sqrt{0.000455} = 0.0213$$

4. **Significance of the Slope:** At a 5% significance level, the critical t-value with 22 degrees of freedom is 2.074. Since the calculated t-statistic for the slope is $7.988 > 2.074$, we reject the null hypothesis $H_0 : \beta_1 = 0$. The slope is statistically significant.

10.3 Using Linear Regression Models for Prediction

c | calculate and interpret predicted values, the standard error of the estimate, and prediction intervals for the dependent variable in a simple linear regression model and describe different functional forms

10.3.1 Prediction Using a Simple Regression

For a given value of the independent variable (X_p), the predicted value of the dependent variable ($\hat{Y}_p$) is:

$$\hat{Y}_p = \hat{\beta}_0 + \hat{\beta}_1 X_p \tag{10.13}$$

To capture the uncertainty of our prediction, we calculate the **Prediction Interval** for the dependent variable. The variance of the prediction error (s^2_{pred}) is:

$$s^2_{pred} = s^2_e \left[1 + \frac{1}{n} + \frac{(X_p - \bar{X})^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}\right] \tag{10.14}$$

The $1 - \alpha$ confidence level prediction interval for the estimated value is:

$$\hat{Y}_p \pm t_{crit} \times s_{pred} \tag{10.15}$$

where t_{crit} is the critical value of the t-distribution with $n - 2$ degrees of freedom at the significance level $\alpha/2$.

Exam Tip

Notice in Equation 10.14 that the variance of the prediction error increases as X_p moves further away from the sample mean $\bar{X}$. This means our predictions are much more precise when X_p is close to the average of our historical data, and become highly uncertain for extreme inputs!

10.3.2 Modeling Non-Linear Relationships

If the relationship between variables is non-linear, we can apply logarithmic transformations to achieve linearity. Table 10.2 summarizes the three primary logarithmic functional forms.

Table 10.2: Logarithmic Regression Functional Forms

Model	Regression Equation	Slope Coefficient Interpretation
Log-Lin	$\ln(Y_i) = \beta_0 + \beta_1 X_i + \epsilon_i$	A 1-unit change in X is associated with a constant percentage ($g = e^{\beta_1} - 1$) change in Y.
Lin-Log	$Y_i = \beta_0 + \beta_1 \ln(X_i) + \epsilon_i$	A 1% relative change in X is associated with an approximate $\beta_1/100$ unit change in Y.
Log-Log	$\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + \epsilon_i$	A 1% relative change in X is associated with an approximate β_1% relative change in Y (constant elasticity β_1).

*Làm thế nào để chọn đúng dạng hàm? 1. Sử dụng **Log-Lin** khi biến phụ thuộc tăng trưởng theo cấp số nhân (ví dụ: quy mô doanh thu của doanh nghiệp qua các năm). 2. Sử dụng **Lin-Log** khi tốc độ tăng trưởng giảm dần theo quy mô (hiệu ứng lợi ích cận biên giảm dần). 3. Sử dụng **Log-Log** để tìm độ co giãn của cầu theo giá hoặc tìm mức độ nhạy cảm không đổi giữa hai đại lượng tỷ lệ.*

Example: Log-Lin Regression of Swiss Company Cap on Revenue

An analyst collects a sample of 34 Swiss companies to evaluate the relationship between revenue per share (X, in CHF) and the natural logarithm of market capitalization ($Y = \ln(\text{Market Cap})$, in CHF billions). The OLS regression yields:

$$\ln(\text{Market Capitalization}_i) = 0.5987 + 0.2951 \times \text{Revenue per Share}_i$$

with $s_{\hat{\beta}_1} = 0.0077$.

Analysis:

1. **Interpretation of the Slope ($\hat{\beta}_1 = 0.2951$):** For every 1 CHF increase in revenue per share, the market capitalization increases by approximately 29.51%. Specifically, the exact relative change is:
$$e^{0.2951} - 1 \approx 0.3432\,(34.32\%)$$
2. **Hypothesis Testing:** To test if the relationship is statistically significant at the 1% significance level ($t_{crit} \approx 2.75$ with 32 df):
$$t = \frac{0.2951}{0.0077} = 38.32$$
Since $38.32 > 2.75$, we reject H_0. The slope is highly statistically significant.

10.4 The Capital Asset Pricing Model (CAPM)

d | calculate and interpret the variable estimates of the capital asset pricing model (CAPM)

CAPM là ứng dụng kinh điển của hồi quy tuyến tính đơn. Đường đặc trưng chứng khoán (SCL) hồi quy phần bù rủi ro tài sản ($R_i - R_f$) theo phần bù rủi ro thị trường ($R_m - R_f$). Hệ số góc chính là Beta (β), đo lường rủi ro hệ thống.

Cập nhật chương trình 2027

Ứng dụng hồi quy tuyến tính đơn biến để ước lượng hệ số Beta của cổ phiếu trong khung lý thuyết CAPM là phần ứng dụng thực tế được nhấn mạnh hơn trong chương trình 2027. Học viên cần hiểu rõ cách lập mô hình hồi quy và diễn giải các hệ số.

10.4.1 CAPM Regression Framework

The CAPM asserts that the expected excess return on an asset is linearly related to the market risk premium. We express it as a simple linear regression:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \epsilon_{i,t} \tag{10.16}$$

where:

- $R_{i,t} - R_{f,t}$ is the excess return of asset i over the risk-free rate at time t (dependent variable Y).
- $R_{m,t} - R_{f,t}$ is the market risk premium at time t (independent variable X).
- β_i is the systematic risk (or market beta) of the asset, reflecting its sensitivity to market movements.
- α_i is the intercept, representing the abnormal return (Jensen's alpha). Under market efficiency, the expected value of α_i is zero.

Case Study: Calculating LVMH's Beta

An analyst estimates the beta of LVMH Moët Hennessy Louis Vuitton (LVMH) using 104 weeks of historical data. The CAC 40 Index excess return is used as the proxy for the market risk premium (X), and LVMH's excess return is the dependent variable (Y). The OLS regression yields the following results:

Source	df	Sum of Squares	Mean Square
Regression	1	0.108622	0.108622
Residual	102	0.041994	0.000412
Total	103	0.150616	–

Parameter	Coefficient	Standard Error	t-statistic
Intercept (α)	0.0010	0.0020	0.5250
Market Slope (β)	1.3684	0.0840	16.2430

Interpretation & Key Estimates:

1. **Estimated Beta ($\hat{\beta}_{LVMH} = 1.3684$):** LVMH has a beta of approximately 1.37. Since $\beta > 1.0$, LVMH is more volatile than the market CAC 40. A 1% increase in the CAC 40 market risk premium is expected to lead to a 1.37% increase in the excess return of LVMH.
2. **Statistical Significance of Beta:** The calculated t-statistic for the slope coefficient is 16.243. Compared to the critical t-value at the 5% level of significance ($t_{crit} \approx 1.96$), $16.243 > 1.96$. Thus, LVMH's beta is statistically significant.
3. **Model R-squared (R^2):**
$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{0.108622}{0.150616} = 0.7212\,(72.1\%)$$
This indicates that 72.1% of the variation in LVMH's weekly excess returns is explained by the market risk premium. The remaining 27.9% is due to firm-specific (unsystematic) factors.
4. **Intercept Alpha ($\hat{\alpha} = 0.0010$):** The intercept represents the abnormal excess return of LVMH. The calculated t-statistic is $0.525 < 1.96$, showing that the alpha is not statistically different from zero, which is consistent with the CAPM.

Module 11

Introduction to Financial Data Science

LEARNING OUTCOMES	
a	describe how big data, machine learning, and artificial intelligence are used in financial data science, fintech, and investment management

11.1 Financial Data and Big Data

a	describe how big data, machine learning, and artificial intelligence are used in financial data science, fintech, and investment management

Nhận xét: Bên cạnh dữ liệu định lượng, dữ liệu định tính (qualitative) và dữ liệu thay thế (alternative data) ngày càng quan trọng. Nhờ số hóa, ta có thể chuyển tin tức, mạng xã hội thành số liệu phân tích để tìm lợi thế cạnh tranh.

11.1.1 Quantitative vs. Qualitative Data

The investment decision-making process analyzes both quantitative and qualitative data:

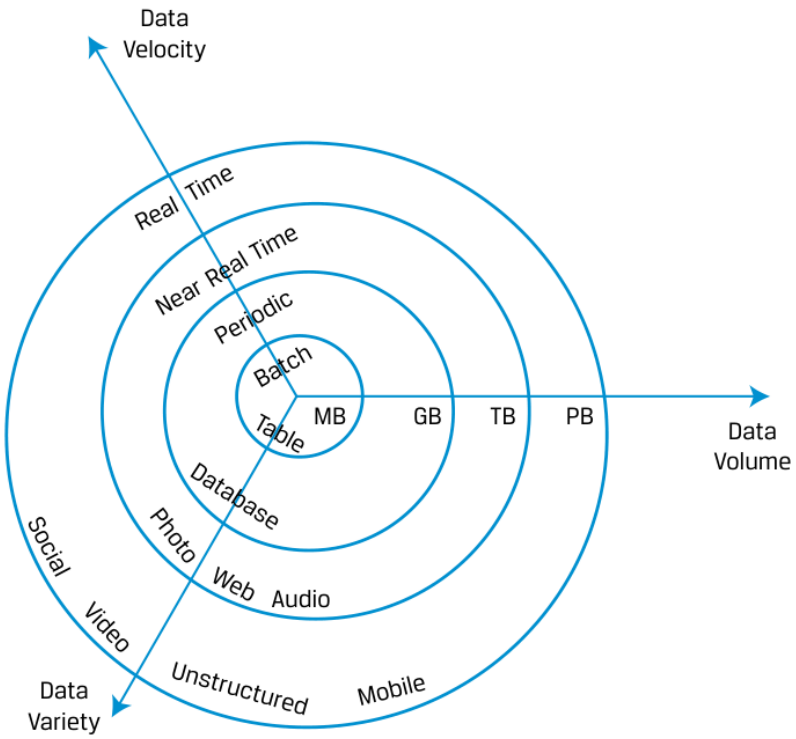
- **Quantitative data** describes numeric information, such as prices, returns, interest rates, currency exchange rates, economic indicators, and accounting numbers, that can be analyzed using statistical methods.
- **Qualitative data** describes non-numeric information, such as market sentiment, management quality, industry trends, environmental factors (ESG), and geopolitical risks.

With the advance of digitization, qualitative data are increasingly captured digitally and processed algorithmically (e.g., using sentiment analysis), allowing for the integration of qualitative insights with quantitative analysis.

11.1.2 Characteristics of Big Data (The 4 Vs)

Datasets available for analysis are growing rapidly. Big data are characterized by four key dimensions:

- **Volume:** The vast amount of data collected (ranging from megabytes to gigabytes, terabytes, and petabytes), representing millions or billions of observations.
- **Velocity:** The speed at which data are communicated and updated. Real-time or low-latency data feeds have become the standard in financial applications.
- **Variety:** The diversity of data formats and sources, which can be structured, semi-structured, or unstructured.
- **Veracity:** The credibility, authenticity, and reliability of different data sources. Veracity is critical in big data because of the varying quality of large, non-traditional datasets.



Data	Volume Key	Bytes of Information
MB	Megabyte	One Million
GB	Gigabyte	One Billion
TB	Terabyte	One Trillion
PB	Petabyte	One Quadrillion

Figure 11.1: Characteristics of Big Data: Volume, Variety, and Velocity

11.1.3 Data Types and Sources

Data can be classified into three types based on structure:

- **Structured data:** Organized in tables (rows and columns) and commonly stored in databases where fields are defined (e.g., SQL tables, CSV files).
- **Unstructured data:** Disparate, unorganized data that do not fit into traditional tables. Examples include social media posts, emails, voice recordings, pictures, and video feeds.
- **Semi-structured data:** Blends structured and unstructured elements. The data do not fit into traditional tables but contain tags or markers (e.g., HTML/XML code, JSON files, tagged financial news).

Table 11.1 outlines common sources of big data in finance:

Table 11.1: Examples of Big Data Sources in Finance

Source	Examples
Financial markets	Equity, fixed income, futures, options, other derivatives, and commodities.
Businesses	Corporate financial statements, commercial transactions, credit card purchases, and supply chain data.
Governments	Trade, economic indicators, regulatory filings, taxation, and payroll data.
Individuals	Personal websites, online product reviews, internet search logs, email, and social media posts.
Sensors	Satellite imagery, aircraft tracking, shipping cargo data, and vehicle traffic patterns.
Internet of Things	Data from smart buildings (climate control, energy usage) and connected commercial devices.

11.1.4 Alternative Data and Risks

Alternative data refers to information collected from non-traditional sources, such as satellite imagery, web traffic, social media sentiment, mobile device geolocation, and retail point-of-sale scanners.

- **Data generated by individuals:** Personal activities like social media clicks, product reviews, and web searches. This data tends to be unstructured and qualitative.
- **Data generated by business processes:** Direct transactions like credit card purchases or point-of-sale scanner data. This structured data acts as a leading indicator of performance, whereas traditional corporate filings are lagging indicators.
- **Data generated by sensors:** Satellite imagery (e.g., counting cars in retail parking lots to predict earnings or monitoring flare gas at oil rigs) and Internet of Things (IoT) streams. It is largely unstructured and grows exponentially.

Lưu ý pháp lý: Web scraping (khai thác dữ liệu web) có thể vi phạm bảo mật thông tin (GDPR) hoặc điều khoản dịch vụ. Khi dùng dữ liệu phi chính thống, tính tuân thủ pháp lý của nguồn dữ liệu là yếu tố sống còn.

11.2 Financial Data Science and Fintech

11.2.1 Definitions

- **Data science** is an interdisciplinary field combining statistical methods, mathematical models, computational tools, and domain-specific knowledge to clean, process, and interpret big data.
- **Financial data science** applies data science principles specifically to financial datasets (prices, interest rates, returns, accounting statements, and macroeconomic variables) to support investment, portfolio management, trading, and risk decisions.

11.2.2 Unique Characteristics of Financial Data

Financial data exhibits unique challenges that require specialized analytical tools:

1. **High volume and speed:** Market transactions occur in microseconds, producing a massive continuous stream of tick data.
2. **Sequential and time-dependent structure:** Observations are chronologically ordered, requiring time-series analysis rather than standard independent assumptions.
3. **Noise and non-stationarity:** Financial data are highly noisy (containing random fluctuations) and non-stationary (statistical properties like mean and variance change over time).
4. **Interdependencies and non-linearity:** Financial variables have complex, non-linear relationships. Advanced tools like **Copulas** (mathematical functions joining multivariate distributions to capture dependency structures) and neural networks are used.
5. **Seasonal and cyclic patterns:** Recurring periods, such as quarterly corporate earnings releases or macro business cycles.
6. **Extreme events and fat tails:** Returns exhibit fat-tailed distributions, meaning extreme market crashes or jumps occur far more frequently than predicted by a standard Gaussian distribution.
7. **Data sparsity and missing values:** Emerging markets or private assets often lack complete historical data, requiring robust imputation methods to avoid estimation bias.

11.2.3 Fintech Evolution

Fintech (financial technology) represents technological innovation in the design and delivery of financial services and products. It has evolved through several stages:

- **Early stage:** Routine automation and simple data processing (e.g., calculating bookkeeping records).
- **Intermediate stage:** Rule-based decision execution systems ("if-then" systems).
- **Advanced stage:** Complex algorithmic and machine learning decision-making (e.g., robo-advisors that automatically construct, balance, and manage diversified portfolios based on client risk profiles and target return requirements).

11.3 Machine Learning and Artificial Intelligence

11.3.1 Dataset Splitting and Time-Series Validation

In machine learning (ML), datasets are divided into three distinct subsets:

- **Training dataset:** The largest portion (typically 60%–80%) used by the algorithm to identify historical relationships between inputs and targets.
- **Validation dataset:** Used to validate and tune the model's parameters and refine its performance (10%–20%).
- **Testing dataset:** Used at the very end to evaluate the model's performance on unseen, out-of-sample data (10%–20%).

For cross-sectional data, data is split randomly. This randomization prevents bias and helps prevent **data leakage** (when information from outside the training set accidentally influences the model). For time-series data, simple randomization cannot be used because it disrupts the chronological sequence and introduces forward data leakage (predicting the past using future data). Instead, analysts use:

- **Time-based splits:** Earlier data is allocated to the training set and subsequent chronological blocks are used for validation and testing.
- **Rolling-window validation:** Shifts a fixed-size historical window forward in time, training on one window and validating on the next sequential period.



Figure 11.2: Time-Based Split vs. Rolling-Window Validation

11.3.2 Overfitting vs. Underfitting

- **Overfitting:** The model learns the training data too precisely, capturing random noise as if it were a true parameter. This leads to extremely low training error but high out-of-sample test error ($Error_{\text{train}} \ll Error_{\text{test}}$).
- **Underfitting:** The model is too simplistic and treats true relationships as noise, failing to learn the underlying structure in the training data.

11.3.3 Machine Learning Approaches

Machine learning is classified into four main approaches, summarized in Table 11.2:

11.3.4 Neural Networks and Generative Models

Neural networks are computational models inspired by biological brains. A neural network consists of layers of interconnected nodes (neurons):

- **Input layer:** Receives the raw input features (e.g., historical returns, financial ratios).
- **Hidden layers:** Intermediate layers where non-linear data transformations occur. Multiple hidden layers constitute a "deep" neural network.
- **Output layer:** Provides the final classification or predicted continuous value.

Training is completed using **backpropagation**:

1. **Forward pass:** Data flows through the network to generate a prediction.
2. **Loss calculation:** Computes the error (loss) between the predicted and actual target.
3. **Backward pass (Backpropagation):** Calculates gradients of the loss function with respect to the network weights.
4. **Weight update:** Weights are updated slightly in the opposite direction of the gradient using **gradient descent**.

Table 11.2: Comparison of Machine Learning Approaches

Approach	Description	Applications	Advantages	Limitations
Supervised learning	Models learn from labeled historical training data where target output is known.	Spam detection Stock return prediction Default classification	Highly accurate Well-understood methods	Labeled data is costly Fails to generalize to new scenarios
Unsupervised learning	Models analyze unlabeled data to discover hidden structures and groupings.	Customer segmentation Anomaly detection Dimension reduction	Discovers unexpected patterns Reduces data dimensionality	Subjective evaluation Difficult to interpret results
Reinforcement learning	Models learn by trial-and-error interaction with an environment to maximize rewards.	Portfolio optimization Algorithmic execution	Good for sequential decisions Adapts to dynamics	High training time Sensitive to reward design
Deep learning	Models use multilayered artificial neural networks to extract abstract patterns.	Image recognition NLP	Automatically extracts features Models non-linear structures	"Black box" nature Requires massive datasets

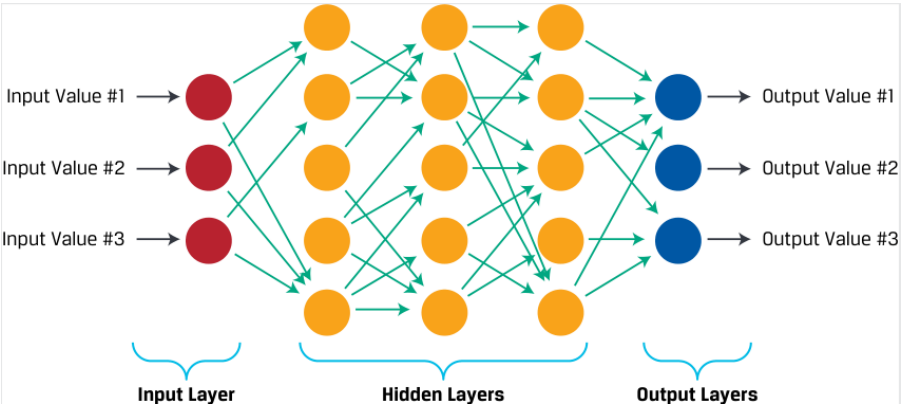


Figure 11.3: Structure of an Artificial Neural Network

This process is repeated over multiple cycles (**epochs**) until the network converges.

Two advanced generative neural network architectures are:

- **Generative Adversarial Networks (GANs):** Combines a generator network (creates synthetic data) and a discriminator network (evaluates authenticity) in an adversarial loop. Used to generate synthetic financial scenarios for stress-testing.
- **Variational Autoencoders (VAEs):** Compresses inputs into a latent, lower-dimensional space (encoder) and reconstructs them (decoder). Useful for dimensionality reduction and anomaly detection.

11.4 Natural Language Processing and Generative AI

11.4.1 Natural Language Processing (NLP)

- **Text analytics** extracts information and identifies patterns from unstructured text- or voice-based datasets (filings, transcripts, emails, news, etc.).
- **Natural language processing (NLP)** is a subfield at the intersection of computer science, AI, and linguistics focused on analyzing human language.

Common applications in finance include sentiment analysis (e.g., parsing earnings call transcripts to assign a positive/negative sentiment rating to detect subtle manager outlooks before analyst rating updates) and monitoring central bank communication (parsing Fed or ECB statements to extract tone shifts on interest rates).

11.4.2 Large Language Models (LLMs)

Large Language Models (LLMs), such as GPT models, represent a significant advancement in generative AI.

- **Architecture:** Rely on **transformers** to process and capture long-range contextual relationships within textual sequences.
- **Huấn luyện song miền (Dual-domain training):** Sourced via self-supervised learning on massive general corpora (to learn grammar, semantics) and specialized financial corpora (filings, earnings call transcripts to master domain jargon).
- **Tinh chỉnh (Fine-tuning):** Adapted via transfer learning for specialized applications (e.g., text generation, report summarization, financial sentiment analysis).
- **Rủi ro và hạn chế (Risks):** LLMs excel at language prediction but are prone to **hallucinations** (generating false information) and lack numerical calculation capabilities. Over-reliance can result in strategic and regulatory errors. A hybrid model combining automated outputs with human review is recommended.

11.4.3 Generative AI vs. Traditional Simulations

Generative AI creates new content or scenarios based on representations learned during training. Unlike Monte Carlo simulations (which assume predefined parametric distributions) or bootstrapping (which simply shuffles or resamples historical records), Generative AI learns complex underlying data structures to produce realistic, non-replication-constrained synthetic scenarios.

Appendices

Appendix A

Index Definition and Calculation

A.1 Công thức chung tính Index Value

Index là một chỉ số tượng trưng cho quy mô của thị trường chứng khoán. Nhìn vào Index, nhà đầu tư nhận biết được quy mô của thị trường đã tăng trưởng bao nhiêu, kể từ khi thị trường được thành lập. Để tính Index, người ta xây dựng một rổ cổ phiếu với số lượng giả định n_i cho mỗi cổ phiếu i trong rổ, và tính tổng giá trị rổ, sau đó quy chuẩn thành điểm chỉ số bằng cách chia cho một divisor D_t.

Công thức tổng quát nhất của một chỉ số là:

$$V_t = \frac{A_t}{D_t}$$

Trong đó:

$$V_t = \text{giá trị chỉ số tại thời điểm } t$$

$$A_t = \text{aggregate index value, tức giá trị thô của rổ chỉ số}$$

$$D_t = \text{divisor, dùng để chuẩn hóa giá trị thô thành điểm chỉ số}$$

Với cách viết tổng quát:

$$A_t = \sum_{i=1}^{N} n_{i,t} P_{i,t}$$

Trong đó:

$$n_{i,t} = \text{số lượng cổ phiếu giả định mà rổ chỉ số nắm giữ}$$

$$P_{i,t} = \text{giá cổ phiếu } i \text{ tại thời điểm } t$$

Liên hệ với Return:

$$V_t = \frac{\sum_{i=1}^{N} n_{i,t} P_{i,t}}{D_t} = V_{t-1} \times \sum_{i=1}^{N} \times \frac{n_i \times P_{i,t-1}}{D_t V_{t-1}} \left(\frac{P_{i,t}}{P_{i,t-1}} \right) = V_{t-1} \times \sum_{i=1}^{N} w_{i,t-1} \left(\frac{P_{i,t}}{P_{i,t-1}} \right)$$

$$\text{Do } D_t V_{t-1} = A_{t-1} \Rightarrow \frac{n_i \times P_{i,t-1}}{D_t V_{t-1}} = \frac{n_i \times P_{i,t-1}}{A_{t-1}} = w_{i,t-1}$$

Index t = Index (t-1)×Tổng của (Weight của cổ phiếu i tại thời điểm t-1 nhân với Price Relative của cổ phiếu i)

Lưu ý, weight thường được lấy ở đầu kỳ, tức là: $w_{i,t-1}$, không phải weight cuối kỳ.

A.2 Cách tính Return

Return của từng cổ phiếu có thể được tính theo hai cách.

A.2.1 Price Return

Cách này chỉ phản ánh biến động giá cổ phiếu.

$$R_{i,t}^{price} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

nên:

$$V_t = V_{t-1} \times \sum_{i=1}^{N} w_{i,t-1}(1 + R_{i,t}) = V_{t-1} \times \left(1 + \sum_{i=1}^{N} w_{i,t-1} R_{i,t}\right) = V_{t-1}(1 + R_{index,t})$$

A.2.2 Total Return

$$R_{i,t}^{total} = \frac{P_{i,t} - P_{i,t-1} + Dividend_{i,t}}{P_{i,t-1}}$$

Cách này phản ánh cả biến động giá và cổ tức.

$$V_{TR,t} = V_{TR,t-1} \times \sum_{i=1}^{N} w_{i,t-1} \left(\frac{P_{i,t} + Dividend_{i,t}}{P_{i,t-1}}\right)$$

Hoặc nếu dùng adjusted price đã phản ánh cổ tức:

$$V_{TR,t} = V_{TR,t-1} \times \sum_{i=1}^{N} w_{i,t-1} \left(\frac{Adjusted\ Price_{i,t}}{Adjusted\ Price_{i,t-1}}\right)$$

A.3 Các loại weight của index

A.3.1 Market-cap Weight

$$w_{i,t}^{M} = \frac{Q_{i,t} P_{i,t}}{\sum_{j=1}^{N} Q_{j,t} P_{j,t}}$$

Nếu có điều chỉnh free float:

$$w_{i,t}^{M} = \frac{Q_{i,t} f_{i,t} P_{i,t}}{\sum_{j=1}^{N} Q_{j,t} f_{j,t} P_{j,t}}$$

$f_{i,t}$ = tỷ lệ free float của cổ phiếu i

Cổ phiếu nào có vốn hóa lớn hơn thì có trọng số lớn hơn trong index.

A.3.2 Price Weight

$$w_{i,t}^{P} = \frac{P_{i,t}}{\sum_{j=1}^{N} P_{j,t}}$$

Cổ phiếu nào có giá cao hơn thì có trọng số lớn hơn.

A.3.3 Equal Weight

Tại ngày rebalance r, mỗi cổ phiếu có cùng trọng số mục tiêu:

$$w_{i,r}^{target} = \frac{1}{N}$$

Ví dụ, nếu index có 5 cổ phiếu thì $w_i = 20\%$

Sau ngày rebalance, giá cổ phiếu biến động khác nhau nên weight thực tế sẽ lệch khỏi mức target.

A.3.4 Fundamental Weight

Với fundamental-weighted index, weight được tính theo một chỉ tiêu cơ bản F_i, ví dụ: doanh thu; lợi nhuận; book value; cash flow; dividend; hoặc một fundamental score tổng hợp.

Công thức target weight là:

$$w_{i,r}^{target} = \frac{F_i}{\sum_{j=1}^{N} F_j}$$

A.4 Diễn giải Divisor

Công thức chung:

$$V_t = \frac{A_t}{D_t}$$

Nếu không có sự kiện kỹ thuật:

$$D_t = D_{t-1}$$

Khi đó:

$$V_t = V_{t-1} \times \frac{A_t}{A_{t-1}}$$

Tức là index tăng hoặc giảm theo tốc độ tăng hoặc giảm của aggregate index value.
Khi có sự kiện kỹ thuật làm A_t thay đổi cơ học, divisor được điều chỉnh để index không bị nhảy giả tạo.
Công thức điều chỉnh divisor là:

$$D_t = D_{t-1} \times \frac{A_t^*}{A_{t-1}}$$

Trong đó:

$$A_{t-1} = \text{aggregate index value trước sự kiện}$$

$A_t^* =$ aggregate index value ngay sau sự kiện kỹ thuật, nhưng chưa tính biến động giá thị trường thật

Logic xuất phát từ yêu cầu:

$$V_t^* = V_{t-1}$$

Tức là:

$$\frac{A_t^*}{D_t} = \frac{A_{t-1}}{D_{t-1}}$$

Suy ra:

$$D_t = D_{t-1} \times \frac{A_t^*}{A_{t-1}}$$

A.5 Cách xác định số lượng cổ phiếu mà rổ index nắm giữ

Giá trị rổ cổ phiếu phụ thuộc vào số lượng cổ phiếu giả định n_i của từng cổ phiếu thành phần.

$$A_t = \sum_{i=1}^{N} n_{i,t} P_{i,t}$$

A.5.1 Market-cap-weighted Index

$$n_{i,t} = Q_{i,t}$$

Nếu có free float:

$$n_i = Q_i f_i$$

A.5.2 Price-weighted Index

Với price-weighted index:

$$n_i = 1$$

A.5.3 Equal-weighted Index

Tại ngày rebalance r:

$$w_{i,r}^{target} = \frac{1}{N}$$

Số lượng cổ phiếu giả định được tính bằng:

$$n_{i,r} = \frac{A_r \times w_{i,r}^{target}}{P_{i,r}} = \frac{A_r \times \frac{1}{N}}{P_{i,r}}$$

A.5.4 Fundamental-weighted Index

Tính target weight theo yếu tố cơ bản:

$$w_{i,r}^{target} = \frac{F_i}{\sum_{j=1}^{N} F_j}$$

Sau đó tính số lượng cổ phiếu giả định:

$$n_{i,r} = \frac{A_r \times w_{i,r}^{target}}{P_{i,r}}$$

A.5.5 A_r lấy từ đâu?

Với equal-weighted index và fundamental-weighted index, A_r không phải là tổng vốn hóa thật ngoài thị trường, mà là quy mô danh nghĩa của rổ index.

Nếu index mới được lập:

$$A_0 = V_0 \times D_0$$

Nếu index đã tồn tại và đang rebalance:

$$A_r^- = \sum_{i=1}^{N} n_{i,r^-} P_{i,r}$$

Tức là tổng giá trị rổ ngay trước rebalance.

Sau đó thường đặt:

$$A_r^+ = A_r^-$$

để index không nhảy điểm vì rebalance.

Rồi tính lại:

$$n_{i,r^+} = \frac{A_r^- \times w_{i,r}^{target}}{P_{i,r}}$$

A.6 Rebalance và Reconstitution

A.6.1 Rebalance

Rebalance là điều chỉnh lại tỷ trọng của các cổ phiếu trong index về mức target.

Sau một thời gian, do giá cổ phiếu biến động khác nhau, weight thực tế sẽ khác weight target.

Công thức weight thực tế là:

$$w_{i,t}^{actual} = \frac{n_{i,r} P_{i,t}}{\sum_{j=1}^{N} n_{j,r} P_{j,t}}$$

Với equal-weighted index, tại ngày rebalance:

$$w_i^{target} = \frac{1}{N}$$

Với fundamental-weighted index, target weight dựa trên fundamental factor:

$$w_i^{target} = \frac{F_i}{\sum F_i}$$

Đến kỳ rebalance, index cập nhật F_i, tính lại target weight mới, rồi điều chỉnh n_i.

Khi nào cần rebalance?

Tùy methodology của index.

Equal-weighted index thường rebalance theo kỳ: monthly, quarterly, semiannual, annual.

Fundamental-weighted index thường rebalance theo kỳ: quarterly, semiannual, annual.

Ngoài lịch định kỳ, rebalance cũng có thể xảy ra khi có sự kiện đặc biệt, chẳng hạn: thêm hoặc loại cổ phiếu; hủy niêm yết; sáp nhập; chia tách hoặc gộp cổ phiếu; corporate actions lớn; thay đổi số lượng cổ phiếu trong index; vi phạm điều kiện index; áp dụng weight cap.

A.6.2 Reconstitution

Reconstitution là thay đổi thành phần index. Ví dụ: loại cổ phiếu A khỏi index; thêm cổ phiếu E vào index; cập nhật lại danh sách cổ phiếu đủ điều kiện.

Cách nhớ:

$$\text{Rebalance} = \text{chỉnh tỷ trọng}$$

$$\text{Reconstitution} = \text{chỉnh danh sách cổ phiếu}$$

Trên thực tế, reconstitution thường kéo theo rebalance, vì khi danh sách cổ phiếu thay đổi, ta phải tính lại: w_i^{target} và n_i

Cách xử lý reconstitution

Trước sự kiện:

$$V^- = \frac{A^-}{D^-}$$

Sau khi thay đổi thành phần index, tính aggregate value mới:

$$A^+ = \sum_{i=1}^{N} n_i^+ P_i$$

Để index không nhảy giả tạo, divisor mới là:

$$D^+ = \frac{A^+}{V^-} = D^- \times \frac{A^+}{A^-}$$

A.7 Kết luận

Có thể nhớ toàn bộ logic bằng bốn lớp.

Lớp 1: Index Level

$$V_t = \frac{A_t}{D_t}$$

Lớp 2: Aggregate Index Value

$$A_t = \sum_{i=1}^{N} n_{i,t} P_{i,t}$$

Cách chọn n_i quyết định index là market-cap weighted, price weighted, equal weighted hay fundamental weighted.

Lớp 3: Return

$$R_{index,t} = \sum_{i=1}^{N} w_{i,t-1} R_{i,t}$$

$$V_t = V_{t-1}(1 + R_{index,t})$$

Hoặc:

$$V_t = V_{t-1} \times \sum_{i=1}^{N} w_{i,t-1} \left(\frac{P_{i,t}}{P_{i,t-1}} \right)$$

Lớp 4: Divisor, Rebalance và Reconstitution

$$D_t = D_{t-1} \times \frac{A_t^*}{A_{t-1}}$$

Divisor giúp index không bị nhảy điểm giả tạo.
Rebalance đưa weight thực tế về weight target.
Reconstitution thay đổi danh sách cổ phiếu, và thường kéo theo rebalance hoặc divisor adjustment.

Appendix B

Proof of Statistical Distributions Formula

B.1 Discrete Uniform Distribution

Giả sử biến ngẫu nhiên rời rạc X nhận các giá trị nguyên trong đoạn $[a, b]$ với xác suất bằng nhau.

- Số lượng giá trị có thể nhận là $N = b - a + 1$
- Xác suất X nhận một giá trị bất kỳ trong đoạn là: $P(X = k) = \frac{1}{n}$ với mọi $k \in \{a, a + 1, ..., b\}$

Kỳ vọng

$$E(X) = \sum_{k=a}^{b} k \cdot P(X = k) = \sum_{k=a}^{b} k \cdot \frac{1}{n} = \sum_{k=a}^{b} k$$

Do $\sum_{k=a}^{b} k$ là tổng của cấp số cộng từ a đến b gồm n phần tử, nên:

$$\sum_{k=a}^{b} = \frac{n(a+b)}{2} \Rightarrow E(X) = \frac{1}{n} \cdot \frac{n(a+b)}{2} = \frac{a+b}{2}$$

Phương sai

Để tính phương sai, ta sử dụng tính chất tịnh tiến: Phương sai không thay đổi khi ta dịch chuyển toàn bộ tập dữ liệu. Ta đặt biến mới $Y = X - a + 1$ (gọi là biến chuẩn hóa). Khi đó:

- Y sẽ nhận các giá trị từ 1 đến $b - a + 1$ (tức là từ 1 đến N)
- $Var(X) = Var(Y) = E(Y^2) - [E(Y)]^2$

$$E(Y) = \frac{1+N}{2} \Rightarrow [E(Y)]^2 = \frac{(N+1)^2}{4}$$

$$E(Y^2) = \sum_{i=1}^{N} i^2 \cdot \frac{1}{N} = \frac{1}{N} \sum_{i=1}^{N} i^2 = \frac{1}{N} \cdot \frac{N(N+1)(2N+1)}{6} = \frac{(N+1)(2N+1)}{6}$$

Suy ra:

$$Var(Y) = \frac{(N+1)(2N+1)}{6} - \frac{(N+1)^2}{4} = (N+1) \cdot \frac{4N + 2 - 3N - 3}{12} = \frac{(N+1)(N-1)}{12} = \frac{N^2 - 1}{12}$$

Do $N = b - a + 1$ nên ta có:

$$Var(X) = \frac{(b-a+1)^2 - 1}{12}$$

B.2 Binomial Distribution

Phân phối Nhị thức đếm số lần thành công trong n lần thử độc lập. Phân phối này được xây dựng dựa trên phân phối Bernoulli, với mỗi lần thử chỉ có 2 kết quả: Thành công (xác suất p) hoặc Thất bại (xác suất $1-p$).

Giải nghĩa các thành phần trong công thức (6.12):

- $\binom{n}{x} = \frac{n!}{x!(n-x)!}$: Tổ hợp chập x của n, đại diện cho số cách sắp xếp x lần thành công trong n lượt thử.
- p^x: Xác suất xảy ra đúng x lần thành công.
- $(1-p)^{n-x}$: Xác suất xảy ra đúng $n-x$ lần thất bại còn lại.

Để tránh các công thức tính toán phức tạp, ta xây dựng biến nhị thức X thực chất là tổng của n phép thử Bernoulli độc lập với nhau (ký hiệu là $X_1, X_2, ..., X_n$). Trong đó mỗi X_i chỉ nhận giá trị 1 (thành công) hoặc 0 (thất bại).

Đối với từng biến độc lập X_i:

- $E(X_i) = 1 \cdot p + 0 \cdot (1-p) = p$
- $E(X_i^2) = 1^2 \cdot p + 0^2 \cdot (1-p) = p$
- $Var(X_i) = E(X_i^2) - [E(X_i)]^2 = p - p^2 = p(1-p)$

Vì $X = X_1 + X_2 + ... + X_n$, áp dụng tính chất của kỳ vọng và phương sai cho các biến độc lập:

Kỳ vọng

$E(X) = E(X_1) + E(X_2) + ... + E(X_n) = np$

Phương sai

$Var(X) = Var(X_1) + Var(X_2) + ... + Var(X_n) = np(1-p)$

B.3 Poisson Distribution

PMF

Giả sử $X \sim \text{Bin}(n,p)$ với PMF là $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$.

Poisson được xây dựng để đếm số lần sự kiện xảy ra trong một khoảng thời gian chia nhỏ thành n sự kiện và trung bình có λ sự kiện xảy ra trong khoảng thời gian đó.

Như vậy xác suất của sự kiện sẽ là $p = \frac{\lambda}{n}$, hay $\lambda = np$. Ta xét khi $n \to \infty$ và $p \to 0$ sao cho $\lambda = np$ không đổi. Suy ra

$$P(X=k) = \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}.$$

Khai triển tổ hợp

$$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!},$$

Do đó

$$\binom{n}{k} \left(\frac{\lambda}{n}\right)^k = \frac{\lambda^k}{k!} \cdot \frac{n(n-1)\cdots(n-k+1)}{n^k}.$$

Ta có

$$\lim_{n\to\infty} \frac{n(n-1)\cdots(n-k+1)}{n^k} = 1,$$

nên

$$\lim_{n\to\infty} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k = \frac{\lambda^k}{k!}.$$

Xét thành phần còn lại,

$$\left(1 - \frac{\lambda}{n}\right)^{n-k} = \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}.$$

Sử dụng

$$\lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^{n}=e^{-\lambda},$$

và

$$\lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^{-k}=1,$$

nên

$$\lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^{n-k}=e^{-\lambda}.$$

Kết hợp lại ta được,

$$\begin{aligned}P(X=k)&=\lim_{n\to\infty}\binom{n}{k}\left(\frac{\lambda}{n}\right)^{k}\left(1-\frac{\lambda}{n}\right)^{n-k}\\&=\frac{\lambda^{k}}{k!}e^{-\lambda}.\end{aligned}$$

Điểm đặc biệt nhất của Poisson là Kỳ vọng đúng bằng Phương sai ($\mu=\sigma^2=\lambda$).

Kỳ vọng

Giả sử $X\sim\text{Poisson}(\lambda)$, với PMF

$$P(X=k)=\frac{e^{-\lambda}\lambda^{k}}{k!},\qquad k=0,1,2,\dots$$

Ta có,

$$E(X)=\sum_{k=0}^{\infty}kP(X=k)=e^{-\lambda}\sum_{k=1}^{\infty}k\frac{\lambda^{k}}{k!}.$$

Giản ước cho k, suy ra

$$E(X)=e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k}}{(k-1)!}.$$

Đưa 1 hệ số λ ra ngoài,

$$=\lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!}.$$

Đặt $j=k-1$.

$$E(X)=\lambda e^{-\lambda}\sum_{j=0}^{\infty}\frac{\lambda^{j}}{j!}.$$

Sử dụng khai triển Taylor của hàm mũ,

$$e^{\lambda}=\sum_{j=0}^{\infty}\frac{\lambda^{j}}{j!},$$

Ta có

$$E(X)=\lambda e^{-\lambda}e^{\lambda}=\boxed{\lambda}.$$

Phương sai

Đối với phương sai, ta sử dụng công thức

$$Var(X) = E(X^2) - [E(X)]^2,$$

Trước tiên,

$$E(X^2) = E[X(X-1)] + E(X).$$

Với giá trị kỳ vọng thứ hai, ta có

$$E[X(X-1)] = e^{-\lambda} \sum_{k=2}^{\infty} k(k-1) \frac{\lambda^k}{k!}.$$

Giản ước cho $k(k-1)$, suy ra

$$E[X(X-1)] = e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!}.$$

Đưa hệ số λ^2 ra ngoài,

$$= \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!}.$$

Đặt $j = k - 2$.

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2.$$

Suy ra,

$$E(X^2) = \lambda^2 + \lambda.$$

$$\begin{aligned} Var(X) &= E(X^2) - [E(X)]^2 \\ &= (\lambda^2 + \lambda) - \lambda^2 \\ &= \boxed{\lambda}. \end{aligned}$$

Hay,

$$\boxed{E(X) = Var(X) = \lambda.}$$

B.4 Continuous Uniform Distribution

Giả sử biến ngẫu nhiên liên tục X nhận giá trị trong đoạn $[a, b]$ với xác suất bằng nhau, tức là X có phân phối đều liên tục (Continuous Uniform Distribution):

$$X \sim U(a, b),$$

với PDF:

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

Kỳ vọng

Theo định nghĩa,

$$E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx.$$

$f(x) = 0$ với những giá trị ngoài khoảng $[a, b]$,

$$E(X) = \frac{1}{b-a} \int_a^b x\, dx. = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}.$$

Phương sai

$$Var(X) = E(X^2) - [E(X)]^2,$$

Theo định nghĩa,

$$E(X^2) = \frac{1}{b-a}\int_a^b x^2\,dx.$$

Do đó,

$$E(X^2) = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{b^3-a^3}{3(b-a)}.$$

$$\boxed{E(X^2) = \frac{a^2+ab+b^2}{3}.}$$

Suy ra,

$$\begin{aligned} Var(X) &= \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{4(a^2+ab+b^2) - 3(a+b)^2}{12} \\ &= \frac{a^2-2ab+b^2}{12} \\ &= \boxed{\frac{(b-a)^2}{12}.} \end{aligned}$$

Tóm lại,

$$\boxed{E(X) = \frac{a+b}{2}, \qquad Var(X) = \frac{(b-a)^2}{12}.}$$

B.5 Logistic Distribution

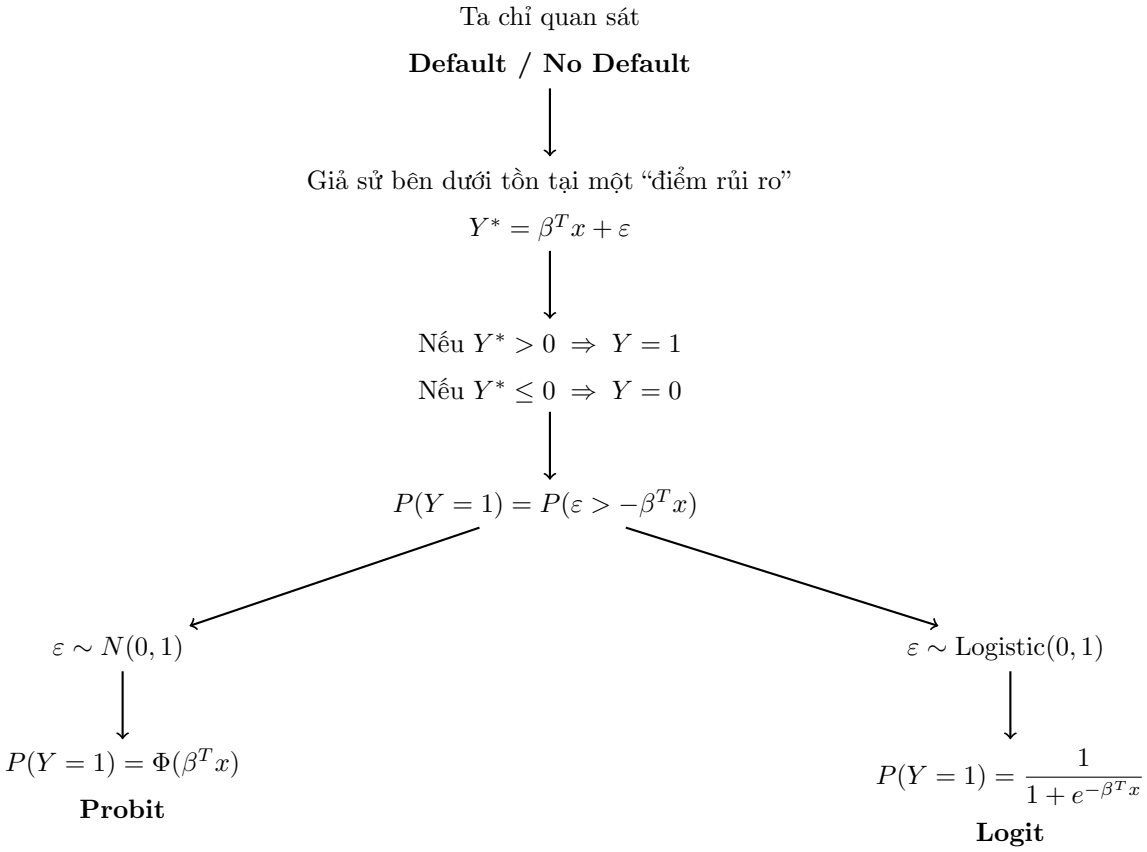


Figure B.1: Logic hình thành Probit và Logit từ mô hình biến tiềm ẩn (Latent Variable Model).

Tại sao mô hình Logit được sử dụng phổ biến hơn Probit?

Mặc dù mô hình Logit và Probit được xây dựng trên các giả định khác nhau về phân phối của sai số, nhưng trong thực tế hai mô hình thường cho kết quả dự báo xác suất rất tương đồng. Sự khác biệt chủ yếu nằm ở tính chất toán học, hiệu quả tính toán và khả năng diễn giải, chứ không phải ở độ chính xác dự báo.

1. Hàm phân phối tích lũy (CDF) có dạng công thức tường minh

Mô hình Logit giả định sai số tuân theo phân phối Logistic,

$$\varepsilon \sim Logistic(0, 1),$$

khi đó xác suất được tính thông qua hàm phân phối tích lũy

$$F(z) = \frac{1}{1 + e^{-z}},$$

là một công thức tường minh (closed-form expression). Ngược lại, mô hình Probit giả định

$$\varepsilon \sim N(0, 1),$$

với hàm phân phối tích lũy

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

Do hàm phân phối tích lũy của phân phối chuẩn không có biểu thức nguyên hàm dưới dạng các hàm sơ cấp, việc tính toán phải dựa trên phương pháp tích phân số hoặc các thuật toán xấp xỉ. Vì vậy, mô hình Logit thuận tiện hơn về mặt tính toán.

2. Đạo hàm có dạng rất đơn giản

Đặt

$$p = \frac{1}{1 + e^{-z}}.$$

Đạo hàm của hàm Logistic là

$$\boxed{\frac{dp}{dz} = p(1 - p).}$$

Đạo hàm chỉ phụ thuộc vào chính xác suất dự báo p, do đó các biểu thức gradient và Hessian đều gọn.

Trong khi đó, đối với mô hình Probit, $p = \Phi(z)$, thì

$$\boxed{\frac{dp}{dz} = \phi(z),}$$

trong đó

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

là hàm mật độ xác suất của phân phối chuẩn.

Mặc dù cả hai mô hình đều có thể tính toán dễ dàng bằng máy tính, nhưng đạo hàm của Logistic có dạng đại số đơn giản hơn, giúp việc xây dựng các thuật toán tối ưu trở nên thuận tiện hơn.

3. Tối ưu hóa Maximum Likelihood đơn giản hơn

Các tham số của Logistic Regression thường được ước lượng bằng phương pháp Ước lượng Hợp lý Tối đa (Maximum Likelihood Estimation – MLE).

Đối với mô hình Logit, gradient của hàm hợp lý có dạng

$$\nabla L = X^T (y - p),$$

và ma trận Hessian là

$$H = -X^T W X,$$

trong đó

$$W = diag\left(p_i(1 - p_i)\right).$$

Các biểu thức trên đều có dạng đơn giản và thuận lợi cho các thuật toán tối ưu lặp như Newton-Raphson hoặc Iteratively Reweighted Least Squares (IRLS).

4. Dễ diễn giải thông qua Odds Ratio

Một ưu điểm rất quan trọng của mô hình Logit là khả năng diễn giải trực tiếp ý nghĩa của các hệ số hồi quy.

Mô hình Logit có thể được viết dưới dạng

$$\boxed{\log\left(\frac{p}{1-p}\right) = \beta^T x,}$$

trong đó $\frac{p}{1-p}$ được gọi là **Odds** (tỷ lệ cược).

Do đó,e^{β_j} được gọi là **Odds Ratio**, cho biết khi biến giải thích x_j tăng thêm một đơn vị thì Odds sẽ thay đổi bao nhiêu lần, trong điều kiện các biến khác được giữ nguyên.

Khả năng diễn giải trực tiếp này khiến Logistic Regression trở thành mô hình tiêu chuẩn trong các lĩnh vực như chấm điểm tín dụng (Credit Scoring), quản trị rủi ro, y học, bảo hiểm và marketing.

5. Phù hợp với các thuật toán Machine Learning

Hàm Logistic (Sigmoid Function) là một trong những hàm kích hoạt (Activation Function) quan trọng nhất trong Machine Learning.

Do đạo hàm

$$\frac{d}{dz}\sigma(z) = \sigma(z)(1-\sigma(z)),$$

có dạng rất đơn giản nên việc lan truyền ngược sai số (Backpropagation) và tối ưu hóa bằng Gradient Descent được thực hiện hiệu quả.

Chính vì vậy, Logistic Regression được xem là nền tảng của nhiều thuật toán phân loại nhị phân hiện đại cũng như các mạng nơ-ron nhân tạo (Neural Networks).

Khi nào nên sử dụng mô hình Probit?

Mặc dù Logit phổ biến hơn, mô hình Probit vẫn là lựa chọn phù hợp trong một số trường hợp:

- Khi mô hình lý thuyết giả định sai số hoặc biến tiềm ẩn (Latent Variable) tuân theo phân phối chuẩn.
- Trong các nghiên cứu kinh tế lượng (Econometrics), nơi giả định phân phối chuẩn của sai số thường xuất phát từ cơ sở lý thuyết của mô hình.
- Trong sinh thống kê (Biostatistics) và tâm lý học lượng hóa (Psychometrics), khi các đặc tính tiềm ẩn của đối tượng nghiên cứu được giả định tuân theo phân phối chuẩn.
- Trong các mô hình Bayes có biến tiềm ẩn, do việc sử dụng phân phối chuẩn giúp việc suy luận hậu nghiệm (Posterior Inference) trở nên thuận tiện hơn.
- Khi cần duy trì tính nhất quán với các nghiên cứu học thuật trước đó đã sử dụng mô hình Probit.

B.6 Law of Total Variance

Công thức chung

Giả sử R là biến ngẫu nhiên cần nghiên cứu và S là biến trạng thái (state variable). Theo *Law of Total Variance*,

$$\boxed{\mathrm{Var}(R) = E[\mathrm{Var}(R \mid S)] + \mathrm{Var}(E[R \mid S])} \tag{B.1}$$

Công thức trên phát biểu rằng phương sai tổng thể (*unconditional variance*) gồm hai thành phần:

- **Within-group variance**
$$E[\mathrm{Var}(R|S)],$$
là trung bình của phương sai trong từng nhóm (hay từng trạng thái).
- **Between-group variance**
$$\mathrm{Var}(E[R|S]),$$
là phương sai của các giá trị kỳ vọng giữa các nhóm.

Do đó,

$$\boxed{\text{Total Variance} = \text{Within-group Variance} + \text{Between-group Variance.}}$$

Algebraic Proof (2-State Case)

Giả sử biến trạng thái

$$S \in \{A, B\},$$

với $P(S = A) = p, \qquad P(S = B) = 1 - p$. Đặt

$$E(R|A) = \mu_A, \qquad Var(R|A) = \sigma_A^2,$$

$$E(R|B) = \mu_B, \qquad Var(R|B) = \sigma_B^2.$$

Theo Law of Total Expectation, $\mu = E(R) = p\mu_A + (1-p)\mu_B$. Theo định nghĩa phương sai, $Var(R) = E[(R-\mu)^2]$. Áp dụng kỳ vọng có điều kiện,

$$Var(R) = pE[(R-\mu)^2|A] + (1-p)E[(R-\mu)^2|B].$$

Xét trạng thái A. Ta cộng và trừ μ_A: $R - \mu = (R - \mu_A) + (\mu_A - \mu)$. Khai triển bình phương:

$$(R-\mu)^2 = (R-\mu_A)^2 + 2(R-\mu_A)(\mu_A - \mu) + (\mu_A - \mu)^2.$$

Lấy kỳ vọng có điều kiện,

$$\begin{aligned} E[(R-\mu)^2|A] = {} & E[(R-\mu_A)^2|A] \\ & + 2(\mu_A - \mu)E[R - \mu_A|A] \\ & + (\mu_A - \mu)^2. \end{aligned}$$

Do $E(R|A) = \mu_A$, nên $E[R - \mu_A|A] = 0$.
Vì vậy, $E[(R-\mu)^2|A] = \sigma_A^2 + (\mu_A - \mu)^2$
Tương tự,$E[(R-\mu)^2|B] = \sigma_B^2 + (\mu_B - \mu)^2$
Thế trở lại biểu thức phương sai,

$$\begin{aligned} Var(R) = {} & p\left(\sigma_A^2 + (\mu_A - \mu)^2\right) \\ & + (1-p)\left(\sigma_B^2 + (\mu_B - \mu)^2\right). \end{aligned}$$

Nhóm các hạng tử,

$$\begin{aligned} Var(R) = {} & \underbrace{p\sigma_A^2 + (1-p)\sigma_B^2}_{E[Var(R|S)]} \\ & + \underbrace{p(\mu_A - \mu)^2 + (1-p)(\mu_B - \mu)^2}_{Var(E[R|S])}. \end{aligned}$$

Do đó,

$$\boxed{Var(R) = E[Var(R|S)] + Var(E[R|S]).}$$

Thành phần đầu tiên phản ánh mức độ phân tán của dữ liệu trong từng nhóm, còn thành phần thứ hai phản ánh sự khác biệt giữa các giá trị trung bình của các nhóm. Khi tất cả các nhóm có cùng giá trị kỳ vọng, thành phần *between-group variance* bằng 0 và phương sai tổng thể chỉ còn là trung bình có trọng số của các phương sai trong từng nhóm.

Appendix C

Simulation

C.1 Sử dụng Cholesky Decomposition để mô phỏng các biến ngẫu nhiên tương quan

Mục tiêu của Monte Carlo Simulation là sinh ra các biến ngẫu nhiên có đúng các đặc tính thống kê đã biết. Giả sử cần mô phỏng hai biến ngẫu nhiên $X_1,\ X_2$ thỏa mãn

$$E(X_1) = \mu_1, \qquad E(X_2) = \mu_2,$$

$$Var(X_1) = \sigma_1^2, \qquad Var(X_2) = \sigma_2^2,$$

và

$$Corr(X_1, X_2) = \rho.$$

Khi đó vector kỳ vọng là

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix},$$

và ma trận hiệp phương sai là

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

Đây là toàn bộ thông tin đầu vào của bài toán mô phỏng.

Bước 1. Sinh các biến chuẩn độc lập

Các bộ sinh số ngẫu nhiên trên máy tính có thể dễ dàng tạo ra các biến chuẩn độc lập

$$Z_1, Z_2 \sim N(0, 1),$$

hay viết dưới dạng vector

$$\mathbf{Z} \sim N(\mathbf{0}, I),$$

với

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Do đó

$$Cov(\mathbf{Z}) = I.$$

Tuy nhiên, các biến này hoàn toàn độc lập và chưa phản ánh mối tương quan mong muốn.

Bước 2. Tìm phép biến đổi tuyến tính

Giả sử tồn tại một ma trận A sao cho

$$\boxed{\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z}.}$$

Khi đó

$$E(\mathbf{X}) = \boldsymbol{\mu},$$

nghĩa là vector kỳ vọng đã được đảm bảo.
Để xác định ma trận A, xét ma trận hiệp phương sai của $\mathbf{X}$

$$Cov(\mathbf{X}) = Cov(A\mathbf{Z}).$$

Theo tính chất của ma trận hiệp phương sai,

$$Cov(A\mathbf{Z}) = A\,Cov(\mathbf{Z})\,A^T.$$

Vì

$$Cov(\mathbf{Z}) = I,$$

nên

$$Cov(\mathbf{X}) = AA^T.$$

Trong khi đó, mục tiêu của bài toán là

$$Cov(\mathbf{X}) = \Sigma.$$

Do đó, ma trận A phải thỏa mãn

$$\boxed{AA^T = \Sigma.}$$

Bước 3. Cholesky Decomposition

Theo định lý Cholesky, với mọi ma trận hiệp phương sai đối xứng xác định dương, luôn tồn tại duy nhất một ma trận tam giác dưới L sao cho

$$\boxed{\Sigma = LL^T.}$$

So sánh với điều kiện

$$AA^T = \Sigma,$$

ta chỉ cần chọn

$$\boxed{A = L.}$$

Do đó, công thức mô phỏng trở thành

$$\boxed{\mathbf{X} = \boldsymbol{\mu} + L\mathbf{Z}.}$$

Đây chính là công thức Monte Carlo tổng quát cho các biến ngẫu nhiên có tương quan.

Ví dụ với hai biến

Giả sử hệ số tương quan giữa hai biến là

$$Corr(X_1, X_2) = \rho.$$

Khi đó ma trận tương quan là

$$R = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

Giả sử ma trận Cholesky có dạng

$$L = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix}.$$

Theo định nghĩa,

$$LL^T = R.$$

Ta có

$$\begin{aligned} LL^T &= \begin{bmatrix} a & 0 \\ b & c \end{bmatrix} \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \\ &= \begin{bmatrix} a^2 & ab \\ ab & b^2 + c^2 \end{bmatrix}. \end{aligned}$$

So sánh từng phần tử với

$$R = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix},$$

thu được

$$a^2 = 1,$$

$$ab = \rho,$$

$$b^2 + c^2 = 1.$$

Suy ra

$$a = 1,$$

$$b = \rho,$$

$$c = \sqrt{1 - \rho^2}.$$

Do đó

$$\boxed{L = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{bmatrix}.}$$

Công thức mô phỏng cuối cùng

Nếu chỉ xét ma trận tương quan, ta thu được

$$X_1 = Z_1,$$
$$X_2 = \rho Z_1 + \sqrt{1-\rho^2}\, Z_2.$$

Nếu bổ sung thêm vector kỳ vọng và độ lệch chuẩn, đặt

$$D = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix},$$

thì

$$\boxed{\mathbf{X} = \boldsymbol{\mu} + DL\mathbf{Z}.}$$

Hay viết tường minh,

$$\boxed{X_1 = \mu_1 + \sigma_1 Z_1,}$$

$$\boxed{X_2 = \mu_2 + \sigma_2 \left(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2\right).}$$

Như vậy, bằng cách áp dụng Cholesky Decomposition lên ma trận hiệp phương sai (hoặc ma trận tương quan), ta có thể biến đổi các biến chuẩn độc lập thành các biến ngẫu nhiên có đúng kỳ vọng, phương sai và cấu trúc tương quan mong muốn. Đây là nền tảng của Monte Carlo Simulation trong quản trị rủi ro, định giá phái sinh và mô hình hóa danh mục đầu tư.

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